

# Algebra III

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## 1 Rings

Questions, comments, corrections are all welcome, please feel free to E-mail me!

Lecture 1

### 1.1 Definitions and examples

There are a few definitions of ring in the literature.

**Definition 1.1.** A monoid  $(M, \cdot)$  is a set  $M$  equipped with a binary operation  $\cdot : M \times M \rightarrow M$  and an element  $1_M \in M$  (called the multiplicative identity) such that

- $\cdot$  is associative, meaning  $x \cdot (y \cdot z) = (x \cdot y) \cdot z$  for all  $x, y, z \in M$ .

- $1_M \cdot m = m \cdot 1_M = m$  for all  $m \in M$ .

A monoid  $M$  is commutative if  $x \cdot y = y \cdot x$  for all  $x, y \in M$ .

**Example 1.2.** Any group is a monoid.

**Example 1.3.** The natural numbers  $\mathbb{N} = \{1, 2, 3, \dots\}$  forms a commutative monoid under multiplication, with multiplicative identity  $1_{\mathbb{N}} = 1$ .

The non-negative integers  $\mathbb{Z}_{\geq 0}$  similarly forms a commutative monoid under multiplication.

The set of  $n \times n$  real matrices  $M_n(\mathbb{R})$  forms a monoid under matrix multiplication, with multiplicative identity the unit matrix  $I$ , but for  $n > 1$  this is not a commutative monoid.

**Definition 1.4.** A ring is a set  $R$  together with functions  $+: R \times R \rightarrow R$  and  $\cdot: R \times R \rightarrow R$ , and given elements  $0_R, 1_R \in R$ , such that

- $(R, +)$  is an abelian group with identity  $0_R$ .
- $(R, \cdot)$  is a monoid with multiplicative identity  $1_R$ .
- The distributivity property holds:  $a(b+c) = ab+ac$  and  $(b+c)a = ba+ca$  for all  $a, b, c \in R$ .

A ring  $R$  is commutative if  $(R, \cdot)$  is a commutative monoid.

$+$  was a function  $R \times R \rightarrow R$  so it would make sense to write  $+(x, y)$  for  $x, y \in R$ , but for sanity we'll always write  $x + y$  for  $+(x, y)$  (and similarly  $xy$  for  $\cdot(x, y)$ ). We'll refer to  $+$  as addition and  $\cdot$  as multiplication.

Similarly  $r \in R$ , we write  $-r$  for the additive inverse of  $r$  in the group  $(R, +)$ , and will often just write  $1 = 1_R$  and  $0 = 0_R$ .

**Definition 1.5.** A subset  $S \subseteq R$  is a subring if  $0_R, 1_r \in S$ , and for all  $r, s \in S$ ,  $-r, r+s, rs \in S$  too. This implies that  $S$  is itself a ring with the operations  $+, \cdot$ .

We write  $S \leq R$  to denote that  $S$  is a subring of  $R$ .

**Example 1.6.** The usual sets of numbers are all rings:

$$\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$$

However  $\mathbb{N} \subseteq \mathbb{Z}$  is not a subring as  $-1 \notin \mathbb{N}$ .

The mod-2 integers  $\mathbb{Z}/2\mathbb{Z}$  is a ring.

**Example 1.7.** The set  $\mathbb{Z}[i] := \{a + ib \in \mathbb{C} \mid a, b \in \mathbb{Z}\}$  is called the Gaussian integers, and is a subring of  $\mathbb{C}$ .

We also have the subring  $\mathbb{Q}[\sqrt{2}] := \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$  of  $\mathbb{R}$ .

**Proposition 1.8.** Let  $R$  be a ring and  $r \in R$ . Then  $r \cdot 0_R = 0_R \cdot r = 0$ .

*Proof.*  $0_R$  is the identity element in the group  $(R, +)$  so  $0_R + 0_R = 0_R$ .

$$\begin{aligned} r \cdot 0_R &= r \cdot (0_R + 0_R) \\ &= r \cdot 0_R + r \cdot 0_R \end{aligned}$$

Adding  $(-r \cdot 0_R)$  to both sides implies  $r \cdot 0_R = 0_R$ , as desired.  $\square$

**Example 1.9.** The trivial ring has  $R = \{0\}$  with  $0 \cdot 0 := 0$  and  $0 + 0 := 0$ . It is the unique ring with a single element.

**Proposition 1.10.** *Let  $R$  be a ring. Then  $1_R = 0_R$  if and only if  $R = \{0\}$  is the trivial ring.*

*Proof.* If  $R = \{0\}$ , since there's only one element we must have  $1_R = 0_R$ .

Now suppose  $1_R = 0_R$ , and let  $r \in R$ . Then

$$\begin{aligned} r &= r \cdot 1_R \\ &= r \cdot 0_R \\ &= 0 \end{aligned}$$

□

Throughout the rest of the course, we will assume any ring we encounter is non-trivial (since the theory of the trivial ring isn't very interesting).

**Definition 1.11.** *An element  $u \in R$  is a unit if there is another element  $v \in R$  such that  $u \cdot v = v \cdot u = 1_R$ . We write  $R^\times \subseteq R$  for the set of units in  $R$ .*

*Note in the above scenario that  $v$  also lies in  $R^\times$ .*

**Definition 1.12.** *A division ring is a non-trivial ring where every non-zero element is a unit, i.e.  $R^\times = R \setminus \{0\}$ . A field is a commutative division ring.*

**Example 1.13.**  $\mathbb{Z}$  isn't a field, as 2 isn't invertible. In fact  $\mathbb{Z} = \{\pm 1\}$ .  $\mathbb{Q}, \mathbb{R}, \mathbb{C}$  are all fields.

*Similarly,  $\mathbb{Z}[i]$  isn't a field, and  $\mathbb{Z}[i]^\times = \{\pm 1, \pm i\}$ .  $\mathbb{Q}[\sqrt{2}]$  is a field.*

**Example 1.14.** Define the quaternions  $\mathbb{H}$  to be the abelian group  $\mathbb{R}^4$ , with standard basis vectors labelled  $\{1, i, j, k\}$ . Then  $\mathbb{H}$  is a ring with 1 the unit, and multiplication determined by  $ij = -ji = k$ ,  $i^2 = j^2 = -1$ , and  $(r \cdot 1) \cdot s = rs$  for any  $r \in \mathbb{R}$  and  $s \in \mathbb{H}$ . This determines all other products, such as  $k^2 = -1$ :

$$\begin{aligned} k^2 &= (ij)(-ji) \\ &= -ijji \\ &= i^2 \\ &= -1 \end{aligned}$$

and similarly  $jk = i$ .

*This ring is not commutative since  $ij \neq ji$ . However it is a division ring since*

$$(a1 + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2$$

*so if  $a1 + bi + cj + dk \neq 0$ , it has inverse*

$$\frac{1}{a^2 + b^2 + c^2 + d^2} \cdot (a - bi - cj - dk) \in \mathbb{H}$$

**Proposition 1.15.** *Inverses are unique: if  $r \in R$  has inverses  $u$  and  $v$ , then  $u = v$ .*

*Proof.* We have that  $1 = ru = rv$ , so  $r(u - v) = 0$ . Since  $ur = 1$ ,  $0 = ur(u - v) = u - v$  so  $u = v$ . □

**Proposition 1.16.** *For any ring  $R$ ,  $R^\times$  is a group under multiplication.*

*Proof.* First we show that  $R^\times$  is closed under multiplication. If  $a, b \in R^\times$ , there are some  $c, d \in R^\times$  such that  $ac = ca = bd = db = 1$ . Then  $(ab)(dc) = a(bd)c = ac = 1$ , and similarly  $(dc)(ab) = 1$ . So  $ab \in R^\times$ .

The element  $1 \in R^\times$  is a unit, and inverses exist: the inverse of  $a$  (as above) is  $c$ , which also lies in  $R^\times$ . □

## 1.2 Constructions of rings

**Example 1.17.** Let  $R, S$  be rings. Then their product  $R \times S$  is a ring with addition and multiplication defined via

$$(r, s) + (r', s') := (r + r', s + s')$$

and

$$(r, s) \cdot (r', s') := (r \cdot r', s \cdot s')$$

The zero is  $(0, 0)$  and the multiplicative unit is  $(1, 1)$ . This is indeed a ring, and is commutative if and only if both  $R$  and  $S$  are.

**Definition 1.18.** Let  $R$  be a ring. Then a polynomial  $f$  with coefficients in  $R$  is a sequence  $f = (a_0, a_1, a_2, \dots)$  in  $R$  which is eventually 0, i.e.  $a_i = 0$  for  $i$  sufficiently large.

If  $a_i = 0$  for  $i > N$ , we will write this polynomial in the following way:

$$f = a_0 + a_1X + a_2X^2 + \dots + a_NX^N$$

Note that in this notation,  $f$  is also equal to  $a_0 + a_1X + a_2X^2 + \dots + a_NX^N + 0X^{N+1}$ .

If there is some  $i$  such that  $a_i \neq 0$ , we define the degree of  $f$ ,  $\deg(f)$ , to be the maximal  $i$  such that  $a_i \neq 0$  (note this exists by assumption). If  $a_i = 0$  for all  $i$ , then  $f = 0$ , and we define  $\deg(0)$  to be  $-\infty$ .

**Definition 1.19.** Let  $R$  be a ring. Then its polynomial ring,  $R[X]$ , is defined to be the set of polynomials with coefficients in  $R$ , with operations defined as follows.

Let  $f = a_0 + \dots + a_nX^n$  and  $g = b_0 + \dots + b_mX^m$ . We may assume  $n = m$  (by adding copies of  $0X^i$  if necessary). Then

$$f + g := (a_0 + b_0) + (a_1 + b_1)X + \dots + (a_n + b_n)X^n$$

and

$$\begin{aligned} f \cdot g &:= (a_0b_0) + (a_1b_0 + a_0b_1)X + (a_2b_0 + a_1b_1 + a_0b_2)X^2 + \dots \\ &= \sum_{i=0}^{n+m} \left( \sum_{j=0}^i a_j b_{i-j} \right) X^i \end{aligned}$$

The subset of polynomials of constant polynomials, i.e. those of the form  $f = a_0$ , with  $a_i = 0$  for  $i > 0$ . This is a subring, which we identify with  $R$ .

We can check that  $R[X]$  is a ring, which is commutative if and only if  $R$  is.

Lecture 2

**Definition 1.20.** If  $f = a_0 + \dots + a_nX^n$  is a polynomial of degree  $n$ , we say  $f$  is monic if  $a_n = 1$ .

For example,  $f = 1 + X^2$  is monic, but  $f = 2$  isn't.

**Example 1.21.** A polynomial  $f = a_0 + a_1X + \dots \in R[X]$  determines a function  $R \rightarrow R$ , sending  $r$  to  $f(r) := a_0 + a_1r + \dots$ . However the polynomial  $f$  itself is just a sequence of elements of  $R$ , and  $X$  some formal symbol: we don't identify the polynomial with the function, and in fact different polynomials can have the same function.

For example, let  $R = \mathbb{Z}/2\mathbb{Z}$ , and let  $f = X \in R[X]$  and  $g = X^2 \in R[X]$ . Then  $f$  and  $g$  are distinct polynomials, but  $f(r) = g(r)$  for all  $r \in R$ .

We can take the polynomial ring of a polynomial ring, to get  $(R[X])[Y]$ . We will write this as  $R[X, Y]$ . We can iterate this any number of times, to get e.g.

$$R[X_1, \dots, X_n] := (\dots((R[X_1])[X_2])\dots)[X_n]$$

**Definition 1.22.** A Laurent polynomial  $f$  in  $R$  consists of a sequence  $f = (\dots, a_{-1}, a_0, a_1, \dots)$ , with only finitely many  $a_i$  non-zero. We write this in the form

$$f = \sum_{i \in \mathbb{Z}} a_i X^i$$

We define  $R[X, X^{-1}]$  to be set of Laurent polynomials on  $R$ . This has a ring structure defined the same as in  $R[X]$ .

**Definition 1.23.** A power series  $f$  in  $R$  consists of a sequence  $f = (a_0, a_1, \dots)$  in  $R$ , where we don't assume only finitely many  $a_i$  are non-zero. We define  $R[[X]]$  to be the ring of power series in  $R$ .

This has a ring structure defined the same as in  $R[X]$ , and contains  $R[X]$  as a subring.

**Definition 1.24.** Let  $M$  be a monoid, and  $R$  a ring. The monoid ring of  $M$  over  $R$ ,  $R[M]$ , is the set of tuples  $f = \{a_m\}_{m \in M}$ , where each  $a_m \in R$ , and only finitely many are non-zero. We write such a tuple  $f$  in the form

$$f = \sum_{m \in M} a_m m$$

We define addition and multiplication similarly to before. Let  $f = \sum_{m \in M} a_m m$  and  $g = \sum_{m \in M} b_m m$  be in  $R[M]$ , and define

$$f + g := \sum_{m \in M} (a_m + b_m) m$$

and

$$f \cdot g := \sum_{m \in M} \left( \sum_{\substack{k, l \in M \\ k \cdot l = m}} a_k b_l \right) m$$

We can check that  $R[M]$  is a ring, which is commutative if and only if both  $R$  and  $M$  are. If  $M$  is a group (i.e. it contains inverses), we call this the group ring of  $M$  over  $R$ .

**Example 1.25.** Let  $M$  be the monoid  $\mathbb{Z}_{\geq 0}$  of non-negative integers (with addition as the binary operation). Then we can identify  $R[X]$  with  $R[M]$ , by identifying  $aX^n$  with  $a \cdot n$ , where  $a \in R$  and  $n \in M$ .

If we take  $M = \mathbb{Z}$  (with addition), we can identify  $R[X, X^{-1}]$  with  $R[M]$  in a similar way.

**Example 1.26.** If  $R$  is a ring and  $n \geq 1$ , the set of  $n \times n$  matrices  $M_n(R)$  forms a ring, under the usual rules for matrix addition and multiplication.

When  $n = 2$  (and  $R$  is non-trivial),  $M_n(R)$  is never commutative, as we see from the following example:

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

whereas if we multiply the same matrices the other way around, we get

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

A similar example shows  $M_n(R)$  is never commutative when  $n \geq 2$ . However we may identify  $M_1(R)$  with  $R$ , and we see that  $M_1(R)$  is commutative if and only if  $R$  is.

### 1.3 Homomorphisms and ideals

**Definition 1.27.** Let  $R, S$  be rings. A function  $\phi : R \rightarrow S$  is a (ring) homomorphism if for all  $a, b \in R$ :

1.  $\phi(a + b) = \phi(a) + \phi(b)$
2.  $\phi(ab) = \phi(a)\phi(b)$
3.  $\phi(0) = 0$
4.  $\phi(1) = 1$

A bijective homomorphism is an isomorphism. The inverse of an isomorphism is also an isomorphism.

**Definition 1.28.** Let  $\phi : R \rightarrow S$  be a homomorphism.

- Its kernel is

$$\ker(\phi) := \{r \in R \mid \phi(r) = 0\} \subseteq R$$

- Its image is

$$\text{Im}(\phi) := \{s \in S \mid s = \phi(r) \text{ for some } r \in R\} \subseteq S$$

**Proposition 1.29.** Let  $\phi : R \rightarrow S$  be a homomorphism. Then  $\text{Im}(\phi) \subseteq S$  is a subring.

*Proof.* Since  $0 = \phi(0)$  and  $1 = \phi(1)$ , we have  $0, 1 \in \text{Im}(\phi)$ .

Suppose  $s, s' \in \text{Im}(\phi)$ . Then  $s = \phi(r)$  and  $s' = \phi(r')$  for some  $r, r' \in R$ .

Then  $s + s' = \phi(r + r') \in \text{Im}(\phi)$ .

Similarly  $0 = s - s$ , but also  $0 = \phi(0) = \phi(r - r) = s + \phi(-r)$ . So  $-s = \phi(-r) \in \text{Im}(\phi)$ .  $\square$

But if  $R$  is non-trivial, since  $\phi(1) = 1$ ,  $1 \notin \ker(\phi)$ , so  $\ker(\phi)$  isn't a subring of  $R$ .

**Proposition 1.30.** A homomorphism  $\phi : R \rightarrow S$  is injective if and only if  $\ker(\phi) = \{0\}$ .

*Proof.* If  $\phi$  is injective, since  $\phi(0) = 0$ , there is no other  $r \in R$  with  $\phi(r) = 0$  so  $\ker(\phi) = \{0\}$ .

If  $\ker(\phi) = \{0\}$ , suppose  $r, r' \in R$  have  $\phi(r) = \phi(r')$ . Then  $\phi(r - r') = 0$  so  $r - r' \in \ker(\phi) = \{0\}$ , so  $r = r'$ .  $\square$

**Definition 1.31.** Let  $I \subseteq R$ .

- $I$  is a left ideal if it's an additive subgroup of  $R$ , and if  $i \in I$  and  $r \in R$ , then  $ri \in I$ .
- $I$  is a right ideal if it's an additive subgroup of  $R$ , and if  $i \in I$  and  $r \in R$ , then  $ir \in I$ .
- $I$  is a two-sided ideal if it's an additive subgroup of  $R$ , and if  $i \in I$  and  $r \in R$ , then  $ri \in I$ , and  $ir \in I$ .

We will normally just say an *ideal* to mean a left ideal. Note that if  $R$  is commutative, all three definitions coincide.

**Example 1.32.** For any ring  $R$ ,  $\{0\}, R \subseteq R$  are both ideals.

**Example 1.33.** Suppose  $I \subseteq R$  is an ideal, and  $1 \in R$ . Then for any  $r \in R$ ,  $r \cdot 1 = r$  also lies in  $I$ . So  $I = R$ .

In particular, the only subring which is also an ideal is the whole of  $R$ .

We say an ideal  $I \subseteq R$  such that  $I \neq R$  is a proper ideal.

**Example 1.34.** We can generalise this: suppose  $I \subseteq R$  is an ideal containing some unit  $u$  with inverse  $v$ , then  $I$  contains  $vu = 1$  and hence  $I = R$ .

**Lemma 1.35.** Let  $\phi : R \rightarrow S$  be a homomorphism. Then  $\ker(\phi) \subseteq R$  is a two-sided ideal.

*Proof.* Since  $\phi$  is a homomorphism of abelian groups,  $\ker(\phi) \subseteq R$  is a subgroup of  $R$ .

Now suppose  $i \in \ker(\phi)$  and  $r \in R$ . Then

$$\phi(ri) = \phi(r)\phi(i) = \phi(r) \cdot 0 = 0$$

so  $ri \in \ker(\phi)$ . Similarly  $ir \in \ker(\phi)$ . □

**Example 1.36.** Let  $R = \mathbb{Z}$ , and  $n \in \mathbb{Z}$ . Then  $n\mathbb{Z} \subseteq \mathbb{Z}$  is an ideal.

We claim all ideals are of this form. Let  $I \subseteq \mathbb{Z}$  be an ideal. If  $I = \{0\}$ , then  $I = 0\mathbb{Z}$ , so suppose it's not zero. Let  $n \in I$  be its smallest positive element.

$rn \in I$  for all  $r \in \mathbb{Z}$  so  $n\mathbb{Z} \subseteq I$ .

Now let  $i \in I$ . By the Euclidean algorithm, we can write

$$i = an + b$$

with  $0 \leq b < n$ .  $i \in I$ , and since  $n \in I$  also  $an \in I$ , so  $b = i - an \in I$ .

Since  $n$  was the smallest positive element of  $I$ , we must have  $b = 0$ , and so  $i \in n\mathbb{Z}$ . So  $I = n\mathbb{Z}$ .

Lecture 3

Similarly to groups, we have

**Lemma 1.37.** Let  $\phi : R \rightarrow S$  and  $\psi : S \rightarrow T$  be homomorphisms of rings. Then their composition  $\psi \circ \phi$  is also a homomorphism.

**Definition 1.38.** For an element  $a \in R$ , the ideal generated by  $a$  is

$$(a) := R \cdot a = \{ra \mid r \in R\} \subseteq R$$

An ideal  $I \subseteq R$  is principal if  $I = (a)$  for some  $a$ .

If  $S \subseteq R$  is any subset, the ideal generated by  $S$  is

$$(S) := R \cdot S = \left\{ \sum_{s \in S} r_s s \mid r_s \in R, \text{ and only finitely many } r_s \text{ are non-zero} \right\}$$

In  $\mathbb{Z}$ ,  $n\mathbb{Z} = (n)$ , so we have shown that all ideals in  $R$  are principal.

**Example 1.39.** The subset

$$\{f \in R[X] \mid \text{the constant coefficient of } f \text{ is zero}\}$$

is an ideal (we won't check this). It's a principal ideal, generated by the polynomial  $X$ .

**Definition 1.40.** Let  $I \subseteq R$  be a two-sided ideal. The quotient ring  $R/I$  consists of additive cosets of the form  $r + I$ , with zero given by  $0 + I$  and one given by  $1 + I$ , and sum and product given by

$$(r + I) + (s + I) := (r + s) + I$$

and

$$(r + I) \cdot (s + I) := (rs) + I$$

**Proposition 1.41.** *The quotient ring  $R/I$  is a ring, and the function  $R \rightarrow R/I$  sending  $r$  to  $r + I$  is a (surjective) ring homomorphism.*

The motivation for the definition of a two-sided ideal is that it's exactly the definition that makes this proposition true, much like the definition of a normal subgroup.

*Proof.* We know addition is well-defined (since  $(R/I, +)$  is the quotient of  $(R, +)$  by a normal subgroup), so we need to check multiplication is well-defined.

Let  $r, s \in R$ , and let  $i \in I$ . So  $r + I = r + i + I$ . Then

$$\begin{aligned}(r + i + I) \cdot (s + I) &= rs + is + I \\ &= rs + I\end{aligned}$$

since  $is \in I$ . Similarly  $(r + I)(s + i + I) = (rs + I)$ . So multiplication is well-defined.

Associativity and distributivity follow from those of  $R$ , and one can check similarly that  $0 + I$  and  $1 + I$  are the zero and one respectively.  $\square$

**Example 1.42.** *Consider the ideal  $n\mathbb{Z} \subseteq \mathbb{Z}$ . Elements of the quotient ring  $\mathbb{Z}/n\mathbb{Z}$  are cosets of the form  $r + n\mathbb{Z}$  for  $r = 0, \dots, (n-1)$ , and addition and multiplication are the usual addition and multiplication and addition mod  $n$ .*

**Example 1.43.** *Let  $R$  be a ring, and  $\phi : G \rightarrow H$  a group homomorphism. Then  $\phi$  induces a ring homomorphism  $\phi_* : R[G] \rightarrow R[H]$ , defined by*

$$\phi_* \left( \sum_{g \in G} a_g g \right) := \sum_{g \in G} a_g \phi(g)$$

*What's  $\ker(\phi_*)$ ? Let  $N = \ker(\phi) \trianglelefteq G$ , a normal subgroup of  $G$ . Consider the (left) ideal  $(N-1) \subseteq R[G]$  generated by elements of  $R[G]$  of the form  $g-1$  for  $g \in N$ . Then  $\phi_*(g-1) = 0$  for all  $g \in N$ , so  $(N-1) \subseteq \ker(\phi_*)$ .*

*One can show that  $(N-1)$  is in fact a two-sided ideal, and that  $\ker(\phi_*) = (N-1)$ . This will be on the first problem sheet.*

**Example 1.44.** *Let  $R$  be a commutative ring. Consider the ideal  $(X) \subseteq R[X]$ . Let's compute  $R[X]/(X)$ .*

*Elements of this quotient ring are of the form*

$$a_0 + a_1X + \dots + a_nX^n + (X)$$

*Everything but the first term is in  $(X)$ . So any such element is equivalent to  $a_0 + (X)$ , and this representation is in fact unique.*

*So there is an isomorphism  $R \rightarrow R[X]/(X)$  sending  $a$  to  $a + (X)$ .*

What about  $\mathbb{R}[X]/(X^2 + 1)$ ? This is harder; we'll need the following result, whose proof is on the first problem sheet.

**Proposition 1.45** (Euclidean algorithm for polynomials over a field). *Let  $F$  be a field and  $f, g \in F[X]$ , with  $g \neq 0$ . Then there are some  $r, q \in F[X]$  such that*

$$f = gq + r$$

*with  $\deg(r) < \deg(g)$ .*

This says that we can use the Euclidean algorithm on polynomials over a field. We'll revisit this property later.

**Example 1.46.** Let  $R = \mathbb{R}[X]/(X^2 + 1)$ . Elements of  $R$  are of the form

$$a_0 + a_1X + \dots + a_nX^n + (X^2 + 1)$$

Letting  $f = a_0 + \dots + a_nX^n$ , we can apply the Euclidean for polynomials, and find  $q, r \in R$  such that

$$f = q(X^2 + 1) + r$$

with  $\deg(r) < 2$ , so  $r = b_0 + b_1X$  for some  $b_0, b_1 \in \mathbb{R}$ .

So any element of  $R$  is of the form  $a + bX + (X^2 + 1)$  for some  $a, b \in \mathbb{R}$ .

This representation is unique: if  $a + bX + (X^2 + 1) = a' + b'X + (X^2 + 1)$ , then  $(a - a') + (b - b')X = (X^2 + 1)g$  for some  $g \in R$ , but if  $g \neq 0$ ,  $\deg((X^2 + 1)g) > 1$ . So we must have  $g = 0$  and so  $a = a'$  and  $b = b'$ .

So every element of  $R$  is of the form  $a + bX + (X^2 + 1)$ , and the element  $X + (X^2 + 1)$  squares to  $-1$ . This looks a lot like the complex numbers.

Define a function  $\phi : R \rightarrow \mathbb{C}$  by

$$\phi(a + bX + (X^2 + 1)) := a + bi \in \mathbb{C}$$

We can check manually that this is well-defined, bijective and a ring homomorphism.

This was a lot of work. The following theorem will make results like this much easier.

**Theorem 1.47** (First isomorphism theorem for rings). Let  $\phi : R \rightarrow S$  be a ring homomorphism. Recall that  $\ker(\phi) \subseteq R$  is a two-sided ideal.

Then there is an isomorphism of rings

$$R/\ker(\phi) \cong \text{Im}(\phi)$$

*Proof.* Define  $\psi : R/\ker(\phi) \rightarrow \text{Im}(\phi)$  by

$$\psi(r + \ker(\phi)) := \phi(r)$$

This is well-defined, since if  $r + \ker(\phi) = r' + \ker(\phi)$  then  $r - r' \in \ker(\phi)$  so  $\phi(r) = \phi(r')$ ; it's a ring homomorphism too.

This is an isomorphism of abelian groups, by the first isomorphism theorem for groups.  $\square$

**Example 1.48.** We can define a homomorphism  $\phi : \mathbb{R}[X] \rightarrow \mathbb{C}$  by setting  $\phi(f) := f(i)$ . Then we can check that  $\phi$  is surjective and that  $\ker(\phi) = (X^2 + 1)$ . Therefore  $\mathbb{C} \cong \mathbb{R}(X)/(X^2 + 1)$ .

**Theorem 1.49** (Second isomorphism theorem). Let  $S$  be a ring and  $R \leq S$  a subring, and  $I \subseteq S$  a two-sided ideal. Then

1.  $R + I := \{r + i \mid r \in R, i \in I\} \leq S$  is a subring.
2.  $I \subseteq R + I$  and  $R \cap I \subseteq R$  are both two-sided ideals.
3.  $(R + I)/I = \{r + I \mid r \in R\} \leq S/I$  is a subring, and

$$R/(R \cap I) \cong (R + I)/I$$

*Proof.* We first show  $R + I \leq S$  is indeed a subring. Since  $R \subseteq R + I$ ,  $R + I$  contains 0 and 1.

Let  $r, s \in R$  and  $i, j \in I$ , so  $r+i, s+j \in R+I$ . Then  $(r+i)+(s+j) = (r+s)+(i+j) \in R+I$ , and

$$(r+i) \cdot (s+j) = (rs) + (is + rj + ij)$$

which is in  $R + I$  (the second term is in  $I$  since  $I$  is a two-sided ideal).

$I \subseteq R + I$  is a two-sided ideal since  $I \subseteq S$  is one. We define a homomorphism  $\phi : R \rightarrow S/I$  by  $\phi(r) := r + I$ . Then  $\ker(\phi)$  consists of elements  $r \in R$  such that  $r + I = I$ , i.e.  $r \in I$ . So  $\ker(\phi) = R \cap I$  which is therefore a two-sided ideal in  $R$ .

$\text{Im}(\phi) = \{r + I \mid r \in R\} = (R + I)/I$ , so this is a subring of  $S/I$ . But by the first isomorphism theorem,  $\text{Im}(\phi) \cong R/(R \cap I)$ . So

$$R/(R \cap I) \cong (R + I)/I$$

□

**Theorem 1.50** (Third isomorphism theorem). *Let  $R$  be a ring, and  $I, J \subseteq R$  two-sided ideals such that  $I \subseteq J$ .*

*Then  $J/I \subseteq R/I$  is a two-sided ideal and*

$$(R/I)/(J/I) \cong R/J$$

*Proof.* Define a homomorphism  $\phi : R/I \rightarrow R/J$  by

$$\phi(r + I) := r + J$$

This is a well-defined and surjective ring homomorphism.

$\ker(\phi)$  consists of those  $r + I \in R + I$  such that  $r + J = 0$ , i.e.  $r \in J$ . So  $\ker(\phi) = J/I$ . The result then follows from the first isomorphism theorem. □

There's a similar important result, which doesn't have a standard name:

**Proposition 1.51.** *Let  $R$  be a ring, and  $I \subseteq R$  a two-sided ideal. Then there is a natural bijection*

$$\begin{aligned} \{\text{two-sided ideals of } R/I\} &\longleftrightarrow \{\text{two-sided ideals of } R \text{ containing } I\} \\ \alpha : J \subseteq R/I &\longmapsto \{r \in R \mid r + I \in J\} \\ K/I \subseteq R/I &\longleftarrow I \subseteq K \subseteq R : \beta \end{aligned}$$

*Proof.* Let  $\alpha$  be the map going right and  $\beta$  the map going left. One can check that  $\alpha(J) \subseteq R$  and  $\beta(K) \subseteq R/I$  are two-sided ideals, so  $\alpha$  and  $\beta$  are well-defined functions.

We need to check these are inverse to each other. First let  $K \subseteq R$  be a two-sided ideal containing  $I$ . Then

$$\begin{aligned} \alpha(\beta(K)) &= \alpha(K/I) \\ &= \{r \in R \mid r + I \in K/I\} \\ &= K \end{aligned}$$

where the last equality is because  $I \subseteq K$ .

Let  $J \subseteq R/I$  be a two-sided ideal. Then

$$\begin{aligned}\beta(\alpha(J)) &= \beta\left(\{r \in R \mid r + I \in J\}\right) \\ &= \{r + I \mid r + I \in J\} \\ &= J\end{aligned}$$

So  $\alpha$  and  $\beta$  must be bijections.  $\square$

Let  $R$  be any ring. There's a (unique) homomorphism  $\iota : \mathbb{Z} \rightarrow R$ , where  $\iota(n) = 1_R + \dots + 1_R$  ( $n$  times) if  $n \geq 0$ , and  $\iota(n) = -(1_R + \dots + 1_R)$  ( $-n$  times) if  $n \leq 0$ . This is a homomorphism by distributivity.

Any homomorphism  $\mathbb{Z} \rightarrow R$  must be equal to  $\iota$ , since it must send 1 to  $1_R$ .

$\ker(\iota) \subseteq \mathbb{Z}$  is an ideal so we must have  $\ker(\iota) = n\mathbb{Z}$  for some  $n \in \mathbb{Z}$ .

**Definition 1.52.** The characteristic of  $R$  is the unique  $n \geq 0$  such that  $\ker(\iota) = n\mathbb{Z}$ , where  $\iota$  is the unique ring homomorphism  $\mathbb{Z} \rightarrow R$ .

**Example 1.53.**  $\mathbb{Q}$  has characteristic 0 since  $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$  is injective so its kernel is 0. Similar  $\mathbb{R}, \mathbb{C}, \mathbb{Z}[i]$  all have characteristic 0.

The homomorphism  $\iota : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$  sends  $r$  to  $r + n\mathbb{Z}$  and has kernel  $n\mathbb{Z}$ , so  $\mathbb{Z}/n\mathbb{Z}$  has characteristic  $n$ , for any  $n \geq 1$ .

In particular, any  $n \geq 0$  is the characteristic of some commutative ring.

## 2 Integral domains

In this section, we will assume all rings are commutative and non-trivial.

The archetypal ring is the integers,  $\mathbb{Z}$ . However it has lots of special properties (e.g. unique factorisation), that we'd like to study in a more general context. We'll introduce several types of ring and study their properties.

### 2.1 Integral domains and ideals

**Definition 2.1.** Let  $R$  be a (commutative) ring. An element  $r \in R$  is a zero divisor if  $r \neq 0$  and there is some  $s \neq 0$  such that  $rs = 0$ .

$R$  is an integral domain (ID) if it contains no zero divisors, i.e. if  $rs = 0$ , then either  $r = 0$  or  $s = 0$ .

**Example 2.2.** The ring of integers  $\mathbb{Z}$  is an integral domain.

Any field is an integral domain, since if  $rs = 0$  with  $r \neq 0$ , then if we let  $r^{-1}$  be an inverse of  $r$ , then

$$0 = r^{-1}rs = s$$

**Example 2.3.**  $\mathbb{Z}/6\mathbb{Z}$  isn't an integral domain, since  $2 + 6\mathbb{Z}$  and  $3 + 6\mathbb{Z}$  are non-zero, but their product is 0.

More generally:

**Lemma 2.4.** Let  $n \geq 1$ .  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain if and only if  $n$  is prime.

*Proof.* Suppose  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain, and suppose  $n|rs$ , for some  $r, s \in \mathbb{Z}$ . This means  $(r + n\mathbb{Z})(s + n\mathbb{Z}) = 0$ , so since  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain,  $r + n\mathbb{Z} = 0$  or  $s + n\mathbb{Z} = 0$ . So  $n|r$  or  $n|s$ , so  $n$  is prime.

Suppose  $n$  is prime, and suppose  $(r + n\mathbb{Z})(s + n\mathbb{Z}) = 0$ . Then  $rs + n\mathbb{Z} = 0$  so  $n|rs$ . So  $n|r$  or  $n|s$  so  $r + n\mathbb{Z} = 0$  or  $s + n\mathbb{Z} = 0$ .  $\square$

**Example 2.5.** If  $R$  is an integral domain and  $S \leq R$  is a subring, then  $S$  is also an integral domain, since a zero divisor in  $S$  would also be one in  $R$ .

$\mathbb{C}$  is an integral domain since it's a field. So  $\mathbb{Z}[i] \leq \mathbb{C}$  is an integral domain.

**Lemma 2.6.** Let  $R$  be an integral domain. Then  $R[X]$  is also an integral domain.

Iterating this, we see that if  $R$  is an integral domain, then  $R[X_1, \dots, X_n]$  is an integral domain for all  $n$ .

*Proof.* Let  $f, g \in R[X]$  be non-zero, and write them as

$$f = a_0 + \dots + a_n X^n$$

and

$$g = b_0 + \dots + b_m X^m$$

where  $a_n$  and  $b_m$  are non-zero (so  $n = \deg(f)$  and  $m = \deg(g)$ ).

Then the coefficient of  $X^{n+m}$  in  $fg$  is  $a_n b_m$ , which is non-zero since  $R$  is an integral domain. So  $fg \neq 0$ .  $\square$

So in particular  $\mathbb{Z}[X]$  is an integral domain.

It is sometimes useful to have alternative descriptions of these notions in terms of ideals.

**Lemma 2.7.** A (non-trivial, commutative) ring  $R$  is a field if and only if its only ideals are  $\{0\}$  and  $R$ .

Since  $\{0\}$  and  $R$  are always ideals in  $R$ , this is equivalent to  $R$  only having two ideals.

*Proof.* ( $\Rightarrow$ ) Suppose  $R$  is a field and  $I \subseteq R$  is a non-zero ideal, so it contains some non-zero element  $r$ . Then since  $R$  is a field,  $r$  is a unit, so  $I = R$ .

( $\Leftarrow$ ) Let  $r \in R$  be non-zero. Then  $(r) \subseteq R$  is a non-zero ideal, and so must be  $R$  by assumption. So there is some  $s$  such that  $rs = 1$ , so  $r$  is a unit. So every non-zero  $r \in R$  is a unit, therefore  $R$  is a field.  $\square$

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**Definition 2.8.** An ideal  $I \subseteq R$  is maximal if  $I \neq R$  and any proper ideal  $J \subseteq R$  containing  $I$  is equal to  $I$ .

**Lemma 2.9.** An ideal  $I \subseteq R$  is maximal if and only if  $R/I$  is a field.

*Proof.* ( $\Rightarrow$ ) Suppose  $I$  is maximal. Then the only ideals in  $R$  containing  $I$  are  $I$  and  $R$ , by the bijection between such ideals and ideals in  $R/I$ , we see that the only ideals in  $R/I$  are  $0$  and  $R/I$ , so  $R/I$  is a field.

( $\Leftarrow$ ) Suppose  $R/I$  is a field, and suppose  $J \subseteq R$  is a proper ideal containing  $I$ . Then  $J + I \subseteq R/I$  is a proper ideal of a field, and so must be  $\{0\}$ . So  $J = I$ .  $\square$

We can give a similar description for integral domains.

**Definition 2.10.** An ideal  $I \subseteq R$  is prime if  $I \neq R$  and whenever there are  $r, s \in R$  such that  $rs \in I$ , we have that  $r \in I$  or  $s \in I$ .

**Example 2.11.** Let  $n \in \mathbb{Z}$  be non-zero. Then the ideal  $n\mathbb{Z} \subseteq \mathbb{Z}$  is prime if and only if  $n$  is prime:

Suppose  $n$  is prime. If  $rs \in n\mathbb{Z}$ , then  $n|rs$ . So  $n|r$  or  $n|s$ , i.e.  $r \in n\mathbb{Z}$  or  $s \in n\mathbb{Z}$ .

Conversely suppose  $n = uv$  is composite (where  $u, v \neq 0, \pm 1$ ). Then  $n = uv \in n\mathbb{Z}$  but  $u, v \notin n\mathbb{Z}$  (since  $0 < |u|, |v| < |n|$ ).

**Lemma 2.12.** An ideal  $I \subseteq R$  is prime if and only if  $R/I$  is an integral domain.

*Proof.* ( $\Rightarrow$ ) First suppose  $I \subseteq R$  is prime. Suppose we have elements  $r + I, s + I \in R/I$  such that  $(r + I)(s + I) = 0 + I$ . This means  $rs \in I$ . Since  $I$  is prime, one of  $r$  and  $s$  is in  $I$ , so one of  $r + I$  and  $s + I$  is 0.

( $\Leftarrow$ ) Suppose  $R/I$  is an integral domain, and suppose we have  $r, s \in R$  such that  $rs \in I$ . Then  $(r + I)(s + I) = 0 + I$ , so one of  $r + I, s + I$  is 0, so one of  $r, s$  is in  $I$ .  $\square$

**Example 2.13.** Let  $R$  be a ring. Then  $R[X]/(X) \cong R$ , so the ideal  $(X) \subseteq R[X]$  is prime iff  $R$  is an integral domain, and it's maximal iff  $R$  is a field.

**Corollary 2.14.** If  $I \subseteq R$  is a maximal ideal, it is a prime ideal.

*Proof.* Since  $I$  is maximal,  $R/I$  is a field, so it is an integral domain, so  $I$  is a prime ideal.  $\square$

(We could have proved this directly from the definitions if we wanted).

**Example 2.15.** The converse isn't true.  $(0) \subseteq \mathbb{Z}$  is a prime ideal but isn't maximal.

Recall any  $n \geq 0$  is the characteristic of some commutative ring.

**Lemma 2.16.** Let  $R$  be an integral domain. Then its characteristic is either 0 or prime.

*Proof.* Let  $\iota : \mathbb{Z} \rightarrow R$  be the unique ring homomorphism, so  $\ker(\iota) = n\mathbb{Z}$ , where  $n \geq 0$  is the characteristic of  $R$ .

Then by the first isomorphism theorem,  $\text{Im}(\iota) \leq R$  is isomorphic to  $\mathbb{Z}/n\mathbb{Z}$ . Since  $R$  is an integral domain,  $\mathbb{Z}/n\mathbb{Z}$  is one too. We proved earlier that if  $n > 0$  and  $\mathbb{Z}/n\mathbb{Z}$  is an integral domain,  $n$  must be prime.  $\square$

## 2.2 Factorisation

Throughout this subsection, we'll only work with rings that are integral domains. We'll explore some number-theoretic properties of certain classes of rings.

**Definition 2.17.** Let  $R$  be an integral domain, and  $r, s \in R$ .

- We say that  $r$  divides  $s$ , written  $r|s$ , if there is some  $u \in R$  such that  $s = ru$ .

This is equivalent to having  $(s) \subseteq (r)$ .

- We say  $r$  and  $s$  are associates if there is some unit  $u \in R^\times$  such that  $s = ru$ .

This is equivalent to having  $(r) = (s)$ , or that both  $r|s$  and  $s|r$ .

**Example 2.18.** In  $\mathbb{Z}$ ,  $r$  and  $s$  are associates if and only if  $r = \pm s$ .

This isn't true in general: in  $\mathbb{Z}[i]$ ,  $2i$  and  $2$  are associates.

**Definition 2.19.** Let  $R$  be an integral domain.  $r \in R$  is irreducible if  $r \neq 0$ ,  $r$  is not a unit, and if  $r = uv$ , then  $u$  or  $v$  is a unit.

**Definition 2.20.** Let  $R$  be an integral domain.  $r \in R$  is prime if  $r \neq 0$ ,  $r$  is not a unit, and whenever  $r|uv$ , either  $r|u$  or  $r|v$  (or both).

**Example 2.21.** These properties depend on the ring, not just the element. For example, 2 is prime in  $\mathbb{Z}$ , but not in  $\mathbb{Q}$  (since it's a unit in  $\mathbb{Q}$ ).

Similarly,  $2X$  is irreducible in  $\mathbb{Q}[X]$ , but not in  $\mathbb{Z}[X]$ .

In  $\mathbb{Z}$  primality and irreducibility coincide, but not in general. Furthermore, another important number-theoretic property of  $\mathbb{Z}$  is the fact that any  $n \in \mathbb{Z}$  has an essentially unique prime factorisation (unique up to reordering and signs), but in the following example, we see an example of factorisation into irreducibles being non-unique.

**Example 2.22.** Consider the ring

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$$

This is a subring of  $\mathbb{C}$  so it's an integral domain. We claim 3 is irreducible but not prime in  $R$ .

2 and 3 both divide  $6 = (1 - \sqrt{-5})(1 + \sqrt{-5})$  but neither 2 nor 3 divides either factor on the right, so 2 and 3 both aren't prime.

Suppose  $3 = (a + b\sqrt{-5})(u + v\sqrt{-5})$ ,  $a, b, u, v \in \mathbb{Z}$ . Applying  $|\cdot|^2$  to both sides, we see that

$$9 = (a^2 + 5b^2)(u^2 + 5v^2)$$

The only solutions are  $a + b\sqrt{-5} = \pm 3$  and  $u + v\sqrt{-5} = \pm 1$ , and the other way around. So 3 is irreducible.

A similar argument shows 2 and  $1 \pm \sqrt{-5}$  are all irreducible. So 6 can be factored into irreducibles in 2 distinct ways.

**Lemma 2.23.** A principal ideal  $(r)$  in an integral domain  $R$  is a prime ideal if and only if  $r = 0$  or  $r$  is prime.

*Proof.* ( $\Rightarrow$ ) Suppose  $(r)$  is a prime ideal. If  $r = 0$ , we're done, so assume  $r \neq 0$ . Since prime ideals aren't the whole ring,  $r$  isn't a unit.

Now suppose  $r|uv$ . So  $uv \in (r)$ . Since  $(r)$  is prime,  $u \in (r)$  or  $v \in (r)$ . So  $r|u$  or  $r|v$ . So  $r$  is prime.

( $\Leftarrow$ ) Conversely, if  $r = 0$ , then the ideal  $(r) = \{0\}$  is prime since  $R$  is an integral domain.

Suppose  $r \neq 0$  is prime. If  $uv \in (r)$ , this means  $r|uv$ , so  $r|u$  or  $r|v$ . So  $u \in (r)$  or  $v \in (r)$ . So  $(r)$  is prime. □

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**Lemma 2.24.** If  $r \in R$  is prime, then  $r$  is irreducible.

We saw in the example  $3 \in \mathbb{Z}[\sqrt{-5}]$  that the converse isn't true.

*Proof.* Let  $r \in R$  be prime, and suppose  $r = uv$ . So  $r|uv$ , so since  $r$  is prime,  $r|u$  or  $r|v$ ; WLOG we assume  $r|u$ . So there is some  $s \in R$  such that  $rs = u$ . So  $r = uv = r(sv)$ . Since  $R$  is an integral domain, this implies  $sv = 1$ , so  $v$  was a unit. □

**Definition 2.25.** An integral domain  $R$  is a Euclidean domain (ED) if there is a function  $\theta : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$  (called a Euclidean function) such that

- $\theta(rs) \geq \theta(r)$  for all  $r, s \neq 0 \in R$ .

- For all  $a, b \in R$  with  $b \neq 0$ , there are  $q, r \in R$  such that

$$a = qb + r$$

and  $r = 0$  or  $\theta(r) < \theta(b)$ .

**Example 2.26.**  $\mathbb{Z}$  is a Euclidean domain with  $\theta(n) = |n|$ .

**Example 2.27.** If  $F$  is a field, then  $F[X]$  is a Euclidean domain, with  $\theta(f) = \deg(f)$ .

**Example 2.28.** If  $F$  is a field, then  $F$  is a Euclidean domain, with  $\theta(r) = 0$  for all  $r$ .

Here's a much more interesting example.

**Proposition 2.29.** The Gaussian integers  $\mathbb{Z}[i]$  is a Euclidean domain with  $\theta(r) = |r|^2$ .

*Proof.* First, note that  $\theta(rs) = \theta(r)\theta(s)$ , and if  $r \neq 0 \in \mathbb{Z}[i]$ , then  $\theta(r) \geq 1$ . So if  $r, s \neq 0$  then  $\theta(rs) \geq \theta(r)$ .

Suppose we choose  $a, b \in \mathbb{Z}[i]$  such that  $b \neq 0$ .

Choose some  $q \in \mathbb{Z}[i]$  such that  $|\frac{a}{b} - q| < 1$ . We can do this; every complex number has distance at most 1 from some Gaussian integer. We can rearrange this to

$$\frac{a}{b} = q + c$$

where  $|c| < 1$  (so we let  $c = \frac{a}{b} - q$ ). Multiplying by  $b$  and setting  $r = bc = a - bq \in \mathbb{Z}[i]$ , we have

$$a = qb + r$$

Since  $|c| < 1$ ,  $r = bc$  either has  $r = 0$  or  $\theta(r) = |r|^2 < \theta(b)$ , so we're done.  $\square$

This strategy doesn't just work for  $\mathbb{Z}[i]$ . This would have worked for any subring  $R \leq \mathbb{C}$  such that for any  $r \in \mathbb{C}$ , there is a point in  $R$  of distance less than 1 from  $r$ . This doesn't hold for  $\mathbb{Z}[\sqrt{-5}]$ , and in fact this ring isn't a Euclidean domain, but this does hold for some other rings of interest.

**Definition 2.30.** Let  $R$  be an integral domain. Then  $R$  is a principal ideal domain (PID) if every ideal is principal, i.e. for any ideal  $I \subseteq R$ ,  $I = (r)$  for some  $r \in R$ .

**Example 2.31.**  $\mathbb{Z}$  is a principal ideal domain.

**Theorem 2.32** (ED  $\Rightarrow$  PID). Let  $R$  be a Euclidean domain. Then  $R$  is a principal ideal domain.

The proof is essentially identical to the case  $R = \mathbb{Z}$ , which we already saw, but using an arbitrary Euclidean function instead of  $|\cdot|$ .

*Proof.* Let  $\theta$  be the Euclidean function for  $R$ . Let  $I \subseteq R$  be an ideal, which we can assume to be non-zero.

Choose  $b \neq 0 \in I$  with  $\theta(b)$  minimised, and let  $a \in I$  be another element. Then there are some  $q, r \in R$  with  $a = qb + r$ , with  $r = 0$  or  $\theta(r) < \theta(b)$ .

Since  $I$  is an ideal and  $a, b \in I$ ,  $r = a - qb$  also lies in  $I$ . We can't have that  $\theta(r) < \theta(b)$  and  $r \neq 0$  (because of how we chose  $b$ ), so  $r = 0$ . So  $a = qb$ , and so  $a \in (b)$ . Since  $a \in I$  was arbitrary, the ideal  $I = (b)$  is principal.  $\square$

**Example 2.33.** For any field  $F$ ,  $F[X]$  is a Euclidean domain and hence a principal ideal domain. Similarly  $\mathbb{Z}[i]$  is also a principal ideal domain.

**Proposition 2.34.** The ideal  $(2, X) \subset \mathbb{Z}[X]$  is not generated by a single element.

Therefore  $\mathbb{Z}[X]$  is not a principal ideal domain (and hence not a Euclidean domain either).

*Proof.* Suppose there is some  $f \in \mathbb{Z}[X]$  such that  $(2, X) = (f)$ .

$2 \in (f)$  so there is some  $g \in \mathbb{Z}[X]$  such that  $fg = 2$ . So  $f$  must have degree 0, and  $f$  must be  $\pm 1$  or  $\pm 2$ .

$f \neq \pm 1$  since  $1 \notin (2, X)$ . Similarly  $f \neq \pm 2$  since 2 does not divide  $X$ . So  $(2, X)$  can't be a principal ideal.  $\square$

**Example 2.35.** Let  $F$  be a field and  $A \in M_n(F)$ . Consider the set

$$I := \{f \in F[X] \mid f(A) = 0\} \subseteq F[X]$$

This is an ideal: if  $f, g \in I$ ,  $(f + g)(A) = f(A) + g(A) = 0$ ; if  $f \in I$  and  $h \in F[X]$ ,  $(fh)(A) = f(A)h(A) = 0$ .

Since  $F[X]$  is a principal ideal domain,  $I = (m)$  for some  $m \in F[X]$ .

Then  $m(A) = 0$  and for any  $f \in F[X]$  such that  $f(A) = 0$ ,  $m \mid f$ .  $m$  is called the minimal polynomial of  $A$ .

Existence of minimal polynomials can be proven much more directly but would be much longer.

**Definition 2.36.** An integral domain  $R$  is a unique factorisation domain (UFD) if

- Every non-zero non-unit  $r \in R$  is a product of irreducibles.
- If  $p_1 \dots p_n = q_1 \dots q_m$  with each  $p_i, q_j \in R$  irreducible, then  $n = m$ , and they can be reordered such that each  $p_i$  is an associate of  $q_i$  (i.e. they're related by multiplication by a unit).

This says that factorisations into irreducibles exist, and hence are unique (so the name is very sensible).

Our next goal will be to prove that principal ideal domains are unique factorisation domains, but we need a few preliminaries first.

**Lemma 2.37.** Let  $R$  be a principal ideal domain. Then a non-zero principal ideal  $(r) \subseteq R$  is maximal if and only if  $r$  is irreducible or  $(r = 0 \text{ and } R \text{ is a field})$ .

*Proof.*  $(\Rightarrow)$  Suppose  $(r)$  is a maximal ideal, and that  $r$  is not irreducible. If  $r = 0$ , then we know  $R = R/(r)$  must be a field. Otherwise, we can assume that  $r = xy$  for  $x, y \in R$  non-units. So  $(r) \subseteq (x) \subseteq R$ .

Since  $x$  isn't a unit,  $(x) \neq R$ . Since  $(r)$  is maximal,  $(r) = (x)$ . So  $r = xz$  for some unit  $z \in R^\times$ .

So  $r = xz = xy$  so  $x(z - y) = 0$  so  $z = y$  (since  $x \neq 0$  and  $R$  is an integral domain). However  $y$  isn't a unit whereas  $z$  is, which is a contradiction.

$(\Leftarrow)$  If  $r = 0$  and  $R$  is a field, then  $(r)$  is maximal. So we instead assume  $r$  is irreducible and that  $(r)$  is non-maximal. So there is some ideal  $(r) \subseteq I \subseteq R$  such that  $I \neq (r), R$ .

Since  $R$  is a principal ideal domain,  $I = (s)$  for some non-unit  $s \in R$ . Since  $(r) \subseteq (s)$ ,  $r = sz$  for some non-unit  $z \in R$ , which contradicts irreducibility of  $r$ .  $\square$

Lecture 7

**Lemma 2.38.** Let  $R$  be a principal ideal domain. If  $r \in R$  is irreducible, it is prime.

Recall primes are always irreducible in any integral domain, but in the example  $\mathbb{Z}[\sqrt{-5}]$ , the element 2 was irreducible but not prime. So the lemma shows  $\mathbb{Z}[\sqrt{-5}]$  isn't a principal ideal domain.

*Proof.* If  $r \in R$  is irreducible, then  $(r) \subseteq R$  is a maximal ideal, by the previous lemma. Therefore it is also a prime ideal. Since the ideal  $(r)$  is prime and  $r \neq 0$ , a previous lemma implies that  $r$  is prime.  $\square$

**Corollary 2.39.** *Let  $R$  be a principal ideal domain. Then every non-zero prime ideal is maximal.*

We already know that maximal ideals are prime. Therefore this corollary says that in a principal ideal domain, prime ideals and maximal ideals are the same (with the exception of the zero ideal, which is always prime if  $R$  is a PID but maximal iff  $R$  is a field).

*Proof.* Let  $I \subseteq R$  be a non-zero prime ideal. Since  $R$  is a PID,  $I = (r)$  for some  $r \in R$ . Since  $I \neq 0$  was a prime ideal,  $r$  is prime (by a previous lemma). So  $r$  is irreducible (recall primes are always irreducible). Therefore by another earlier lemma,  $(r)$  is maximal.  $\square$

We need one more property of PIDs.

**Proposition 2.40.** *Let  $R$  be a principal ideal domain, and  $I_1 \subseteq I_2 \subseteq \dots \subseteq R$  an increasing sequence of ideals in  $R$ . Then this sequence is eventually constant, i.e. there is some integer  $N$  such that  $I_n = I_{n+1}$  for each  $n \geq N$ .*

Rings which have this property are called *Noetherian*. They're a very important class of rings, and are studied in more depth in the commutative algebra course.

*Proof.* Let  $I = \cup_{i \geq 1} I_i \subseteq R$ . This is an ideal in  $R$  (proof is on the problem sheet). Since  $R$  is a PID,  $I = (r)$  for some  $r \in I = \cup_i I_i$ , so there is some  $N$  such that  $r \in I_N$ . Then

$$(r) \subseteq I_N \subseteq I_{N+1} \subseteq \dots \subseteq I = (r)$$

So for  $n \geq N$ ,  $I_n = (r)$ .  $\square$

We can finally prove the theorem.

*Proof that (PID  $\Rightarrow$  UFD).* We first show uniqueness. Suppose  $p_1 \dots p_n = q_1 \dots q_m$ , where each  $p_i, q_j \in R$  is irreducible.

In particular,  $p_1 | q_1 \dots q_m$ . Since  $p_1$  is irreducible, it's prime, so  $p_1 | q_j$  for some  $j$ . Reordering if necessary, we assume  $p_1 | q_1$ , so  $p_1 a = q_1$  for some  $a$ . Since  $q_1$  is irreducible,  $a$  is a unit, so  $p_1, q_1$  are associates.

Since  $R$  is an integral domain, we can cancel  $p_1$  to obtain

$$p_2 \dots p_n = (aq_2)q_3 \dots q_m$$

and, replacing  $q_2$  with  $aq_2$ , we get

$$p_2 \dots p_n = q_2 \dots q_m$$

and repeating the process, we find that (possibly after reordering)  $p_i$  and  $q_i$  are associates for all  $i$ .

We also find that  $n = m$ : if not, WLOG  $n > m$ , and we find that  $p_{m+1} \dots p_n = 1$ , so  $p_n$  is a unit, which is a contradiction.

We next show existence. Suppose  $r \in R$  is a non-unit which can't be factored as a product of irreducibles. In particular,  $r$  isn't irreducible. So we can write  $r = r_1 s_1$ , with  $r_1, s_1 \in R$  non-units.

Since  $r$  isn't a product of irreducibles, at least one of  $r_1, s_1$  isn't either, WLOG it's  $r_1$ . So we write  $r_1 = r_2 s_2$  with  $r_2, s_2 \in R$  non-units. Again, WLOG  $r_2$  isn't a product of irreducibles, so  $r_2 = r_3 s_3$  with  $r_3, s_3 \in R$  non-units, and we may continue this process indefinitely.

By assumption this process doesn't end, so we have an increasing sequence of ideals:

$$(r) \subseteq (r_1) \subseteq (r_2) \subseteq \dots \subseteq R$$

Since  $R$  is a PID and hence Noetherian, there is some  $N$  such that  $(r_N) = (r_{N+1}) = (r_{N+2}) = \dots$

Now since  $(r_N) = (r_{N+1})$  and  $r_N = r_{N+1} s_{N+1}$ ,  $s_{N+1}$  must be a unit, which is a contradiction ( $s_{N+1}$  was a non-unit by construction). So  $r$  must be a product of irreducibles.  $\square$

**Example 2.41.**  $\mathbb{Z}[i]$  is a Euclidean domain, and therefore a PID and also a UFD.

**Definition 2.42.** Let  $R$  be a ring. An element  $d \in R$  is a greatest common divisor (GCD) of a finite sequence  $a_1, \dots, a_n \in R$  if  $d|a_i$  for all  $i$ , and if any other  $d' \in R$  such that  $d'|a_i$  for all  $i$ , then  $d'|d$ .

In general, these don't always exist. If a GCD exists, it's not unique, but it is unique up to multiplication by units.

**Proposition 2.43.** Let  $R$  be a unique factorisation domain. Then for any  $a_1, \dots, a_n \in R$  not all zero, they have a greatest common divisor  $d$ , and any other greatest common divisor  $d'$  is an associate of  $d$ .

The proof is essentially the same as in  $\mathbb{Z}$ .

*Proof.* Let  $p_1, \dots, p_m \in R$  be a collection of irreducibles in  $R$  such that any irreducible factor of any  $a_i$  is an associate of some  $p_j$ , and no  $p_i, p_j$  are associates of each other for  $i \neq j$ .

We can write

$$a_i = u_i \prod_{j=1}^m p_j^{k_{ij}}$$

where each  $k_{ij} \in \mathbb{Z}_{\geq 0}$  and each  $u_i$  is a unit. We let  $l_j = \min_i \{k_{ij}\}$  for all  $j$ , and define

$$d := \prod_{j=1}^m p_j^{l_j}$$

Since  $l_j \leq k_{ij}$  for all  $i$ , it follows that  $d|a_i$  for all  $i$ .

Suppose  $d' \in R$  also satisfies  $d'|a_i$  for all  $i$ . Then we can write

$$d' = v \prod_{j=1}^m p_j^{c_j}$$

for some unit  $v$ , and some  $c_j \in \mathbb{Z}_{\geq 0}$ . Since  $d'|a_i$  for all  $i$  we must have  $c_j \leq l_j$  for all  $j$ . So  $d'|d$ , so  $d$  is a GCD.

Suppose  $d, d'$  are both GCDs of  $a_1, \dots, a_n$ . Then they must divide each other and hence are associates.  $\square$

We can summarise some of the main things we've proved so far in the following chain of implications:

$$\mathbb{Z} \Rightarrow \text{ED} \Rightarrow \text{PID} \Rightarrow \text{UFD} \Rightarrow \text{ID} \Rightarrow \text{Commutative ring} \Rightarrow \text{Ring}$$

However neither of these implications goes both ways. Counterexamples to the other direction of each implication respectively are: Lecture 8

1.  $\mathbb{Q}, \mathbb{Z}[i]$
2.  $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ . We won't justify that this ring is a PID but not an ED as it would be quite a long digression.
3.  $\mathbb{Z}[X]$ . We saw already this isn't a PID; we'll see in the next section that it is a UFD.
4.  $\mathbb{Z}[\sqrt{-5}]$
5.  $\mathbb{Z}/6\mathbb{Z}$
6.  $M_2(\mathbb{R})$

### 2.3 Factorisation in polynomial rings

Recall that if  $R$  is an integral domain,  $R[X]$  also is. This forms an important class of rings.

An important special case of this is  $F[X]$  when  $F$  is a field.  $F[X]$  is a Euclidean domain, principal ideal domain and unique factorisation domain, which by results of the previous section implies:

1. If  $I \subseteq F[X]$  is a non-zero ideal, then  $I = (f)$  for some non-zero  $f \in F[X]$ , and  $I$  is maximal if and only if  $I$  is prime.
2. An element  $f \in F[X]$  is irreducible if and only if it's prime.

We saw that the following are equivalent for  $f \in F[X]$ :

1.  $f$  is irreducible
2.  $f$  is prime
3.  $F[X]/(f)$  is an integral domain
4.  $F[X]/(f)$  is a field

**Definition 2.44.** Let  $R$  be an integral domain. Its field of fractions  $F$  is defined as follows.

As a set, let

$$F := \{(a, b) \in R \times R \mid b \neq 0\} / \sim$$

where  $(a, b) \sim (c, d)$  if  $ad = bc$ .

**Claim 2.45.**  $\sim$  is an equivalence relation.

*Proof of claim.* Symmetry and reflexivity are straightforward. We still need to show transitivity.

Suppose  $(a, b) \sim (c, d)$  and  $(c, d) \sim (e, f)$ .

This means  $ad = bc$  and  $cd = de$ . Multiplying these equations by  $f$  and  $b$  respectively, this tells us that  $adf = bcf$  and  $bcf = bde$  respectively. So  $adf = bde$ .

Since  $d \neq 0$  and  $R$  is an integral domain, we can cancel the  $d$ , and find that  $af = be$ , meaning  $(a, b) \sim (e, f)$ . □

We write the equivalence class of  $(a, b)$  as  $a/b$  or  $\frac{a}{b}$ .  
We equip  $F$  with a ring structure as follows. Define

$$0_F := \frac{0}{1}$$

and

$$1_F := \frac{1}{1}$$

and for  $a/b, c/d \in F$ , we define

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}$$

and

$$\frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}$$

**Lemma 2.46.** *The operations  $+$  and  $\cdot$  on  $F$  defined above are well-defined, i.e. they respect the equivalence relation  $\sim$ , and  $F$  is a ring.*

Proving this is straightforward; here are some examples of what needs to be checked, and the rest is similar.

**Example 2.47.** *Let  $a/b, c/d \in F$ . Then*

$$\begin{aligned} 0_F + \frac{a}{b} &= \frac{0}{1} + \frac{a}{b} \\ &= \frac{0 \cdot b + 1 \cdot a}{1 \cdot b} \\ &= \frac{a}{b} \end{aligned}$$

so  $0_F$  is an additive identity for  $F$ .

$$\begin{aligned} \left( \frac{a}{b} \cdot \frac{c}{d} \right) \cdot \frac{e}{f} &= \left( \frac{ac}{bd} \right) \cdot \frac{e}{f} \\ &= \frac{ace}{bdf} \\ &= \frac{a}{b} \cdot \left( \frac{c}{d} \cdot \frac{e}{f} \right) \end{aligned}$$

so the product  $\cdot$  on  $F$  is associative.

**Proposition 2.48.** *The field of fractions  $F$  is a field, and the set*

$$\left\{ \frac{r}{1} \mid r \in R \right\} \leq F$$

*is a subring isomorphic to  $R$ .*

*Proof.* First, let  $a/b \in F$  be non-zero. This means that  $a \neq 0$ , so  $b/a \in F$  too. Then

$$\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ba} = 1$$

so  $a/b$  is a unit.

Let  $\phi : R \rightarrow F$  be the homomorphism sending  $r$  to  $r/1$ . This has kernel:

$$\ker(\phi) = \left\{ r \in R \mid \frac{r}{1} = \frac{0}{1} \right\} = \{0\}$$

so  $\phi$  is injective. We see that  $\text{Im}(\phi) \leq F$  is a subring and by the first isomorphism theorem,  $\text{Im}(\phi)$  is isomorphic to  $R$ .  $\square$

**Example 2.49.** *The field of fractions of  $\mathbb{Z}$  is  $\mathbb{Q}$ .*

*The field of fractions of  $\mathbb{C}[X]$  is*

$$\left\{ \frac{p}{q} \mid p, q \in \mathbb{C}[X], q \neq 0 \right\}$$

*which is a ring we haven't seen before.*

*In general, if  $F$  is a field, we write  $F(X)$  for the field of fractions of the polynomial ring  $F[X]$ .*

We recall a definition from earlier:

**Definition 2.50.** *If  $R$  is a ring, and  $f \in R[X]$ , then  $f$  is monic if  $f = a_0 + a_1 + \dots + a_{n-1}X^{n-1} + 1 \cdot X^n$ , i.e. the top coefficient is 1.*

**Definition 2.51.** *Let  $R$  be a UFD, and  $f = a_0 + \dots + a_nX^n \in R[X]$ . Then the content of  $f$ , written  $c(f)$ , is the greatest common divisor of the coefficients  $a_0, \dots, a_n$ .*

*This is only well-defined up to associates, though note that the ideal it generates  $(c(f))$  is completely well-defined.*

**Definition 2.52.** *Let  $R$  be a UFD and  $f \in R[X]$ . We say  $f$  is primitive if  $c(f)$  is a unit, i.e. the coefficients of  $f$  are all coprime.*

**Example 2.53.** *Let  $f = a_0 \in R[X]$  be a constant polynomial. Then  $c(f) = a_0$  (up to associates).*

*Let  $f = 2 + 3X + 4X^2 \in \mathbb{Z}[X]$ . Then  $c(f)$  is a unit so  $f$  is primitive.*

*Let  $g = 2 + 4X + 6X^2 \in \mathbb{Z}[X]$ . Then  $c(g) = 2$  (up to associates) so  $g$  isn't primitive.*

*However 2 is a unit in  $\mathbb{Q}$  so  $g$  is primitive in  $\mathbb{Q}[X]$ .*

**Lemma 2.54.** *Let  $R$  be a UFD, and  $f \in R[X]$ . Then*

$$f = c(f) \cdot f'$$

*for some primitive  $f' \in R[X]$ .*

*Proof.* By the definition of GCD,  $c(f) \mid a_i$  for all  $i$ , so  $a_i = c(f)b_i$  for some  $b_i \in R$ , such that the GCD of  $b_1, \dots, b_n$  is 1.

Let  $f' = b_0 + \dots + b_nX^n$ . Then  $f'$  is primitive and  $f = c(f) \cdot f'$ .  $\square$

**Lemma 2.55.** *Let  $R$  be a UFD. If  $f, g \in R[X]$  are primitive, so is  $f \cdot g$ .*

*Proof.* Write  $f = a_0 + \dots + a_nX^n$  and  $g = b_0 + \dots + b_mX^m$ , with  $a_n, b_m \neq 0$ , and  $f, g \in R[X]$  primitive.

Suppose  $fg$  isn't primitive. So  $c(fg)$  isn't a unit.

Since  $R$  is a UFD, there's some irreducible  $p \in R$  with  $p \mid c(fg)$ .

By assumption,  $c(f)$  and  $c(g)$  are units, so  $p$  doesn't divide  $c(f)$  or  $c(g)$ .

This implies  $p$  doesn't divide all  $a_i$  or all  $b_j$ . So there are some  $k, l \geq 0$  such that  $p|a_0, p|a_1, \dots, p|a_{k-1}$  but  $p$  doesn't divide  $a_k$ , and similarly  $p|b_0, p|b_1, \dots, p|b_{l-1}$  but  $p$  doesn't divide  $b_l$ .

Consider the  $X^{k+l}$  coefficient in  $f \cdot g$ . This is

$$\sum_{i+j=k+l} a_i b_j = (a_{k+l} b_0 + \dots + a_{k+1} b_{l-1}) + a_k b_l + (a_{k-1} b_{l+1} + \dots + a_0 b_{l+k})$$

Note that in the first set of brackets in the RHS,  $p$  divides each  $b_j$ , and in the final set of brackets,  $p$  divides each  $a_i$ , by assumption. Furthermore  $p$  divides this whole coefficient by assumption. So  $p$  must divide the only other term, which is  $a_k b_l$ .

Since  $p$  is irreducible and hence prime,  $p|a_k$  or  $p|b_l$ , either of which provides a contradiction.

So  $c(f \cdot g)$  is a unit so  $f \cdot g$  is primitive.  $\square$

**Corollary 2.56.** *Let  $R$  be a UFD. Then for  $f, g \in R[X]$ ,  $c(f \cdot g)$  is an associate of  $c(f) \cdot c(g)$ .*

**Remark 2.57.** *First note that this would imply the previous lemma, but we will use the previous lemma to prove this corollary.*

*Also note that we can't really say  $c(f \cdot g) = c(f) \cdot c(g)$ , since both are only well-defined up to associates.*

*Proof.* Write  $f = c(f) \cdot f'$  and  $g = c(g) \cdot g'$  where  $f', g' \in R[X]$  are primitive (we can do this by an earlier lemma).

Then  $f \cdot g = c(f)c(g)(f' \cdot g')$ .

$f'$  and  $g'$  are primitive so by the previous lemma,  $f' \cdot g'$  is primitive, so  $c(f)c(g)$  is a GCD of the coefficients of  $f \cdot g$ .  $\square$

**Lemma 2.58** (Gauss' lemma). *Let  $R$  be a UFD, and  $f \in R[X]$  a primitive polynomial. Let  $F$  be the field of fractions of  $R$ .*

*Then  $f$  is irreducible in  $R[X]$  if and only if it's irreducible in  $F[X]$ .*

This is useful because checking irreducibility in  $F[X]$  is generally harder than checking it in  $R[X]$ . Here is an example to illustrate this.

**Example 2.59.** *Let  $f = 1 + X + X^3 \in \mathbb{Z}[X]$ .*

*$c(f) = 1$  so  $f$  is primitive. Now suppose  $f$  is irreducible in  $\mathbb{Q}[X]$ .*

*By Gauss' lemma,  $f$  is reducible in  $\mathbb{Z}[X]$ . So  $f = g \cdot h$ , where  $g, h \in \mathbb{Z}[X]$  are non-units.*

*Since  $g$  and  $h$  are non-units, they can't be constant since all coefficients of  $f$  are 0 and 1, so they would be either 0 or  $\pm 1$ .*

*$\deg(g) + \deg(h) = \deg(f) = 3$  so WLOG  $\deg(g) = 1$  and  $\deg(h) = 2$ . So we can write  $g = b_0 + b_1 X$  and  $h = c_0 + c_1 X + c_2 X^2$ , where the  $b_i, c_j$  are in  $\mathbb{Z}$ .*

*Since  $f = g \cdot h$ , the  $X^0$  coefficient of  $f$  is*

$$1 = b_0 c_0$$

*and the  $X^3$  coefficient is*

$$1 = b_1 c_2$$

*So since all  $b_i, c_j$  are all integers, all of  $b_0, c_0, b_1, c_2$  must be  $\pm 1$ . In particular,  $g = \pm 1 \pm X$ . Since  $g$  is a factor of  $f$ , this implies  $\pm 1$  is a root of  $f$ .*

*But we can compute  $f(1)$  and  $f(-1)$ , and they're both non-zero, which is a contradiction. So  $f$  is irreducible in  $\mathbb{Q}[X]$ . In particular  $\mathbb{Q}[X]/(f)$  is a field.*

Note that when we concluded  $b_0$  and  $b_1$  are  $\pm 1$ , we really needed to be working in  $\mathbb{Z}$ - if we were in  $\mathbb{Q}$  this would tell us a lot less. It was Gauss' lemma that allowed us to work over  $\mathbb{Z}$  rather than  $\mathbb{Q}$ .

*Proof of Gauss' lemma.* Let  $R$  be a UFD,  $F$  its field of fractions, and  $f \in R[X]$  a primitive polynomial.

( $\Rightarrow$ ) Suppose  $f$  is reducible in  $R[X]$ . Then  $f = gh$  where  $g, h \in R[X]$  are both non-units.  $f$  is primitive so both  $g$  and  $h$  are too.

If  $g$  had degree 0, since it's primitive it would be a unit, which it's not. So  $\deg(g) > 0$ . Similarly  $\deg(h) > 0$ .

Since  $\deg(g), \deg(h) > 0$ ,  $g$  and  $h$  aren't units in  $F[X]$ . So writing  $f = gh$  and viewing  $f, g, h$  as lying in  $F[X]$ , we see that  $f$  is reducible in  $F[X]$ .

( $\Leftarrow$ ) Suppose  $f$  is reducible in  $F[X]$ . So  $f = g \cdot h$ , where  $g, h \in F[X]$  are non-units; this implies that  $\deg(g), \deg(h) > 0$ .

We now clear denominators: we choose  $a, b \in R \setminus \{0\}$  such that  $ag$  and  $bh$  lie in  $R[X]$  (we can do this, for example by taking  $a$  to be the product of all denominators of coefficients of  $g$  and similarly for  $b$ ).

Let  $g' = ag$  and  $h' = bh$ , these both lie in  $R[X]$ . Note that since  $h$  may not be in  $R[X]$ , so writing  $h' = bh$  isn't a factorisation on  $R[X]$ , and similarly for  $g' = ag$ .

Plugging in, we find that

$$abf = g'h' \in R[X]$$

and all factors in this equation lie in  $R[X]$ . So we can write  $g' = c(g') \cdot g''$  and  $h' = c(h') \cdot h''$ , where  $g'', h'' \in R[X]$  are both primitive.

Since  $f$  is primitive,  $c(abf) = ab$ . But by the above equation, this is equal to  $c(g'h') = c(g')c(h')$  (up to associates). So  $ab$  is an associate of  $c(g')c(h')$ , i.e.  $uab = c(g')c(h')$  where  $u \in R^\times$  is some unit. Therefore

$$\begin{aligned} abf &= g'h' \\ &= c(g')c(h')g''h'' \\ &= uabg''h'' \end{aligned}$$

Since  $R[X]$  is an integral domain and  $ab \neq 0$ , we can cancel the factors of  $ab$  from both sides, obtaining

$$f = u \cdot g'' \cdot h''$$

Here  $u \in R^\times$  is a unit and  $g'', h'' \in R[X]$  both lie in  $R[X]$ , and have positive degree and hence are non-units. So  $f$  is reducible in  $R[X]$ .  $\square$

This was quite a long proof, a good exercise is to trace through this for  $R = \mathbb{Z}$  and  $F = \mathbb{Q}$ . We now come to the main theorem of this section.

**Theorem 2.60.** *Let  $R$  be a UFD. Then  $R[X]$  is a UFD.*

Iterating this, this implies that if  $R$  is a UFD,  $R[X_1, \dots, X_n]$  is a UFD, for  $n \geq 1$ .

This also shows that  $\mathbb{Z}[X]$  is a UFD; we showed earlier it's not a PID, which completes the proof that not all UFDs are PIDs.

*Proof.* We first prove existence of factorisations. Let  $f \in R[X]$  be non-zero and not a unit.

We first write  $f = c(f) \cdot f_1$ , with  $f_1 \in R[X]$  primitive. Since  $R$  is a UFD, we can factor the content as

$$c(f) = p_1 \dots p_n$$

where each  $p_i \in R$  is irreducible in  $R$  (and hence irreducible in  $R[X]$ ). Next, we deal with  $f_1$ .

Suppose  $f_1$  isn't the product of irreducibles. In particular,  $f_1$  itself isn't irreducible, so we can write  $f_1 = f_2 g_2$ , with  $f_2, g_2 \in R[X]$  non-units.

Also since  $f_1$  is primitive,  $f_2$  and  $g_2$  must also be primitive, and also can't be constants. So we must have  $\deg(f_2), \deg(g_2) > 0$ .

Since  $\deg(f_1) = \deg(f_2) + \deg(g_2)$ , we must also have  $\deg(f_2), \deg(g_2) < \deg(f_1)$ .

If both  $f_2, g_2$  were products of irreducibles, then  $f_1$  would be too. So we can assume WLOG that  $f_2$  isn't a product of irreducibles. We apply the same argument to  $f_2$  to write  $f_2 = f_3 \cdot g_3$  as a product of non-units.

Continuing like this gives a sequence  $f_1, f_2, f_3 \dots \in R[X]$  of non-zero elements, with  $\deg(f_1) > \deg(f_2) > \dots > 0$ . But there can't be an infinite sequence of positive integers which is strictly decreasing, so we have a contradiction.

So we can write  $f_1 = q_1 \dots q_m$ , with each  $q_i \in R[X]$  irreducible.

So  $f = p_1 \dots p_n \cdot q_1 \dots q_m$ , where all the  $p_i$  are irreducible constant polynomials, and all the  $q_i$  are irreducible non-constant polynomials. So factorisations into irreducibles exist.

We next prove uniqueness of factorisations of a non-zero non-unit  $f \in R[X]$ .

First we deal with the content. We write  $c(f) = p_1 \dots p_n$  as before, a factorisation of  $c(f)$  into irreducibles in  $R$  (noting this is unique up to reordering and associates since  $R$  is a UFD). Cancelling the content out, we only need to show that primitive polynomials can be factored uniquely.

Suppose  $f' \in R[X]$  is primitive and that we are given two factorisations

$$f' = q_1 \dots q_m = r_1 \dots r_l$$

where each  $q_i, r_j \in R[X]$  is irreducible. Note that  $f'$  is primitive, so since each  $c(q_i)$  and  $c(r_j)$  is a factor of  $c(f')$  (which is a unit), each  $q_i$  and  $r_j$  must be primitive.

Let  $F$  be the field of fractions of  $R$ , and consider  $q_i, r_j \in F[X]$ . By Gauss' lemma since the  $q_i$  and  $r_j$  are irreducible in  $R[X]$ , they're also irreducible in  $F[X]$ .

Since  $F[X]$  is an ED, it's a PID and hence a UFD, by uniqueness of factorisation in  $F[X]$ , we find that  $m = l$ , and (possibly after reordering),  $r_i$  and  $q_i$  are associates in  $F[X]$  for all  $i$ , i.e.  $r_i = u_i q_i$  for some units  $u_i \in F[X]$  (we'd like to prove the same thing but with each  $u_i \in R[X]$  instead).

Note each  $u_i$  must be a constant polynomial since it's a unit. So  $u_i = a_i/b_i$ , for some elements  $a_i, b_i \in R$ , which must both be non-zero.

Now  $a_i r_i = b_i q_i$ , and all factors in this equation are in  $R[X]$ .  $r_i$  and  $q_i$  are primitive, so taking contents of both sides, we find that  $a_i, b_i$  are associates.

So we can write  $b_i = v_i a_i$ , for some unit  $v_i \in R^\times$ . Therefore

$$a_i r_i = v_i a_i q_i$$

Cancelling the  $a_i$  factor (noting it's non-zero so we can do this), we find that  $r_i = v_i \cdot q_i$ .  $v_i$  is a unit in  $R^\times$  so it's also a unit in  $R[X]^\times$ . So  $r_i$  and  $q_i$  are associates (in  $R[X]$ ) for each  $i$ , implying factorisations into irreducibles are unique in the appropriate sense.  $\square$

That was quite a long proof. If you're confused, a good exercise is to trace through it in the case  $R = \mathbb{Z}$ . Here are some comments about this proof.

**Remark 2.61.** *Showing existence of factorisations was similar to showing  $(PID \Rightarrow UFD)$ : first we found factorisations  $f_1 = f_2 g_2, f_2 = f_3 g_3, \dots$  into non-units, and then we showed this process must terminate; here we did this second step by showing that at each stage we decrease degrees, whereas earlier we did this using the Noetherian property.*

To show uniqueness, we first took out out factors of the contents to turn everything into primitives, used Gauss' lemma to argue reducibility in  $R[X]$  and  $F[X]$  are the same (noting we're working with primitive polynomials at this stage), and used the fact that  $F[X]$  is a UFD.

**Proposition 2.62** (Eisenstein's criterion). *Let  $R$  be a UFD, and let  $f = a_0 + a_1X + \dots + a_nX^n \in R[X]$  be a primitive polynomial with  $a_n \neq 0$ .*

*Let  $p \in R$  be irreducible (and hence prime) such that*

1.  *$p$  doesn't divide  $a_n$*
2.  *$p$  divides  $a_i$  for all  $0 \leq i < n$*
3.  *$p^2$  doesn't divide  $a_0$ .*

*Then  $f$  is irreducible in  $R[X]$  (and hence also in  $F[X]$ , by Gauss' lemma.*

The proof is on the second problem sheet. These seem like strange conditions, but this is actually really useful, as illustrated in the following examples.

**Example 2.63.** *Consider  $f = X^n - p \in \mathbb{Z}[X]$ , where  $p \in \mathbb{N}$  is prime. Note all the conditions for Eisenstein's criterion hold. So  $\mathbb{Z}[X]$  is irreducible in  $\mathbb{Z}[X]$  and hence by Gauss' lemma also in  $\mathbb{Q}[X]$ .*

*In particular,  $X^n - p$  has no rational zeroes, i.e.  $\sqrt[n]{p}$  is irrational for  $n > 1$ .*

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**Example 2.64.** *Let  $f = X^{p-1} + X^{p-2} + \dots + 1 \in \mathbb{Z}[X]$ , for  $p \in \mathbb{N}$  prime. Note that*

$$f = \frac{X^p - 1}{X - 1}$$

*Since all the coefficients are 1, Eisenstein's criterion can't be applied with any prime.*

*Let  $Y = X - 1$ . Then  $f(X) = \hat{f}(Y)$  where  $\hat{f} \in \mathbb{Z}[Y]$  is given by*

$$\begin{aligned} \hat{f} &= \frac{(Y+1)^p - 1}{Y} \\ &= Y^{p-1} + \binom{p}{1}Y^{p-2} + \binom{p}{2}Y^{p-3} + \dots + \binom{p}{p-1} \end{aligned}$$

*Now Eisenstein's criterion can be applied with the prime  $p$  (note  $\hat{f}$  is primitive since the top coefficient is 1, and  $p \mid \binom{p}{i}$  for  $1 \leq i \leq p-1$ , but  $p \nmid \binom{p}{p-1}$ ). So  $\hat{f} \in \mathbb{Z}[Y]$  is irreducible.*

*If we had a factorisation  $f = gh$  in  $\mathbb{Z}[X]$ , then  $\hat{f}(Y) = g(Y+1)h(Y+1)$ . So  $f \in \mathbb{Z}[X]$  is irreducible.*

*In particular, none of the roots of  $f$  are rational. Note we already knew this: the roots are  $e^{\frac{2\pi ik}{p}}$  for  $1 \leq k \leq p-1$ , which aren't even real for  $p > 2$ .*

## 2.4 Algebraic integers

We defined rings like  $\mathbb{Z}[i], \mathbb{Z}[\sqrt{-5}]$  as e.g.  $\{a + bi \mid a, b \in \mathbb{Z}\}$ . What about  $\mathbb{Z}[e^{\frac{2\pi i}{n}}]$ ?

**Definition 2.65.** *A complex number  $\alpha \in \mathbb{C}$  is called an algebraic integer if it's the root of some monic polynomial  $f \in \mathbb{Z}[X]$ , i.e.  $f(\alpha) = 0$ .*

**Remark 2.66.** *Note not all complex numbers can be algebraic integers: there are only countably many polynomials with integer coefficients, each of which only has finitely many roots, so the set of algebraic integers is countable. But  $\mathbb{C}$  is uncountable.*

**Definition 2.67.** For  $\alpha$  an algebraic integer, we write  $\mathbb{Z}[\alpha] \leq \mathbb{C}$  for the smallest subring containing  $\alpha$ . More explicitly,

$$\mathbb{Z}[\alpha] := \bigcap_{S \leq \mathbb{C}, \alpha \in S} S$$

where we take the intersection over all subrings  $S \leq \mathbb{C}$  containing  $\alpha$ .

Equivalently, we can let  $\phi_\alpha : \mathbb{Z}[X] \rightarrow \mathbb{C}$  be the homomorphism sending  $f$  to  $f(\alpha)$ , and let  $\mathbb{Z}[\alpha] = \text{Im}(\phi_\alpha) \leq \mathbb{C}$ .

Note that by the first isomorphism theorem,

$$\mathbb{Z}[\alpha] \cong \frac{\mathbb{Z}[X]}{\ker(\phi_\alpha)}$$

and since  $\alpha$  is an algebraic integer,  $\ker(\phi_\alpha)$  is non-empty.

**Proposition 2.68.** Let  $\alpha \in \mathbb{C}$  be an algebraic integer, and  $\phi_\alpha : \mathbb{Z}[X] \rightarrow \mathbb{C}$  the homomorphism sending  $f$  to  $f(\alpha)$ . Let  $I = \ker(\phi_\alpha) \subseteq \mathbb{Z}[X]$ .

Then the ideal  $I$  is principal, and  $I = (f_\alpha)$ , for some  $f_\alpha \in \mathbb{Z}[X]$  which is irreducible and monic.

Note that since we require that  $f_\alpha$  is monic, these conditions determine  $f_\alpha$  uniquely.

**Definition 2.69.** We call the polynomial  $f_\alpha$  as above the minimal polynomial of  $\alpha$ .

*Proof of the proposition.* We know there is some monic  $f \in \mathbb{Z}[X]$  such that  $f(\alpha) = 0$ , so  $f \in I$ , so  $I \neq \emptyset$ .

Let  $f_\alpha \in I$  be a non-zero polynomial of minimal degree in  $I$ . If  $f_\alpha$  isn't primitive, we can write  $f_\alpha = c(f_\alpha) \cdot f'_\alpha$  for some primitive  $f'_\alpha \in I$ , so we can in fact assume we have chosen  $f_\alpha$  to be primitive. We claim  $f_\alpha$  generates  $I$ ,  $f_\alpha$  is irreducible, and (up to associates)  $f_\alpha$  is monic.

Let  $h \in I$ . By the Euclidean algorithm in  $\mathbb{Q}[X]$ , we can write

$$h = f_\alpha \cdot q + r$$

for some  $q, r \in \mathbb{Q}[X]$ , with  $r = 0$  or  $\deg(r) < \deg(f_\alpha)$ .

We clear denominators: there is some  $a \neq 0 \in \mathbb{Z}$  such that  $aq, ar \in \mathbb{Z}[X]$ . So

$$ah = f_\alpha \cdot (aq) + (ar)$$

Evaluating at  $\alpha$  shows that  $ar(\alpha) = 0$ , so  $ar \in I$ . Since  $f_\alpha \in I$  has minimal degree among non-zero elements, we must have  $ar = 0$ , so  $r = 0$ .

So  $ah = (aq) \cdot f_\alpha$  is a factorisation in  $\mathbb{Z}[X]$ . But we wanted to factorise  $h$ , not  $ah$ . Taking contents:

$$\begin{aligned} ac(h) &= c(ah) \\ &= c(f_\alpha) \cdot c(aq) \\ &= c(aq) \end{aligned}$$

(the last equality is up to associates) since  $f_\alpha$  is primitive. So  $a|c(aq)$ . So  $aq = a\bar{q}$ , for some  $\bar{q} \in \mathbb{Z}[X]$ . Since  $a \neq 0$ , we must have  $q = \bar{q} \in \mathbb{Z}[X]$ .

So  $h = f_\alpha \cdot q \in I$ , implying  $I = (f_\alpha)$ .

Since  $\mathbb{Z}[\alpha] \leq \mathbb{C}$ ,  $\mathbb{Z}[X]/I$  is an integral domain, so  $I$  is a prime ideal, so  $f_\alpha$  is prime and hence irreducible.

Since  $I$  contained some monic polynomial  $f$  and  $f_\alpha|f$ , the top coefficient of  $f_\alpha$  must be a unit in  $\mathbb{Z}$ , i.e.  $\pm 1$ . If it's  $-1$ , we take  $-f_\alpha$ . Either way, we may assume  $f_\alpha$  is monic.  $\square$

**Example 2.70.** 1.  $\alpha = i$  is an algebraic integer with minimal polynomial  $f_i = X^2 + 1$ .

2.  $\alpha = \sqrt{2}$  is an algebraic integer with minimal polynomial  $X^2 - 2$ .

3.  $\alpha = \frac{1}{2}(1 + \sqrt{-3})$  is an algebraic integer with minimal polynomial  $X^2 - X - 1$ .

**Lemma 2.71.** Let  $\alpha \in \mathbb{Q}$  be an algebraic integer. Then  $\alpha \in \mathbb{Z}$ .

The terminology “algebraic integer” would be rather odd if this lemma weren’t true.

*Proof.* Let  $f_\alpha \in \mathbb{Z}[X]$  be the minimal polynomial of  $\alpha$ . Then  $f_\alpha$  is irreducible and primitive, so by Gauss’ lemma,  $f_\alpha$  is irreducible in  $\mathbb{Q}[X]$ .

Since  $f_\alpha(\alpha) = 0$  and  $\alpha \in \mathbb{Q}$ ,  $X - \alpha | f_\alpha$  in  $\mathbb{Q}[X]$ . Since  $f_\alpha$  is irreducible (and monic),  $f_\alpha = X - \alpha$ . But since  $f_\alpha \in \mathbb{Z}[X]$ , we must have  $\alpha \in \mathbb{Z}$ .  $\square$

**Remark 2.72.** It turns out that the set of algebraic integers is a subring of  $\mathbb{C}$ .

We now turn our attention to a specific subring of algebraic integers:  $\mathbb{Z}[i]$ . This is a UFD. We’ll characterise its prime/irreducible elements.

**Proposition 2.73.** Let  $p \in \mathbb{N}$  be a prime number. Then  $p$  is prime in  $\mathbb{Z}[i]$  if and only if  $p$  can’t be written as  $a^2 + b^2$  for  $a, b \neq 0 \in \mathbb{Z}$ .

*Proof.* ( $\Rightarrow$ ) If  $p = a^2 + b^2$ , then  $p = (a + ib)(a - ib)$  isn’t irreducible, so it isn’t prime.

( $\Leftarrow$ ) Suppose  $p$  is reducible in  $\mathbb{Z}[i]$ . So  $p = uv$ , with  $u, v \in \mathbb{Z}[i]$  non-units. Taking norm squareds, we find that  $p^2 = |u|^2 |v|^2$ . Since  $u, v$  aren’t units, we must have that  $|u|^2, |v|^2 \neq 1$ . So  $|u|^2 = p$ . If  $u = a + ib$ , this says that  $p = a^2 + b^2$ .  $\square$

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Before we say more about primes in  $\mathbb{Z}[i]$ , we need the following technical lemma.

**Lemma 2.74.** Let  $p \in \mathbb{N}$  be a prime number, and let  $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$  be the field with  $p$  elements. Let  $\mathbb{F}_p^\times = \mathbb{F}_p \setminus \{0\}$  be its group of units.

Then  $\mathbb{F}_p^\times \cong C_{p-1}$  is cyclic of order  $p - 1$ .

*Proof.*  $\mathbb{F}_p^\times$  has order  $p - 1$ , and is abelian. From the classification of finite abelian groups, if  $\mathbb{F}_p^\times$  isn’t cyclic, it contains a subgroup isomorphic to  $C_m \times C_m$  for some  $m > 1$ .

Consider the polynomial  $f = X^m - 1$ . All elements of this subgroup are roots of  $f$ , so  $f$  has at least  $m^2$  roots, so it has at least  $m^2$  distinct linear factors. However  $\mathbb{F}_p[X]$  is a UFD so it can’t have more than  $m$  distinct linear factors, which is a contradiction.  $\square$

**Theorem 2.75.** The primes in the Gaussian integers  $\mathbb{Z}[i]$  are, up to associates:

1. Primes numbers  $p \in \mathbb{N} \subseteq \mathbb{Z}[i]$  which are  $3 \bmod 4$

2.  $z \in \mathbb{Z}[i]$  with  $|z|^2 = p$ , with  $p \in \mathbb{N}$  a prime number which is either  $2$  or  $1 \bmod 4$ .

*Proof.* We first show that these are indeed primes.

If  $p \in \mathbb{N}$  is prime and is  $3 \bmod 4$ , then  $p \neq a^2 + b^2$  for  $a, b \in \mathbb{Z}$  as any square number is always  $0$  or  $1 \bmod 4$ . So  $p$  is a prime in  $\mathbb{Z}[i]$ .

If  $|z|^2 = p$  is prime and  $z = uv$ , then  $|u|^2 |v|^2 = p$ . So WLOG  $|u|^2 = 1$ , so  $u$  is a unit. So  $z$  is irreducible.

Now let  $z \in \mathbb{Z}[i]$  be irreducible/prime. It’s not a unit so  $|z|^2 > 1$ . Note that  $\bar{z}$  is also irreducible. So  $|z|^2 = z\bar{z}$  is a factorisation of  $|z|^2$  into irreducibles.

Let  $p \in \mathbb{N}$  be a prime factor of  $|z|^2 > 1$ . So  $p | z\bar{z}$  in  $\mathbb{Z}[i]$ . Consider  $p \bmod 4$ .

If  $p$  is 3 mod 4, then  $p$  is prime in  $\mathbb{Z}[i]$ . So  $p|z$  or  $p|\bar{z}$ . Noting  $p = \bar{p}$ , either way we must have  $p|z$ . Since both  $p$  and  $z$  are irreducible, they must be associates. So  $z$  falls into the first class of primes we found.

Otherwise,  $p = 2$  or  $p$  is 1 mod 4. If  $p$  is 1 mod 4, then  $p - 1 = 4k$  for some  $k \in \mathbb{Z}$ , so  $\mathbb{F}_p^\times \cong C_{4k}$  by the previous lemma.

Let  $a \in \mathbb{F}_p^\times$  be an element of order 4.  $a^2$  is an element of order 2; there's only one, so  $a^2 = -1$ . So there is some  $a \in \mathbb{Z}$  with  $p|a^2 + 1$ . So  $p|(a+i)(a-i) = a^2 + 1$ .

If  $p = 2$ , then similarly  $p|(a+i)(a-i)$  for  $a = 1$ . However  $p \nmid a+i$  for any prime  $p_i$  so  $p$  isn't prime in  $\mathbb{Z}[i]$ .

So we can write  $p = z_1 z_2$ , with  $z_1, z_2 \in \mathbb{Z}[i]$  non-units. Taking norm squared:

$$p^2 = |z_1|^2 |z_2|^2$$

Since  $z_1, z_2$  aren't units, we must have  $|z_1|^2 = |z_2|^2 = p$ . So  $p = z_1 \bar{z}_1 = z_2 \bar{z}_2 = z_1 z_2$ , so  $z_1 = \bar{z}_2$ .

So  $p = z_1 \bar{z}_1$  divides  $|z|^2 = z \bar{z}$ .  $z$  is irreducible so by unique factorisation,  $z = z_1$  or  $z = \bar{z}_1$ . Either way  $|z|^2 = p$ . This implies  $p$  is a sum of two squares, so we must have  $p = 2$  or  $p$  is 1 mod 4.  $\square$

The upshot of this is that finding primes in  $\mathbb{Z}[i]$  is essentially an equivalent problem to finding sums of two squares with are prime.

**Corollary 2.76.**  $n \in \mathbb{N}$  is a sum of two squares  $n = x^2 + y^2$  if and only if when we write  $n = p_1^{n_1} \dots p_k^{n_k}$  as a product of powers of distinct primes,  $n_i$  is even whenever  $p_i$  is 3 mod 4.

*Proof.* ( $\Rightarrow$ ) Suppose  $n = x^2 + y^2$ . So  $n = |x + iy|^2$ ; let  $z = x + iy$ .

Write  $z = \alpha_1 \dots \alpha_q$ , where each  $\alpha_i \in \mathbb{Z}[i]$  is irreducible.

So  $n = z \bar{z} = |\alpha_1|^2 \dots |\alpha_q|^2$ . By the theorem, each  $|\alpha_i|^2$  is either  $p^2$  with  $p$  being 3 mod 4, or is a prime  $p$  which is 2 or 1 mod 4. So each prime which is 3 mod 4 appears an even number of times in the prime factorisation of  $n$ .

( $\Leftarrow$ ) Write  $n = p_1^{n_1} \dots p_k^{n_k}$  as a product of powers of distinct primes. For each  $p_i$ , if  $p$  is 2 or 1 mod 4, then  $p_i = |\alpha_i|^2$  for some  $\alpha_i \in \mathbb{Z}[i]$ . So  $p_i^{n_i} = |\alpha_i^{n_i}|^2$ .

If  $p_i$  is 3 mod 4, then  $p_i^{n_i} = |p_i^{n_i/2}|^2$ . Since  $|\cdot|^2$  is multiplicative, we find that  $n = |\beta|^2$  for some  $\beta \in \mathbb{Z}[i]$ , i.e.  $n$  is the sum of two squares.  $\square$

**Example 2.77.**  $65 = 5 \times 13$ . 5 and 13 are both 1 mod 4 so 65 is a sum of two squares, and the proof tells us how to write it in such a form.

First, factor 5 and 13 into irreducibles in  $\mathbb{Z}[i]$ :

$$5 = (2+i)(2-i)$$

and

$$13 = (2+3i)(2-3i)$$

So  $65 = |2+i|^2 \cdot |2+3i|^2 = |(2+i)(2+3i)|^2 = |1+8i|^2 = 1^2 + 8^2$  but also  $65 = |(2+i)(2-3i)|^2 = |7-4i|^2 = 7^2 + 4^2$  and this is all ways of writing 65 as a sum of two squares.

### 3 Modules

"A module is to a ring what a vector space is to a field". Various parts of the theory of vector spaces will carry over, but many won't, making the theory of modules far richer than that of vector spaces.

In this section, we won't assume our rings are commutative. Let  $R$  be a ring.

**Definition 3.1.** An  $R$ -module is a set  $M$  with functions  $+: M \times M \rightarrow M$  and  $\cdot: R \times M \rightarrow M$ , along with an element  $0 = 0_M \in M$ , such that  $(M, +)$  is an abelian group with  $0 = 0_M$ , and

1.  $(r + r') \cdot m = r \cdot m + r' \cdot m$
2.  $r \cdot (m + m') = r \cdot m + r \cdot m'$
3.  $r \cdot (r' \cdot m) = (r \cdot r') \cdot m$
4.  $1_R \cdot m = m$

for all  $r, r' \in R$  and  $m, m' \in M$ .

**Remark 3.2.** Here we always multiply elements of  $M$  by elements of  $R$  on the left. Because of this, what we call a module is sometimes called a left module; there is an analogous notion of right module too, but the theory of such things is essentially the same.

We can give an alternative characterisation of modules. For an abelian group  $A$ , let  $\text{End}A$  be the set of group homomorphisms  $A \rightarrow A$ . This is a ring.

**Definition 3.3.** An  $R$ -module is an abelian group  $M$  equipped with a ring homomorphism  $\phi: R \rightarrow \text{End}(M)$ .

Given such a  $\phi$ , and  $r \in R$  and  $m \in M$ , we write  $r \cdot m$  for  $(\phi(r))(m)$ .

**Proposition 3.4.** These two definitions of module are equivalent.

We can think of an  $R$ -module like “ $R$  acting on an abelian group”, just like how groups can act on sets.

**Example 3.5.** Let  $F$  be a field. Then an  $F$ -vector space is exactly the same thing as an  $F$ -module (the axioms for both are exactly the same).

**Example 3.6.** Let  $I \subseteq R$  be an ideal. Then  $I$  is an  $R$ -module, via  $r \cdot a := r \cdot_R a$  (for  $r \in R$  and  $a \in I$ ).

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**Example 3.7.**  $R$  is an  $R$ -module.

For  $n \geq 0$ ,  $R^n$  is an  $R$ -module, via  $r \cdot (r_1, \dots, r_n) := (r \cdot r_1, \dots, r \cdot r_n)$ .

**Example 3.8.** If  $I \subseteq R$  is a two-sided ideal, then  $R/I$  is an  $R$ -module via  $r \cdot (a + I) := (r \cdot a) + I$ .

**Example 3.9.** A  $\mathbb{Z}$ -module is the same thing as an abelian group: let  $A$  be an abelian group,  $n \in \mathbb{Z}$  and  $a \in A$ . Then  $n \cdot a := a + \dots + a$ , where we sum  $n$  copies of  $a$  if  $n \geq 0$ , and  $n \cdot a := (-a) + \dots + (-a)$  where we sum  $-n$  copies of  $-a$  if  $n \leq 0$ .

Alternatively, there's a unique ring homomorphism  $\mathbb{Z} \rightarrow \text{End}(A)$ , so we can take this to endow  $A$  with the structure of a  $\mathbb{Z}$ -module.

### 3.1 Constructions

**Definition 3.10.** Let  $M_1, \dots, M_k$  be  $R$ -modules. Their direct sum, written  $M_1 \oplus \dots \oplus M_k$  is the abelian group  $M_1 \times \dots \times M_k$ , with  $r \cdot (m_1, \dots, m_k) := (r \cdot m_1, \dots, r \cdot m_k)$ .

**Example 3.11.**  $R^n = R \oplus \dots \oplus R$ , where we have  $n$  copies of  $R$  on the RHS.

**Definition 3.12.** A subset  $N \subseteq M$  is an  $(R)$ -submodule if it's a subgroup of  $M$  and if  $r \in R$  and  $n \in N$ , then  $r \cdot n \in N$ .

If this holds, we write  $N \leq M$ .

**Example 3.13.**  $\{0\}, M \leq M$  are always submodules.

A subset  $I \subseteq R$  is a submodule if and only if  $I$  is an ideal.

**Example 3.14.** If  $F$  is a field, submodules and sub-vector spaces are the same thing.

**Definition 3.15.** Let  $N \leq M$  be a submodule. The quotient module  $M/N$  is the abelian group  $M/N$ , with  $R$  action given by  $r \cdot (m + N) := (r \cdot m) + N$ .

**Remark 3.16.** Groups have both subgroups and normal subgroups; the latter is a special type of the former. Rings have subrings and ideals, neither of which is a special type of the other. However modules just have submodules  $\odot$ .

**Definition 3.17.** A function  $\phi : M \rightarrow N$  between  $R$ -modules is an  $R$ -module homomorphism if it's a homomorphism of abelian groups, and  $\phi(r \cdot m) = r \cdot \phi(m)$  for all  $r \in R$  and  $m \in M$ .

An isomorphism is once again a bijective homomorphism.

**Example 3.18.** The ideals  $(2), (3) \subseteq \mathbb{Z}$  are different, but they're isomorphic as  $R$ -modules, via the isomorphism  $\phi : (2) \rightarrow (3)$  sending  $2k$  to  $3k$ .

In fact, if  $R$  is an integral domain and  $r \neq 0 \in R$ , then  $R$  and  $(r)$  are always isomorphic as  $R$ -modules.

**Definition 3.19.** Let  $R, R'$  be rings,  $M$  an  $R$ -module, and  $M'$  an  $R'$ -module. Then  $M \times M'$  is an  $R \times R'$ -module, with  $(r, r') \cdot (m, m') := (r \cdot m, r' \cdot m')$ .

### 3.2 Some theory

There are three isomorphism theorems, just like with groups and rings.

**Theorem 3.20** (First isomorphism theorem). Let  $\phi : M \rightarrow N$  be an  $R$ -module homomorphism. Then:

1.  $\ker(\phi) \leq M$  is an  $R$ -submodule.
2.  $\text{Im}(\phi) \leq N$  is an  $R$ -submodule.
3. There is an isomorphism of  $R$ -modules:

$$\frac{M}{\ker(\phi)} \cong \text{Im}(\phi)$$

Here we do find something different to rings and groups:

**Definition 3.21.** The cokernel of  $\phi$ , written  $\text{coker}(\phi)$ , is the  $R$ -module  $N/\text{Im}(\phi)$ .

Compare this to rings/groups, where  $\text{Im}(\phi)$  is a subring/subgroup rather than ideal/normal subgroup respectively. So this is really something we couldn't have defined for rings/groups.

**Theorem 3.22** (Second isomorphism theorem). Let  $A, B \leq M$  be submodules. Then:

1.  $A + B := \{a + b | a \in A, b \in B\} \leq M$  is a submodule.
2.  $A \cap B \leq M$  is a submodule.
3. There is an isomorphism of  $R$ -modules:

$$\frac{A + B}{A} \cong \frac{B}{A \cap B}$$

**Remark 3.23.** In general, if  $\{A_i\}_{i \in S}$  is a collection of submodules, then

$$\sum_{i \in S} A_i \leq M$$

is a submodule, where the LHS is defined to be the set of finite sums of elements of the  $A_i$ . Similarly

$$\bigcap_{i \in S} A_i \leq M$$

is a submodule.

**Theorem 3.24** (Third isomorphism theorem). Let  $N \leq L \leq M$  be submodules. Then

1.  $L/N \leq M/N$  is a submodule.
2. There is an isomorphism of  $R$ -modules

$$M/L \cong \frac{M/N}{L/N}$$

**Proposition 3.25.** Let  $N \leq M$  be a submodule. There is a natural bijection

$$\{\text{submodules of } M/N\} \leftrightarrow \{\text{submodules of } M \text{ containing } N\}$$

**Proposition 3.26.** The composition of two  $R$ -modules homomorphisms is an  $R$ -module homomorphism.

The proofs of the results we've just seen are omitted as they're essentially the same as for rings, but writing a careful proof of these is a good exercise.

**Definition 3.27.** Let  $m \in M$ . The submodule generated by  $m$ ,  $Rm \leq M$ , is  $\{r \cdot m | r \in R\}$ .

Alternatively, there is a homomorphism  $\phi_m : R \rightarrow M$  sending  $r$  to  $r \cdot m$ , and  $R \cdot m := \text{Im}(\phi_m)$ .

Also note that  $\ker(\phi_m) \leq R$  is a submodule. We define the annihilator of  $m$ , written  $\text{Ann}(m)$ , to be

$$\text{Ann}(m) := \{r \in R | r \cdot m = 0\} = \ker(\phi_m)$$

This is a submodule of  $R$  so it's an ideal.

Note that by the first isomorphism theorem,  $R \cdot m \cong R/\text{Ann}(m)$ .

**Definition 3.28.** An  $R$ -module  $M$  is finitely generate (abbreviated f.g.) if there are elements  $m_1, \dots, m_n \in M$ , such that

$$\begin{aligned} M &= Rm_1 + \dots + Rm_n \\ &= \{r_1m_1 + \dots + r_nm_n | r_i \in R\} \end{aligned}$$

**Example 3.29.** If  $F$  is a field, and  $M$  an  $F$ -module, then  $M$  is finitely generated (as an  $F$ -module) if and only if it's finite-dimensional (as an  $F$ -vector space).

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**Lemma 3.30.** An  $R$ -module  $M$  is finitely generated if and only if there's a surjective homomorphism  $R^n \rightarrow M$ .

*Proof.* ( $\Rightarrow$ ) If  $M = Rm_1 + \dots + Rm_n$ , let  $\phi : R^n \rightarrow M$  be the homomorphism sending  $(r_1, \dots, r_n)$  to  $r_1m_1 + \dots + r_nm_n$ . It's surjective by assumption.

( $\Leftarrow$ ) Suppose  $\phi : R^n \rightarrow M$  is a surjective homomorphism. Take  $m_1, \dots, m_n$  to be the images under  $\phi$  of the standard basis vectors. Then  $M = Rm_1 + \dots + Rm_n$ .  $\square$

**Corollary 3.31.** *If  $N \leq M$  is a submodule and  $M$  is finitely generated, then  $M/N$  is finitely generated too.*

*Proof.* Let  $\phi : R^n \rightarrow M$  be a surjective homomorphism. Note the quotient  $M \rightarrow M/N$  is also surjective. So the composition  $R^n \rightarrow M \rightarrow M/N$  is also surjective.  $\square$

**Example 3.32.** *However, if  $N \leq M$  is a submodule and  $M$  is finitely generated, then  $N$  isn't necessarily finitely generated.*

*Let  $R = \mathbb{Z}[X_1, X_2, \dots]$ , and let  $I$  be the ideal  $I = (X_1, X_2, \dots) \subseteq R$ . Then  $R$  is a finitely generated  $R$ -module but  $I \leq R$  is a submodule which isn't finitely generated.*

*Proof.* Suppose  $I = (f_1, \dots, f_k)$  is finitely generated. Let  $p \in \mathbb{N}$  be the largest number such that  $X_p$  appears in any of the  $f_i$ .

Then  $X_{p+1} \in I$  but  $X_{p+1} \notin (f_1, \dots, f_k)$ , a contradiction.  $\square$

### 3.3 Free modules

**Definition 3.33.** *Let  $S$  be a set. The free module over  $S$ , written  $R^{(S)}$ , is*

$$R^{(S)}L = \oplus_{i \in S} R := \{(x_i)_{i \in S} | x_i \in R; x_i = 0 \text{ for all but finitely many } i\}$$

*with coordinate-wise addition and  $R$ -action.*

**Proposition 3.34.** *Let  $R$  be a ring, and  $S$  a set. Then  $R^{(S)}$  is finitely generated if and only if  $S$  is finite.*

*Proof.* ( $\Leftarrow$ ) If  $|S| = n$ , then  $R^{(S)} \cong R^n$ , so there is a surjective homomorphism  $R^n \rightarrow R^{(S)}$ .

( $\Rightarrow$ ) Suppose  $S$  is infinite and  $R^{(S)} = Rm_1 + \dots + Rm_n$ . Write each  $m_k = (x_i^{(k)})_{i \in S} \in R^{(S)}$ . Define the subset  $T \subseteq S$  by

$$T := \{i \in S | x_i^{(k)} \neq 0 \text{ for some } 1 \leq k \leq n\}$$

Note  $T$  is a finite set. However  $S$  is infinite, so we can find some  $s \in S \setminus T$ . Let  $a = (a_i)_{i \in S} \in R^{(S)}$  be the element with

$$a_i := \begin{cases} 1 & \text{if } i = s \\ 0 & \text{if } i \neq s \end{cases}$$

Then  $a \notin Rm_1 + \dots + Rm_n$ , a contradiction.  $\square$

**Theorem 3.35.** *Let  $F$  be a field. Any  $F$ -module is free.*

This is in fact equivalent to the axiom of choice.

We can give an alternative characterisation of freeness.

**Definition 3.36.** *A subset  $S \subseteq M$  generates  $M$  freely if*

1.  *$S$  generates  $M$  as an  $R$ -module, i.e.  $R \cdot S = M$  (where  $R \cdot S$  is defined to be the set of finite sums of elements of the form  $rs$ , with  $r \in R$  and  $s \in S$ ), and*

2. For any other  $R$ -module  $N$ , any function  $\psi : S \rightarrow N$  extends to an  $R$ -module homomorphism  $\phi : M \rightarrow N$ .

**Remark 3.37.** Suppose this holds, and suppose  $\phi, \phi' : M \rightarrow N$  are two such extensions. Then  $\phi - \phi' : M \rightarrow N$  is a homomorphism, sending all of  $S$  to 0.

So  $S \subseteq \ker(\phi - \phi')$ . But since  $S$  generates  $M$ , this implies  $\ker(\phi - \phi') = M$ , i.e.  $\phi = \phi'$ .

So the extension  $\phi$  is unique.

**Definition 3.38.** An  $R$ -module  $M$  is free if it's freely generated by some subset  $S \subseteq M$ .

Such a subset  $S \subseteq M$  is called a basis for  $M$ .

**Proposition 3.39.** These two definitions of free module are equivalent.

*Proof.* First suppose  $M \cong R^{(S)}$ . We view  $S$  as a subset in  $M$  by identifying  $s \in S$  with the element  $(x_i^s)_{i \in S}$  defined by

$$x_i^s = \begin{cases} 1 & \text{if } i = s \\ 0 & \text{otherwise} \end{cases}$$

This subset generates  $M$  as an  $R$ -module.

Let  $\psi : S \rightarrow N$  be any function, for  $N$  some other  $R$ -module. Define a homomorphism  $\phi : M \rightarrow N$  to send  $(x_i)_{i \in S}$  to

$$\phi((x_i)_{i \in S}) := \sum_{i \in S} x_i \cdot \psi(i) \in N$$

This is a homomorphism extending  $\psi$ , so  $S$  generates  $M$  freely.

Conversely, now suppose some subset  $S \subseteq M$  freely generates  $M$ . We claim  $M \cong R^{(S)}$ .

Let  $\psi : S \rightarrow R^{(S)}$  send  $s \in S$  to the element  $(x_i^s)_{i \in S}$  defined above. This extends to a homomorphism  $\phi : M \rightarrow R^{(S)}$ .

This is an isomorphism, with inverse sending  $(x_i)_{i \in S}$  to  $\sum_{i \in S} x_i \cdot i$ . □

**Definition 3.40.** A set of elements  $m_1, \dots, m_n \in M$  are linearly independent (LI) if whenever  $r_1 \cdot m_1 + \dots + r_n \cdot m_n = 0$  for  $r_i \in R$ , we have  $r_1 = \dots = r_n = 0$ .

**Proposition 3.41.** For a subset  $S = \{m_1, \dots, m_n\} \subseteq M$ , the following are equivalent.

1.  $S$  generates  $M$  freely.
2.  $S$  generates  $M$  and  $S$  is linearly independent.
3. Every element of  $M$  is uniquely expressible as  $r_1 \cdot m_1 + \dots + r_n m_n$ , for  $r_i \in M$ .

**Example 3.42.** Consider the  $\mathbb{Z}$ -module  $\mathbb{Z}/2\mathbb{Z}$ . This isn't free: if it were freely generated by  $S \subseteq \mathbb{Z}/2\mathbb{Z}$ , we would have  $S = \{1\}$  (since any set containing 0 can't be linearly independent). But this set isn't linearly independent.

**Example 3.43.** The subset  $\{2, 3\} \subseteq \mathbb{Z}$  generates  $\mathbb{Z}$ , but not freely, as it's not linearly independent:  $3 \cdot 2 + (-2) \cdot 3 = 0$ .

Recall that if  $S$  spans an  $F$ -vector space  $V$  and  $S$  isn't linearly independent, we can throw away some of  $S$  and find some basis  $T \subseteq S$ . However no subset of  $\{2, 3\}$  freely generates  $\mathbb{Z}$ .

**Definition 3.44.** Let  $M$  be a finitely generated  $R$ -module. There is some surjective homomorphism  $\phi : R^n \rightarrow M$ . We call the submodule  $\ker(\phi) \leq R^n$  the relation module for these generators.

**Definition 3.45.** A finitely generated  $R$ -module  $M$  is finitely presented if there's a surjective homomorphism  $R^n \rightarrow M$ , with finitely generated kernel. In other words, this means  $M$  is isomorphic to the cokernel of some homomorphism  $\phi : R^m \rightarrow R^n$  for some  $n, m$ .

**Remark 3.46.** Finitely presented modules are finitely generated, by definition. However, the converse isn't always true; see the problem sheet.

We can make the following construction, similar to vector spaces:

**Proposition 3.47.** Let  $\phi : R^m \rightarrow R^n$  be an  $R$ -module homomorphism.

Let  $u_1, \dots, u_m \in R^m$  and  $v_1, \dots, v_n \in R^n$  be the standard bases. Then we can write  $\phi(u_j) = \sum_{i=1}^n A_{ij}v_i$ , for some matrix  $A \in M_{n \times m}(R)$ , i.e.  $\phi(r) = r \cdot A$  so  $\phi$  is given by multiplying by some matrix on the right.

If  $R$  is commutative, we could similarly write  $\phi(r) = B \cdot r$ , i.e. we could have represented  $\phi$  by multiplying by some matrix on the left.

This provides a bijection

$$\begin{aligned} M_{n \times m}(R) &\leftrightarrow \{R\text{-module homomorphisms } R^m \rightarrow R^n\} \\ A &\mapsto (r \mapsto r \cdot A) \end{aligned}$$

Recall from linear algebra, that if  $F$  is a field and  $F^n \cong F^m$ , then  $n = m$ . The same is true for any commutative ring:

**Theorem 3.48** (Invariant basis number property for commutative rings). Let  $R$  be a commutative ring, and suppose that  $R^n \cong R^m$  as  $R$ -modules. Then  $n = m$ .

We'll assume the following theorem. We won't prove it: in fact, it's equivalent to the axiom of choice (we won't prove this equivalence either):

**Theorem 3.49.** Any commutative ring contains a maximal ideal.

We'll use the existence of maximal ideals to reduce the previous theorem to the case of a field.

*Proof.* Let  $I \subseteq R$  be a maximal ideal, and let  $F = R/I$ , a field.

For  $M$  an  $R$ -module, we define  $IM$  to be the submodule of  $M$  generated by elements  $i \cdot m$ , for  $i \in I$  and  $m \in M$ :

$$IM := \{i_1 \cdot m_1 + \dots + i_k \cdot m_k \mid i_j \in I, m_j \in M\} \leq M$$

Then  $M/IM$  is an  $R$ -module. In fact,  $M/IM$  is an  $R/I = F$ -module: for  $r + I \in F$  and  $m + IM \in M/IM$ , we define

$$(r + I) \cdot (m + IM) := (r \cdot m) + IM$$

If  $R^n \cong R^m$  as  $R$ -modules, this then implies that

$$R^n/IR^n \cong R^m/IR^m$$

as  $F$ -modules.

We claim that  $R^k/IR^k \cong F^k$  are isomorphic  $F$ -modules. Indeed there is an isomorphism  $\psi : R^k/IR^k \rightarrow F^k = (R/I)^k$  sending  $(r_1, \dots, r_k) + IR^k$  to  $(r_1 + I, \dots, r_k + I)$ .

From this, it follows that  $F^n \cong F^m$  as  $F$ -modules. Since  $F$  is a field, this implies that  $n = m$ .  $\square$

**Remark 3.50.** This can fail for non-commutative rings: there exist rings  $R$  such that, for example,  $R^2 \cong R^1$  are isomorphic  $R$ -modules.

### 3.4 Modules over PIDs

Our next goal is to prove the following:

**Theorem 3.51** (Classification of finitely generated modules over a PID). *Let  $R$  be a PID (in particular,  $R$  is commutative), and  $M$  a finitely generated  $R$ -module. Then*

$$M \cong R^n \oplus R/(d_1) \oplus \dots \oplus R/(d_r)$$

*are isomorphic  $R$ -modules, for some  $d_1, \dots, d_r \in R$ , with  $d_1 | \dots | d_r$ .*

*(Furthermore,  $n$  and the  $d_i$  are unique up to associates, but we won't prove this).*

This generalises the classification theorem for finitely generated abelian groups, since they're just  $\mathbb{Z}$ -modules.

Here is an outline of the proof.

1. First, we show  $M$  is in fact finitely presented, so  $M \cong \text{coker}(\phi_A : R^n \rightarrow R^m)$  for some matrix  $A \in M_{n \times m}(R)$ , where  $\phi_A(x) := A \cdot x$ .
2. We then show that if  $B = PAQ$ , for invertible matrices  $P, Q$ , then  $\phi_A$  and  $\phi_B$  have isomorphic cokernels.
3. We show that for any matrix  $A$  over a PID  $R$ , there are invertible matrices  $P, Q$  such that  $PAQ$  is a rectangular diagonal matrix. (This is the hardest step).
4. Then we put all this together to prove the theorem.

First:

**Lemma 3.52.** *Let  $R$  be a PID, and  $N \leq R^n$  a submodule. Then  $N \cong R^k$  for some  $k \leq n$  - so  $N$  is f.g. and free.*

*Proof.* We induct on  $n$ . If  $n = 0$ ,  $R^0 = 0$  so  $n = 0 \cong R^0$ .

If  $n = 1$ , the  $N \subseteq R$  is an ideal so  $N = (\alpha)$  for some  $\alpha \in R$ . Since PIDs are IDs,  $N \cong R^1$  if  $\alpha \neq 0$  and  $N \cong R^0$  if  $\alpha = 0$ .

Now let  $\pi_n : R^n \rightarrow R$  be the homomorphism sending  $(r_1, \dots, r_n)$  to  $r_n$ . Then  $\ker(\pi_n) \cong R^{n-1}$ .

$\pi_n(N) \leq R$  is a submodule, so it's  $(\alpha)$  for some  $\alpha \in R$  since  $R$  is a PID.

If  $\alpha = 0$ , this means  $N \leq \ker(\pi_n) \cong R^{n-1}$ , so by the induction hypothesis we're done.

If  $\alpha \neq 0$ , we can pick some  $\beta \in N$  such that  $\pi_n(\beta) = \alpha$ . By the induction hypothesis, we can choose some  $x_1, \dots, x_k \in N \cap \ker(\pi_n)$  which freely generate  $N \cap \ker(\pi_n)$ , with  $k \leq n - 1$ . Let  $S = \{x_1, \dots, x_k, \beta\} \subseteq N$ ; we claim that  $S$  freely generates  $N$ ; this will complete the proof.

We first show  $S$  generates  $N$ . Let  $y \in N$ . Then  $\pi_n(y) = r\alpha$ , for some  $r \in R$ . So  $y - r\beta \in N \cap \ker(\pi_n)$  is spanned by the  $x_1, \dots, x_k$ . So  $y$  is generated by  $S$ .

Now suppose  $0 = r_1 \cdot x_1 + \dots + r_k \cdot x_k + r' \cdot \beta$ , for  $r_i, r' \in R$ . Then  $0 = \pi_n(0) = r'\alpha$ , so  $r' = 0$ . Since  $x_1, \dots, x_k$  are linearly independent, we must have  $r_1 = \dots = r_k = 0$ , so  $S$  is linearly independent.  $\square$

**Corollary 3.53.** *Let  $M$  be a finitely generated module over a PID  $R$ . Then  $M$  is finitely presented.*

This completes the first step of the proof of the theorem.

*Proof.* There's a surjective homomorphism  $\phi : R^n \rightarrow M$ .  $\ker(\phi) \leq R^n$  is a submodule, so by the lemma,  $\ker(\phi) \cong R^k$  for some  $k$ . Let  $\psi : R^k \rightarrow R^n$  have image  $\ker(\phi)$ . Then  $M \cong \text{coker}(\psi)$ . So  $M$  is finitely presented.  $\square$

**Definition 3.54.** Two matrices  $A, B \in M_{m \times n}(R)$  are equivalent if there are invertible matrices  $P \in GL_m(R), Q \in GL_n(R)$ , such that  $B = PAQ$ .

By considering the homomorphisms  $\phi_A, \phi_B : R^m \rightarrow R^n$ , this is equivalent to there being isomorphisms  $f = \phi_P : R^n \rightarrow R^n$  and  $g = \phi_Q : R^m \rightarrow R^m$  such that  $f \circ \phi_B = \phi_A \circ g$ . In other words,  $A$  and  $B$  are “the same after changing bases”.

This is an equivalence relation.

**Proposition 3.55.** Let  $A, B \in M_{m \times n}(R)$  be matrices. If  $A$  and  $B$  are equivalent, then  $\text{coker}(\phi_A) \cong \text{coker}(\phi_B)$  are isomorphic  $R$ -modules.

This completes the second step of the proof of the theorem.

**Remark 3.56.** Suppose  $R = F$  is a field. Then equivalent matrices have the same rank, so  $\text{coker}(\phi_A)$  and  $\text{coker}(\phi_B)$  have the same dimension, so they’re isomorphic. So we already knew the proposition in this case.

*Proof.* Let  $f : R^n \rightarrow R^n$  and  $g : R^m \rightarrow R^m$  be isomorphisms such that  $f \circ \phi_B = \phi_A \circ g$ .

Note that  $\text{Im}(\phi_A) = \text{Im}(\phi_A \circ g) = \text{Im}(f \circ \phi_B) = f(\text{Im}(\phi_B))$ .

Let  $\psi : \text{coker}(\phi_B) = R^n / \text{Im}(\phi_B) \rightarrow \text{coker}(\phi_A) = R^n / \text{Im}(\phi_A)$  send  $x + \text{Im}(\phi_B)$  to  $f(x) + \text{Im}(\phi_A)$ . This is a well-defined homomorphism, and is an isomorphism since  $f$  is.  $\square$

Next: special types of equivalences.

**Definition 3.57.** An elementary matrix is an  $n \times n$  matrix  $A$  over  $R$  of one of the following forms.

- (i).  $A$  has 1s along the diagonal,  $A_{ij} = c$  for some fixed  $c \in R$  and  $1 \leq i, j \leq n$ , and all other entries are 0.
- (ii). Let  $1 \leq i, j \leq n$  be distinct and fixed. Then  $A$  is  $I$  except in 4 entries:  $A_{ii} = A_{jj} = 0$  and  $A_{ij} = A_{ji} = 1$ .
- (iii). Let  $1 \leq i$  and  $c \in R^\times$  a unit be fixed. Then  $A$  is  $I$  except in 1 entry:  $A_{ii} = c$ .

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Now let  $A \in M_{m \times n}$  be any matrix. An elementary row/column operation on  $A$  is given by replacing  $A$  with  $PA$  or  $AP$  respectively, for  $P$  an elementary matrix.

Elementary matrices of type (i) correspond to adding  $c \in R$  times the  $i^{\text{th}}$  row/column to the  $j^{\text{th}}$  row/column.

Elementary matrices of type (ii) correspond to swapping the  $i^{\text{th}}$  and  $j^{\text{th}}$  rows/columns.

Elementary matrices of type (iii) correspond to multiplying the  $i^{\text{th}}$  row/column by  $c \in R^\times$ .

Note all elementary matrices are invertible. Therefore:

**Proposition 3.58.** Let  $A \in M_{m \times n}(R)$  be a matrix. If  $B$  is obtained from  $A$  by row and column operations, then  $A$  and  $B$  are equivalent matrices.

In particular, if this holds,  $\text{coker}(\phi_A) \cong \text{coker}(\phi_B)$ .

**Definition 3.59.** A matrix  $A \in M_{m \times n}(R)$  is in Smith normal form if  $A$  is a rectangular diagonal matrix, i.e.

$$A = \begin{pmatrix} d_1 & & & & & 0 \\ & \ddots & & & & \\ & & d_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ 0 & & & & & 0 \end{pmatrix}$$

and all  $d_i \neq 0$  and  $d_1 | \dots | d_r$ .

An example is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 6 & 0 \end{pmatrix}$$

**Theorem 3.60** (Smith normal form). *Let  $A \in M_{m \times n}(R)$  be a matrix over a PID  $R$ . Then  $A$  is equivalent to a matrix in Smith normal form.*

Note  $R$  is a PID and so a UFD, so GCDs exist. These are only defined up to units, so each time we take GCDs, we'll just fix some choice, for the sake of our sanity.

*Proof.* Let  $A = (A_{ij}) \in M_{m \times n}(R)$ . If  $A = 0$ , we're done. So assume  $A \neq 0$ .

**Claim 3.61** (Claim 1). *Given two entries  $A_{ij}$  and  $A_{kl}$  in the same row (so  $i = k$ ) or column (so  $j = l$ ), we can modify  $A$  so that  $\text{GCD}(A_{ij}, A_{kl})$  appears in  $A$ .*

*Proof of claim.* Assume they're in the same column (for now).

Since invertible  $2 \times 2$  matrices can be extended to invertible  $m \times m$  matrices, it suffices to prove the claim when  $m = 2$ , i.e. for any vector  $\begin{pmatrix} a \\ b \end{pmatrix} \in R^2$ , we can left-multiply by some matrix  $P \in M_2(R)$  to get a vector with  $\text{GCD}(a, b)$  in.

$R$  is a PID so  $(a, b) = (d) \subseteq R$  are equal ideals, for some  $d \in R$ . So  $d = \text{GCD}(a, b)$ .

$d \in (a, b)$  so there are some  $x, y \in R$  such that  $xa + yb = d$ .

Since  $\text{GCD}(a, b) = d$ , we must have that  $\text{GCD}(x, y) = 1$ ; similarly to above there exist elements  $u, v \in R$  such that  $xu + yv = 1$ .

Now let  $P$  be the matrix  $P := \begin{pmatrix} x & y \\ -v & u \end{pmatrix}$ . Note that  $\det(P) = xu + yv = 1$  so  $P$  is invertible.

Now

$$\begin{aligned} P \cdot \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} x & y \\ -v & u \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} \\ &= \begin{pmatrix} xa + yb \\ -va + ub \end{pmatrix} \\ &= \begin{pmatrix} d \\ \text{stuff} \end{pmatrix} \end{aligned}$$

so we're done.

If instead  $A_{ij}$  and  $A_{kl}$  are in the same column, we can do the same but with the row matrix  $\begin{pmatrix} a & b \end{pmatrix}$  and *right* multiplication by  $P^T$ .  $\square$

**Claim 3.62** (Claim 2). *We can modify  $A$  so that  $A_{11}$  divides the rest of the first row and column, i.e.  $A_{11} | A_{i1}, A_{1j}$  for all  $i, j$ .*

*Proof of claim.* For  $r \in R \setminus \{0\}$ , let  $\delta(r) \in \mathbb{Z}_{\geq 0}$  be the number of (possibly repeated) irreducible factors of  $r$ . So for example if  $r = p_1 \dots p_k$ , with all  $p_i$  irreducible, then  $\delta(r) = k$ ; and  $r$  is a unit if and only if  $\delta(r) = 0$ . Note that  $R$  is a UFD so this is well-defined.

Suppose  $A$  isn't of the required form. First:  $A$  must contain some non-zero entry  $A_{ij}$ ; using row and column operations, we can move this to the top left. Let  $\alpha_1 = A_{11} \neq 0$ .

If  $A$  still isn't of the required form, there are some  $1 \leq i, j \leq n$ , not both equal to 1, such that  $A_{11} \nmid A_{ij}$ . By Claim 1, we can modify  $A$  so that  $\alpha_2 := \text{GCD}(A_{11}, A_{ij})$  appears in our matrix; using row and column operations, we can assume this entry is in the top left.

Note that  $\alpha_2 | \alpha_1$ , but  $\alpha_2$  and  $\alpha_1$  aren't associates since  $\alpha_2 | A_{ij}$  but  $\alpha_1 \nmid A_{ij}$ . So  $\delta(\alpha_2) < \delta(\alpha_1)$ .

So if  $A$  wasn't of the required form, we could modify it so that the top left entry had strictly lower  $\delta$ . So if we do this process repeatedly, it must terminate, and we find that  $A_{11} | A_{ij}$  whenever  $i = 1$  or  $j = 1$ , as required.  $\square$

Now we modify  $A$  to be of the form in Claim 2.

Since  $A_{11} | A_{1j}$  for all  $j > 1$ , subtracting multiples of the first column from the other columns, we can modify  $A$  so that  $A_{1j} = 0$  for all  $j > 1$ :

$$A = \begin{pmatrix} A_{11} & 0 & \dots & 0 \\ A_{21} & A_{22} & \dots & \\ \vdots & & \ddots & \\ 0 & & & A_{mn} \end{pmatrix}$$

Similarly since  $A_{11} | A_{i1}$  for all  $i > 1$ , we can modify  $A$  such that we also have that  $A_{i1} = 0$  for all  $i > 1$ :

$$A = \begin{pmatrix} A_{11} & 0 & \dots & 0 \\ 0 & A_{22} & & \\ \vdots & & \ddots & \\ 0 & & & A_{mn} \end{pmatrix}$$

In other words, we have that

$$A = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix}$$

for some smaller matrix  $A' \in M_{(m-1) \times (n-1)}$ .

We apply the same process to  $A'$ , and repeating this process, we can modify  $A$  (without changing its equivalence class) so that it's in rectangular diagonal form:

$$A = \begin{pmatrix} d_1 & & & & 0 \\ & \ddots & & & \\ & & d_r & & \\ & & & 0 & \\ 0 & & & & \ddots & \\ & & & & & 0 \end{pmatrix}$$

We're not done yet: we still need to modify  $A$  so that  $d_1 | \dots | d_r$  (and then we'll be done).

First, we do the case  $r = 2$ : we can do the following sequence of row operations:

$$\begin{aligned} \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix} &\sim \begin{pmatrix} d_1 & GCD(d_1, d_2) \\ 0 & d_2 \end{pmatrix} && \text{as } GCD(d_1, d_2) = xd_1 + yd_2 \text{ for some } x, y \in R \\ &\sim \begin{pmatrix} d_1 & GCD(d_1, d_2) \\ -\lambda d_1 & 0 \end{pmatrix} && \begin{array}{l} \text{where } d_2 = \lambda \cdot GCD(d_1, d_2) \text{ for some } \lambda \in R. \\ \text{This is by subtracting } \lambda \text{ times the first row from the second row} \end{array} \\ &\sim \begin{pmatrix} GCD(d_1, d_2) & 0 \\ 0 & \lambda d_1 \end{pmatrix} && \begin{array}{l} \text{by swapping columns, the subtracting a multiple} \\ \text{of the first column from the second} \end{array} \end{aligned}$$

We now proceed as in the proof of Claim 2.

Let  $\alpha_1 = d_1$ . If  $d_1 \nmid d_i$  for some  $i \geq 2$ , let  $\alpha_2 = \text{GCD}(d_1, d_i)$ , and use row and column operations to move it to the place of  $d_1$ .

$\alpha_2 \mid \alpha_1$  but  $\alpha_1 \nmid \alpha_2$ , so we have that  $\delta(\alpha_2) < \delta(\alpha_1)$ . So one again this process terminates, and we find that  $d_1 \mid d_i$  for all  $i \geq 2$ .

Repeating, we can modify  $A$  so that  $d_2 \mid d_i$  for all  $i \geq 3$ ; this process preserves  $d_1 \mid d_i$  for  $i \geq 2$ . Iterating this gives the required result.  $\square$

**Remark 3.63.** When we modified  $A$  to an equivalent matrix, we mostly did this by applying row and column operations, but not always: in proving Claim 1, we used an equivalence of matrices that wasn't necessarily of this form.

In fact, if  $R$  is a Euclidean domain, any  $A \in \text{GL}_n(R)$  is a product of elementary matrices. So if  $R$  is a Euclidean domain, and  $A \in M_{m \times n}(R)$  can be put into Smith normal form, just using elementary matrices.

To complete the classification theorem, we just need the following:

**Proposition 3.64.** Let  $A \in M_{m \times n}(R)$  be in Smith normal form, i.e.

$$A = \begin{pmatrix} d_1 & & & & & \\ & \ddots & & & & \\ & & d_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix}$$

with  $d_1, \dots, d_r \in R$  non-zero, and  $d_1 \mid \dots \mid d_r$ . Then there is an isomorphism of  $R$ -modules

$$\text{coker}(\phi_A) \cong \frac{R}{(d_1)} \oplus \dots \oplus \frac{R}{(d_r)} \oplus R^{m-r}$$

where  $\phi_A$  is given by left multiplication by  $A$ .

*Proof.* Let  $\psi : R^m \rightarrow \frac{R}{(d_1)} \oplus \dots \oplus \frac{R}{(d_r)} \oplus R^{m-r}$  be the homomorphism sending  $(x_1, \dots, x_m)$  to

$$(x_1 + (d_1), \dots, x_r + (d_r), x_{r+1}, \dots, x_m)$$

Then  $\text{Im}(\phi_A) = \ker(\psi) = (d_1) \oplus \dots \oplus (d_r) \oplus 0 \dots \oplus 0$  (where there are  $m-r$  0s at the end). So by the first isomorphism theorem, the result follows.  $\square$

Let's put all of this together to prove the theorem:

**Theorem 3.65** (Classification of finitely generated modules over a PID). Let  $R$  be a PID (in particular,  $R$  is commutative), and  $M$  a finitely generated  $R$ -module. Then

$$M \cong R^n \oplus R/(d_1) \oplus \dots \oplus R/(d_r)$$

for some  $d_1, \dots, d_r \in R$  non-zero, with  $d_1 \mid \dots \mid d_r$ .

*Proof.* 1. Since  $R$  is a PID,  $M$  is in fact finitely presented. So  $M \cong \text{coker}(\phi_A)$  for some  $A \in M_{m \times n}(R)$ .

2. If  $A, B \in M_{m \times n}(R)$  are equivalent, then  $\text{coker}(\phi_A) \cong \text{coker}(\phi_B)$ .
3.  $A$  is equivalent to a matrix in Smith normal form, i.e. we can assume that

$$A = \begin{pmatrix} d_1 & & & & & \\ & \ddots & & & & \\ & & d_r & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix}$$

4. Therefore

$$M \cong R^n \oplus R/(d_1) \oplus \dots \oplus R/(d_r)$$

and  $d_1 | \dots | d_r$  are all non-zero.

□

**Remark 3.66.** *There's another type of decomposition that we could consider (but won't go into so much detail about), called prime decomposition:*

$$M \cong R^n \oplus \frac{R}{(P_1^{n_1})} \oplus \dots \oplus \frac{R}{(p_k^{n_k})}$$

for  $n_i \in \mathbb{N}$  and  $p_i \in R$  irreducible.

*This follows from the classification result that we proved, along with the fact that if  $d = p_1^{n_1} \cdot \dots \cdot p_k^{n_k}$ , then*

$$\frac{R}{(d)} \cong \frac{R}{(P_1^{n_1})} \oplus \dots \oplus \frac{R}{(p_k^{n_k})}$$

*The proof of this is on the third problem sheet.*

Note that if we know how to compute GCDs, our proof was entirely constructive.

**Example 3.67.** *Let  $A$  be the abelian group generated by the elements  $a, b, c$ , with relations*

$$\begin{array}{rcl} 2a + 3b & & +c = 0 \\ a + 2b & & = 0 \\ 5a + 6b & & +7c = 0 \end{array}$$

*This means that*

$$A = \frac{\mathbb{Z}^3}{\left\langle \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix} \right\rangle} = \text{coker}(\phi_X)$$

*where  $X$  is the matrix*

$$X = \begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 6 \\ 1 & 0 & 7 \end{pmatrix}$$

We put  $X$  into Smith normal form, using the following sequence of row and column operations:

$$\begin{aligned}
\begin{pmatrix} 2 & 1 & 5 \\ 3 & 2 & 6 \\ 1 & 0 & 7 \end{pmatrix} &\sim \begin{pmatrix} 1 & 2 & 5 \\ 2 & 3 & 6 \\ 0 & 1 & 7 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 2 & 5 \\ 0 & -1 & -4 \\ 0 & 1 & 7 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -4 \\ 0 & 1 & 7 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 1 & 7 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{pmatrix} \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}
\end{aligned}$$

This final matrix is in Smith normal form. So

$$A \cong \frac{\mathbb{Z}}{1\mathbb{Z}} \oplus \frac{\mathbb{Z}}{1\mathbb{Z}} \oplus \frac{\mathbb{Z}}{3\mathbb{Z}} \cong C_3$$

is cyclic of order 3.

### 3.5 Jordan normal form

In this section, we'll study modules over polynomial rings of fields using the theorem from the previous section, and amongst other things deduce Jordan normal form.

Let  $F$  be a field,  $V$  an  $F$ -vector space, and  $\alpha : V \rightarrow V$  a linear map.

Then we can make  $V$  into an  $F[X]$ -module: if  $d \in F[X]$  and  $v \in V$ , then we set

$$f \cdot v := (f(\alpha))(v)$$

We write  $V_\alpha$  for  $V$ , when we want to view it as an  $F[X]$ -module (so  $V$  and  $V_\alpha$  are in particular the same set).

We'll use the structure of  $V_\alpha$  as an  $F[X]$ -module to study the linear map  $\alpha$ .

**Lemma 3.68.** *If  $V$  is a finite dimensional  $F$ -vector space, then  $V_\alpha$  is a finitely generated  $F[X]$ -module.*

*Proof.* If  $v_1, \dots, v_n \in V$  is a basis for  $V$  as a vector space, then they generate  $V_\alpha$  as an  $F[X]$ -module.  $\square$

**Example 3.69.** *Suppose  $V_\alpha \cong F[X]/(X^r)$  as  $F[X]$ -modules. In particular, they're isomorphic as  $F$ -vector spaces.*

$1, X, \dots, X^{r-1} \in F[X]/(X^r)$  are an  $F$ -basis, and the action of multiplication by  $X$  has matrix

$$\begin{pmatrix} 0 & & 0 \\ 1 & 0 & \\ & 1 & \ddots \\ 0 & & 1 & 0 \end{pmatrix}$$

with respect to this basis.

So in  $V_\alpha$ ,  $\alpha$  is represented by the same matrix with respect to the corresponding basis.

**Example 3.70.** Suppose that  $V_\alpha \cong F[X]/((X - \lambda)^r)$ , as  $F[X]$ -modules, for some  $\lambda \in F$ .

Let  $\beta = \alpha - \lambda I$ ; this is a linear map  $\beta: V \rightarrow V$ .

Then  $V_\beta \cong F[Y]/(Y^r)$ , for  $Y = X - \lambda$ . So by the previous example, there is a basis for  $V$  such that  $\beta$  is given by the matrix

$$\begin{pmatrix} 0 & & 0 \\ 1 & 0 & \\ & 1 & \ddots \\ 0 & & 1 & 0 \end{pmatrix}$$

So  $\alpha = \beta + \lambda I$  has matrix

$$\begin{pmatrix} \lambda & & 0 \\ 1 & \lambda & \\ & 1 & \ddots \\ 0 & & 1 & \lambda \end{pmatrix}$$

a Jordan block.

**Example 3.71.** Suppose  $V_\alpha \cong F[X]/(f)$ , for  $f = a_0 + \dots + a_{r-1}X^{r-1} + X^r \in F[X]$  some monic polynomial.

$V_\alpha$  has an  $F$ -basis given by  $1, X, \dots, X^{r-1}$ , in which  $\alpha$  is given by

$$C(f) := \begin{pmatrix} 0 & & 0 & -a_0 \\ 1 & 0 & & -a_1 \\ & 1 & & \vdots \\ & & \ddots & \\ 0 & & & 1 & -a_{r-1} \end{pmatrix}$$

This matrix  $C(f)$  is called the companion matrix of  $f$ .

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**Theorem 3.72** (Rational canonical form). Let  $F$  be any field,  $V$  a finite dimensional  $F$ -vector space, and  $\alpha: V \rightarrow V$  some linear map. Then

$$V_\alpha \cong \frac{F[X]}{(f_1)} \oplus \dots \oplus \frac{F[X]}{(f_r)}$$

with  $f_1 | \dots | f_r$  all non-zero polynomials over  $F$ .

Therefore there is a basis for  $V$  in which  $\alpha$  is given by the block diagonal matrix

$$\begin{pmatrix} C(f_1) & & 0 \\ & \ddots & \\ 0 & & C(f_r) \end{pmatrix}$$

*Proof.*  $F[X]$  is a PID so by the classification theorem for f.g. modules over a PID, we must have

$$V_\alpha \cong \frac{F[X]}{(f_1)} \oplus \dots \oplus \frac{F[X]}{(f_r)} \oplus F[X]^n$$

with  $f_1 | \dots | f_r$ .

Since  $F[X]$  isn't a finite-dimensional  $F$ -vector space, but  $V_\alpha$  is, we must have that  $n = 0$ .

If we write  $b_i$  for each  $\deg(f_i)$ , then with respect to the basis

$$X_1^0, X_1, \dots, X_1^{b_1-1}, \dots, X_r^0, \dots, X_r^{b_r-1}$$

(where  $X_j$  is the  $X$  in the  $j^{\text{th}}$  term in the direct sum  $F[X]/(f_j)$ ),  $\alpha$  is in the required form.  $\square$

If we required all the  $f_i$  to be monic, this is completely canonical- there is no ambiguity.

**Example 3.73.** The  $\mathbb{Q}[X]$ -modules

$$\frac{\mathbb{Q}[X]}{X^2 - 1}$$

and

$$\frac{\mathbb{Q}[X]}{X - 1} \oplus \frac{\mathbb{Q}[X]}{X + 1}$$

are isomorphic, but only the first one is in rational canonical form, since neither of  $X - 1$  and  $X + 1$  divide each other.

We now focus in on the case  $F = \mathbb{C}$ .

**Lemma 3.74.** The primes in  $\mathbb{C}[X]$  are, up to associates,  $X - \lambda$ , for  $\lambda \in \mathbb{C}$ .

*Proof.* First note that any linear polynomial  $X - \lambda \in \mathbb{C}[X]$  is prime.

Now let  $f \in \mathbb{C}[X]$  be prime.  $f$  is non-zero and not a unit so it's non-constant. So by the fundamental theorem of algebra,  $f$  has some root,  $\lambda \in \mathbb{C}$ . So  $(X - \lambda) | f$ .

Since  $f$  is irreducible, we must have that  $X - \lambda$  and  $f$  are associates.  $\square$

Combining previous results of this section, we have the following:

**Theorem 3.75** (Jordan normal form). Let  $\alpha : V \rightarrow V$  be a linear map, where  $V$  is a finite dimensional complex vector space. Then there is an isomorphism of  $\mathbb{C}[X]$ -modules:

$$V_\alpha \cong \frac{\mathbb{C}[X]}{((X - \lambda_1)^{n_1})} \oplus \dots \oplus \frac{\mathbb{C}[X]}{((X - \lambda_r)^{n_r})}$$

for some  $n_i \in \mathbb{N}$ ,  $\lambda_i \in \mathbb{C}$ .

Furthermore there's a  $\mathbb{C}$ -basis for  $V$  in which  $\alpha$  has the form of the following block diagonal matrix:

$$\begin{pmatrix} J_{n_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{n_r}(\lambda_r) \end{pmatrix}$$

where  $J_n(\lambda)$  is the  $n \times n$  Jordan block:

$$J_n(\lambda) := \begin{pmatrix} \lambda & & 0 \\ 1 & \ddots & \\ \vdots & & 1 & \lambda \\ 0 & & & \end{pmatrix}$$

*Proof.* For the first part, we apply the prime decomposition theorem to  $V_\alpha$ , and use the fact that all primes in  $\mathbb{C}[X]$  are of the form  $X - \lambda$ .

Then take a basis as in the proof of rational canonical form, and use the fact that

$$C((X - \lambda)^n) = J_n(\lambda)$$

□

**Lemma 3.76.** *Let  $V$  be a finite-dimensional  $F$ -vector space, and  $\alpha, \beta : V \rightarrow V$  two linear maps. Then  $V_\alpha \cong V_\beta$  are isomorphic as  $F[X]$ -modules if and only if  $\alpha$  and  $\beta$  are conjugate, i.e. there is some isomorphism  $\gamma : V \rightarrow V$  such that  $\gamma \circ \alpha = \beta \circ \gamma$*

*Proof.* ( $\Rightarrow$ ) Let  $\gamma : V_\alpha \rightarrow V_\beta$  be an  $F[X]$ -module isomorphism; so it's in particular an isomorphism of  $F$ -vector spaces. Then for  $v \in V$ ,  $\beta(v) = X \cdot_{V_\beta} v$  (here  $\cdot_{V_\beta}$  denotes the  $F[X]$ -module action on  $V_\beta$ ), and similarly  $\alpha(v) = X \cdot_{V_\alpha} v$ . Then

$$\begin{aligned} \beta \circ \gamma(v) &= X \cdot_{V_\beta} \gamma(v) \\ &= \gamma(X \cdot_{V_\alpha} v) \\ &= \gamma \circ \alpha(v) \end{aligned}$$

So  $\beta \circ \gamma = \gamma \circ \alpha$ , i.e.  $\alpha$  and  $\beta$  are conjugate.

( $\Leftarrow$ ) Suppose  $\alpha$  and  $\beta$  are conjugate, so there is an isomorphism  $\gamma : V \rightarrow V$  such that  $\beta \circ \gamma = \gamma \circ \alpha$ . We claim that  $\gamma : V_\alpha \rightarrow V_\beta$  is an  $F[X]$ -module isomorphism. Indeed for  $f \in F[X]$  and  $v \in V$ :

$$\begin{aligned} \gamma(f \cdot_{V_\alpha} v) &= \gamma(f(\alpha)(v)) \\ &= f(\beta)(\gamma(v)) \end{aligned}$$

where we note that  $\gamma \circ \alpha^i = \alpha^i \circ \beta$  for all  $i \geq 0$ .

So  $\gamma$  is an  $F[X]$ -module homomorphism; it's a bijection since it's an  $F$ -linear isomorphism. □

Reinterpreting this plus rational canonical form, we have a classification result for square matrices over a field; it's weaker than Jordan normal form but works over any field:

**Corollary 3.77.** *Let  $F$  be a field. There's a bijection between conjugacy classes of  $n \times n$  matrices over  $F$  and sequences of monic polynomials  $f_1, \dots, f_r \in F[X]$ , such that  $f_1 | \dots | f_r$ , and  $\sum_i \deg(f_i) = n$ .*

*Proof.* Let  $A$  be an  $n \times n$  matrix over  $F$ , let  $V = F^n$  and  $\alpha : V \rightarrow V$  send  $v$  to  $A \cdot v$ . Then by rational canonical form,

$$V_\alpha \cong \frac{F[X]}{(f_1)} \oplus \dots \oplus \frac{F[X]}{(f_r)}$$

for some unique sequence  $f_i \in F[X]$ , with  $f_1 | \dots | f_r$  all non-zero monic polynomials in  $F[X]$ . By counting dimensions of both sides,  $\sum_i \deg(f_i) = n$ .

Furthermore, if  $A$  and  $B$  are conjugate matrices, then  $V_\alpha$  and  $V_\beta$  are isomorphic  $F[X]$ -modules (by the lemma), so they define the same sequence of polynomials.

If  $f_1 | \dots | f_r$  is any sequence of monic polynomials in  $F[X]$  with  $\sum_i \deg(f_i) = n$ , then

$$V := \frac{F[X]}{(f_1)} \oplus \dots \oplus \frac{F[X]}{(f_r)}$$

is a finite dimensional  $F$ -vector space of dimension  $n$ , with a linear map  $\alpha : V \rightarrow V$  sending  $v$  to  $X \cdot v$ , which determines a matrix up to conjugation, by picking a basis. □

**Example 3.78.** Let's look at conjugacy classes in  $M_{2 \times 2}(F)$ . This corresponds to classifying  $F[X]$ -modules of the form

$$\frac{F[X]}{(f_1)} \oplus \dots \oplus \frac{F[X]}{(f_r)}$$

with  $f_1 | \dots | f_r$  a sequence of monic polynomials, with  $\sum_i \deg(f_i) = 2$ .

So either  $r = 1$  and  $\deg(f_1) = 2$ , or  $r = 2$  and  $\deg(f_1) = \deg(f_2) = 1$ .

In the latter case- so  $r = 2$ - since  $f_1 | f_2$  and both are monic,  $f_1 = f_2 = X - \lambda$ , for some  $\lambda \in F$ . So since the companion matrix is the  $1 \times 1$  matrix  $(\lambda)$ , the corresponding matrix in  $M_{2 \times 2}(F)$  is

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

In the case  $r = 1$ , we write  $f_1 = X^2 + aX + b$ , for some  $a, b \in F$ . Then the matrix we get is the companion matrix of  $f_1$ , which is

$$C(f_1) = \begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix}$$

If  $f_1$  is irreducible, we can't simplify this any further for an arbitrary field  $F$ . However if for example  $F = \mathbb{Z}/3\mathbb{Z}$ , we could find all irreducible degree 2 polynomials, and get a better classification.

If  $f_1$  is reducible, then  $f_1 = (X - \lambda)(X - \mu)$ , for some  $\lambda, \mu \in F$ .

If  $\lambda = \mu$ , then  $f_1 = (X - \lambda)^2$  so we saw in an earlier example that the matrix is conjugate to the Jordan block:

$$\begin{pmatrix} \lambda & 0 \\ 1 & \lambda \end{pmatrix}$$

If  $\lambda \neq \mu$ , the matrix

$$\begin{pmatrix} 0 & -\lambda\mu \\ 1 & \lambda + \mu \end{pmatrix}$$

has distinct eigenvalues so it's conjugate to

$$\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$$

Lecture 18

## 4 Matrix Lie groups

### 4.1 Matrix groups

Let  $R$  be a commutative ring. Recall  $M_n(R)$  is the ring of  $n \times n$  matrices over  $R$ , which is non-commutative for  $n \geq 2$ .

**Definition 4.1.** The general linear group  $GL_n(R)$  over  $R$  is defined to be the unit group of  $M_n(R)$ ,  $M_n(R)^\times$ .

One can define the determinant function  $\det : M_n(R) \rightarrow R$  with the same formula as you saw in linear algebra, and the same proof shows that  $M$  is invertible if and only if  $\det(M) \in R^\times$ .

**Exercise 4.2.** Verify this.

**Proposition 4.3.** The centre of  $GL_n(R)$  is the set of matrices of the form  $\lambda I$ , for  $\lambda$  in  $R^\times$ .

*Proof.* Let  $A$  lie in the centre  $Z(\mathrm{GL}_n(R))$ . For  $1 \leq u, v \leq n$  distinct, let  $E^{uv} \in \mathrm{GL}_n(R)$  be the matrix with 1s on the diagonal,  $E_{uv}^{uv} = 1$ , and all other entries 0. Explicitly,

$$E_{ij}^{uv} = \delta_{ij} + \delta_{ui}\delta_{vj}$$

We must have that  $AE^{uv} = E^{uv}A$ , and so for all  $1 \leq i, j \leq n$ ,

$$\begin{aligned} 0 &= (E^{uv}A - AE^{uv})_{ij} \\ &= E_{ik}^{uv}A_{kj} - A_{ik}E_{kj}^{uv} \\ &= \delta_{ui}A_{vj} - \delta_{vj}A_{ui} \end{aligned}$$

For  $u = i \neq v = j$ , this tells us  $A_{ii} = A_{jj}$ , so all diagonal entries are the same.

If  $i = j = u \neq v$ , this tells us  $A_{vj} = 0$ , so all other entries are 0. So  $A$  must be of the form  $\lambda I$ . For  $A$  to be invertible,  $\lambda$  must be invertible.  $\square$

Lecture 19

## 4.2 Recap of some topology

Here we quickly recap some things from topology.

**Definition 4.4.** A topology  $\mathcal{T}$  on a set  $X$  is a collection of subsets of  $X$  (called the open subsets) such that

- $\emptyset$  is open.
- $X$  is open in  $X$
- If  $\{U_i\}_{i \in I}$  is a collection of open subsets of  $X$  ( $I$  is a possibly infinite set), then their union

$$\bigcup_i U_i$$

is open in  $X$ .

- If  $U_1, \dots, U_n \subseteq X$  are open, their intersection

$$U_1 \cap \dots \cap U_n$$

is open in  $X$  too.

A subset  $C \subseteq X$  is closed if  $X \setminus C$  is open.

**Example 4.5.** If  $X \subseteq \mathbb{R}^n$ , we give it the Euclidean topology: we say that  $U \subseteq X$  is open if for all  $x \in X$ , there is some  $\varepsilon > 0$  such that any  $y \in X$  with  $|y - x| < \varepsilon$  lies in  $U$ .

**Example 4.6.** Consider  $X = \mathbb{Q} \subseteq \mathbb{R}$ . Open intervals  $(a, b) \cap \mathbb{Q}$  are open in  $\mathbb{Q}$ ; any open subset  $U \subseteq \mathbb{Q}$  is a union of these.

**Example 4.7.** The “open square”  $(0, 1)^2 \subseteq \mathbb{R}^2$  is open.

**Definition 4.8.** A function  $f : X \rightarrow Y$  between two topological spaces is continuous if  $f^{-1}(U)$  is open in  $X$  whenever  $U \subseteq Y$  is open.

This implies that if  $f : X \rightarrow Y$  is continuous and  $C \subseteq Y$  is closed,  $f^{-1}(C) \subseteq X$  is closed.

**Lemma 4.9.** *If  $X \subseteq \mathbb{F}^n$  and  $Y \subseteq \mathbb{F}^m$  have the subspace topology, then  $f : X \rightarrow Y$  is continuous with respect to the above definition if and only if it's continuous with respect to the  $\varepsilon$ - $\delta$  definition of continuity: for all  $x \in X$ , for all  $\varepsilon > 0$ , there is some  $\delta > 0$  such that whenever  $y \in X$  satisfies  $|x - y| < \delta$ , we have that  $|f(x) - f(y)| < \varepsilon$ .*

*"y sufficiently close to x implies f(y) close to f(x)".*

**Lemma 4.10.** *The composition of two functions is continuous.*

**Example 4.11.** *A function  $f = (f_1, \dots, f_n) : \mathbb{F}^m \rightarrow \mathbb{F}^n$  is continuous if and only if each component  $f_i : \mathbb{F}^m \rightarrow \mathbb{F}$  are continuous.*

*Projection maps  $\pi_j : \mathbb{F}^n \rightarrow \mathbb{F}$ , sending  $(x_1, \dots, x_n)$  to  $x_j$  are continuous. Also if  $f, g : \mathbb{F}^n \rightarrow \mathbb{F}$  are continuous, then so are  $f + g$  and  $f \cdot g$ .*

*Combining this, any polynomial (in multiple variables)  $f : \mathbb{F}^n \rightarrow \mathbb{F}$  is continuous.*

*This also implies the multiplication map*

$$\mu : M_n(\mathbb{F}) \times M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$$

*sending  $(A, B)$  to  $A \cdot B$  is continuous, since each entry of  $A \cdot B$  is a polynomial in the entries of  $A$  and  $B$ .*

**Definition 4.12.** *If  $Y \subseteq \mathbb{F}^n$  has the Euclidean topology and  $X \subseteq Y$  is a supset,  $X$  is dense in  $Y$  if  $\bar{X}$ , the closure of  $X$  in  $Y$ , is the whole of  $Y$ . Equivalently, this means that for all  $y \in Y$  and for all  $\varepsilon > 0$ , there is some  $x \in X$  with  $|x - y| < \varepsilon$ .*

**Example 4.13.**  $\mathbb{Q} \subseteq \mathbb{R}$  is dense.

$\mathbb{Q}[i] \subseteq \mathbb{C}$  is dense.

We will occasionally use some other topological properties.

**Definition 4.14.** *A topological space  $X$  is path-connected if for all  $x, y \in X$ , there is a path from  $x$  to  $y$ , i.e. there is a continuous function  $f : [0, 1] \rightarrow X$  sending 0 to  $x$  and 1 to  $y$ .*

**Lemma 4.15.** *Let  $x, y, z \in X$ . If there's a path  $\gamma$  from  $x$  to  $y$  and a path  $\delta$  from  $y$  to  $z$ , there's a path from  $x$  to  $z$ .*

*Proof.* Let  $\xi : [0, 1] \rightarrow X$  send  $t \in [0, 1]$  to

$$\begin{cases} \gamma(2t) & \text{if } t \leq \frac{1}{2} \\ \delta(2t - 1) & \text{if } t \geq \frac{1}{2} \end{cases}$$

This is a path from  $x$  to  $z$ . □

**Lemma 4.16.** *If  $X \subseteq \mathbb{F}^n$ ,  $X$  is compact if  $X$  is a closed subset of  $\mathbb{F}^n$  and it's a bounded subset, meaning there is some  $C > 0$  such that all  $x \in X$  have  $|x| \leq C$ .*

All spaces that we'll care about are subspaces of  $\mathbb{F}^n$  for some  $n$  so we can take this to be the definition.

**Example 4.17.** *We'll show  $U(1)$  is compact.*

$$\begin{aligned} U(1) &= \left\{ (z) \in M_1(\mathbb{C}) \mid (z)^\dagger (z) = I = (1) \right\} \\ &= \{ z \in \mathbb{C} \mid \bar{z}z = 1 \} \\ &= \{ z \in \mathbb{C} \mid |z| = 1 \} \end{aligned}$$

*This is bounded as  $|z| \leq 1$ . It's a closed subset of  $\mathbb{C}$  since  $|\cdot|$  is a continuous function.*

**Example 4.18.**  $\mathrm{GL}_n(\mathbb{F}) \subseteq \mathbb{F}^{n \times n}$  isn't bounded, and hence isn't compact. To see this, consider the matrix

$$\begin{pmatrix} 1 & 0 & \cdots & k \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix} \in \mathrm{GL}_n(\mathbb{F})$$

for  $k \in \mathbb{N}$ . As  $k \rightarrow \infty$ , the norm of this matrix goes to  $\infty$  too, so  $\mathrm{GL}_n(\mathbb{F})$  isn't a bounded subset of  $M_n(\mathbb{F})$ .

**Example 4.19.**  $\mathrm{GL}_1(\mathbb{R}) = \mathbb{R}^\times$  isn't path-connected, since there's no path between 1 and -1. In fact, for any  $n$ ,  $\mathrm{GL}_n(\mathbb{R})$  isn't path-connected.

*Proof.* We'll show there's no path in  $\mathrm{GL}_n(\mathbb{R})$  between

$$A := \begin{pmatrix} -1 & \cdots & & 0 \\ \vdots & 1 & & \\ & & \ddots & \\ 0 & & & 1 \end{pmatrix}$$

and  $I$ . Suppose  $\gamma : [0, 1] \rightarrow \mathrm{GL}_n(\mathbb{R})$  is a path from  $I$  to  $A$ . Then  $\det \circ \gamma : [0, 1] \rightarrow \mathbb{R}$  is continuous, and sends 0 to 1 and 1 to -1. So by the intermediate value theorem there would be some  $t$  such that  $\det(\gamma(t)) = 0$ . But  $\gamma(t)$  was an invertible matrix, so this is a contradiction.  $\square$

**Example 4.20.**  $\mathrm{GL}_n(\mathbb{C})$  is always path-connected.

*Sketch.* Let  $A \in \mathrm{GL}_n(\mathbb{C})$ . We can write  $A = PBP^{-1}$ , where  $B$  is in Jordan normal form. We assume for simplicity that  $B$  is a single Jordan block:

$$B = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & & \\ \vdots & & \ddots & & \\ 0 & & & & \lambda \end{pmatrix}$$

$\lambda \neq 0$  so  $\lambda = e^z$  for some  $z \in \mathbb{C}$ . We can now write down the path we want:

$$\gamma(t) := \begin{pmatrix} e^{tz} & t & 0 & \cdots & 0 \\ 0 & e^{tz} & t & & \\ \vdots & & \ddots & & \\ 0 & & & & e^{tz} \end{pmatrix}$$

This is invertible for all  $t$ , and  $\gamma(0) = A$  and  $\gamma(1) = I$ .

The case where  $B$  isn't just a single Jordan block is similar.  $\square$

Next we'll recap some stuff about differentiability.

**Definition 4.21.** A function  $\gamma : \mathbb{R} \rightarrow \mathbb{F}^m$  is continuously differentiable (or  $C^1$ ) if its derivative  $\dot{\gamma}(t)$  exists for all  $t \in \mathbb{R}$ , and is a continuous function  $\dot{\gamma} : \mathbb{R} \rightarrow \mathbb{F}^m$ .

A function  $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$  is continuously differentiable (or  $C^1$ ) if all partial derivatives  $\frac{\partial f_i}{\partial x_j}$  exist, and are continuous. The derivative at  $p$  is the matrix

$$\left( \frac{\partial f_i}{\partial x_j}(p) \right)_{ij}$$

We'll use the following only once or twice.

**Theorem 4.22** (Inverse Function Theorem.). Let  $f : \mathbb{R}^m \rightarrow \mathbb{R}^m$  be continuously differentiable, and  $x_0 \in \mathbb{R}^m$  some point, such that the derivative  $df_{x_0}$  of  $f$  at  $x_0$  is an invertible matrix.

Then there's some open neighbourhood  $U$  of  $x_0$  in  $\mathbb{R}^m$  such that if  $V := f(U)$ , then

$$f|_U : U \rightarrow V$$

is a homeomorphism, and is continuously differentiable with continuously differentiable inverse.

So if the derivative at  $x_0$  is invertible,  $f$  is invertible “locally near  $x_0$ ”.

**Example 4.23.** Consider  $\exp : \mathbb{R} \rightarrow \mathbb{R}$ , sending  $x$  to  $e^x$ . This isn't a homeomorphism as it's not surjective. But the derivative at 0 is  $1 \in M_1(\mathbb{R})$ , which is invertible. We can find the open neighbourhood in the theorem explicitly:

$$\exp|_{(-1,1)} : (-1,1) \rightarrow (e^{-1}, e^1)$$

is a homeomorphism, it's continuously differentiable, and its inverse is  $\log$  which is continuously differentiable.

**Example 4.24.** Consider  $\exp : \mathbb{C} \rightarrow \mathbb{C}$ , sending  $z$  to  $e^z$ . The derivative at 0,  $d\exp_0 : \mathbb{C} = \mathbb{R}^2 \rightarrow \mathbb{C} = \mathbb{R}^2$ , is the identity  $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ , which is invertible. Now it's less obvious what  $U$  and  $V$  should be, though we can still do it; it turns out we can take  $U$  to be the disc of radius 1 around 0.

### 4.3 Matrix Lie groups

For the rest of the course, we will work with matrices over  $\mathbb{R}$  or  $\mathbb{C}$ . We write  $\mathbb{F}$  to mean either  $\mathbb{R}$  or  $\mathbb{C}$ .

We equip  $GL_n(\mathbb{F}) \subseteq M_n(\mathbb{F})$  with the subspace topology. Since  $GL_n(\mathbb{F}) = \{M \in M_n(\mathbb{F}) \mid \det(M) \neq 0\}$ ,  $GL_n(\mathbb{F}) \subseteq M_n(\mathbb{F})$  is an open subset.

**Definition 4.25.** A topological group is a group  $G$  which is also equipped with a topology, such that the multiplication map  $G \times G \rightarrow G$  and the inverse map  $G \rightarrow G$  are both continuous, and  $G$  is Hausdorff.

A homomorphism of topological groups is a homomorphism of groups which is continuous. An isomorphism is an isomorphism of groups which is a homeomorphism.

Since the composition of two continuous functions is continuous and the composition of two homomorphisms is a homomorphism, the composition of two homomorphisms of topological groups is itself a homomorphism of topological groups.

**Example 4.26.** 1. Any group  $G$ , when equipped with the discrete topology, becomes a topological group.

2.  $(\mathbb{R}, +)$  is a topological group.
3.  $(\mathbb{Q}, +)$  is a topological group, and the inclusion  $\mathbb{Q} \hookrightarrow \mathbb{R}$  is a homomorphism of topological groups.
4.  $(\mathbb{R}^\times, \cdot)$  and  $(\mathbb{C}^\times, \cdot)$  are topological groups. There is a continuous homomorphism  $(\mathbb{R}, +) \rightarrow (\mathbb{C}^\times, \cdot)$  sending  $t$  to  $e^{2\pi it}$ .

**Lemma 4.27.**  $\mathrm{GL}_n(\mathbb{F})$  is a topological group (for  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{C}$ ).

This is one of the most important examples for us.

*Proof.* Matrix multiplication  $M_n(\mathbb{F}) \times M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  is polynomial in each entry, and hence is continuous.

For an invertible matrix  $M$ ,  $\det(M)M^{-1}$  has entries which are polynomials in the entries of  $M$ , and so varies as a continuous function of  $M$ .  $\det(M)$  is also a continuous function of  $M$ , so  $M^{-1}$  is a continuous function of  $M$ .

For a subgroup  $G \leq \mathrm{GL}_n(\mathbb{F})$ , the multiplication and inverse maps are the restrictions of those on  $\mathrm{GL}_n(\mathbb{F})$ , and so are also continuous.  $\square$

**Lemma 4.28.** Let  $G$  be a topological group. Any subgroup  $H \leq G$ , when equipped with the subspace topology, is a topological group.

*Proof.* The product  $H \times H \rightarrow H$  is the restriction of the product  $G \times G \rightarrow G$ , so is continuous in the subspace topology; similar for the inverse.  $\square$

**Definition 4.29.** A matrix Lie group is a topological group  $G$ , which is isomorphic (as a topological group) to a closed subgroup  $H \leq \mathrm{GL}_n(\mathbb{F})$  for some  $n$ .

Note that since  $\mathrm{GL}_n(\mathbb{F})$  is Hausdorff and in fact metrisable, so is any subspace. So the same applies to any matrix Lie group.

**Example 4.30.**  $(\mathbb{R}^\times, \cdot) = \mathrm{GL}_1(\mathbb{R})$  is a matrix Lie group. This contains the subgroup  $\mathbb{Q}^\times \subseteq \mathbb{R}^\times$ , but this isn't a closed subgroup. That doesn't immediately imply  $\mathbb{Q}^\times$  isn't a matrix Lie group (since maybe it could be isomorphic to e.g. a closed subgroup of  $\mathrm{GL}_7(\mathbb{C})$ ), but we'll see later on that it is in fact not a matrix Lie group.

If  $H \leq G$  is a closed subgroup, and  $G \leq \mathrm{GL}_n(\mathbb{F})$  is a closed subgroup, then  $H \leq \mathrm{GL}_n(\mathbb{F})$  is a closed subgroup. It follows that a closed subgroup of a matrix Lie group is itself a matrix Lie group.

**Example 4.31.** The following are examples of matrix Lie groups (proofs are exercises on the problem sheet).

1.  $\mathrm{SL}_n(\mathbb{F}) \leq \mathrm{GL}_n(\mathbb{F})$ . It's a closed subset since  $\mathrm{SL}_n(\mathbb{F}) = \{M \in \mathrm{GL}_n(\mathbb{F}) \mid \det(M) = 1\}$ , and  $\det$  is a continuous function.
2. The set of diagonal matrices in  $\mathrm{GL}_n(\mathbb{F})$ .
3.  $O(n) \leq \mathrm{GL}_n(\mathbb{R})$ , the orthogonal group of orthogonal matrices, meaning  $A^T A = I$ .
4.  $SO(n) \leq \mathrm{GL}_n(\mathbb{R})$ , the special orthogonal group of orthogonal matrices with determinant 1.
5.  $U(n) \leq \mathrm{GL}_n(\mathbb{C})$ , the unitary group of unitary matrices, meaning  $A^\dagger A = I$ .

6.  $SU(n) \leq GL_n(\mathbb{C})$ , the special unitary group of unitary matrices with determinant 1.
7.  $\mathbb{Z}$  or any finite group, equipped with the discrete topology.
8. We know that the centre of  $GL_n(\mathbb{F})$  is  $Z(GL_n(\mathbb{F})) = \{\lambda I \mid \lambda \in \mathbb{F}^\times\}$ . We define the projective general linear group  $PGL_n(\mathbb{F})$  to be the quotient  $GL_n(\mathbb{F})/Z(GL_n(\mathbb{F}))$  (equipped with the quotient topology).
9. The Heisenberg group  $H_n(\mathbb{F}) \subseteq GL_n(\mathbb{F})$  is defined to be the set of upper triangular matrices with 1s on the diagonal, i.e. matrices of the form

$$\begin{pmatrix} 1 & ? & \cdots & ? \\ 0 & 1 & & \\ \vdots & & \ddots & \\ 0 & & & 1 \end{pmatrix}$$

**Example 4.32.** There is an isomorphism of topological groups  $\phi : \mathbb{R}/\mathbb{Z} \rightarrow U(1)$ , defined by  $\phi(t) := (e^{2i\pi t})$ .

**Lemma 4.33.** Let  $G$  be a topological group, and let  $H \leq G$  be any subgroup. Then the closure  $\bar{H}$  of  $H$  in  $G$  is a closed subgroup.

*Proof.*  $\bar{H} \subseteq G$  is certainly a closed subgroup. We need to check it's a subgroup.

Let  $g, h \in \bar{H}$ . Then there are sequences  $\{g_i\}_i$  and  $\{h_i\}_i$  in  $H$ , such that  $g_i \rightarrow g$  and  $h_i \rightarrow h$ . Then since multiplication is continuous,  $g_i \cdot h_i \rightarrow g \cdot h$ . Since each  $g_i \cdot h_i \in H$ , the limit of this sequence,  $g \cdot h$ , lies in  $\bar{H}$ . Similarly each  $g_i^{-1} \in H$  so  $g^{-1} \in \bar{H}$  too.  $\square$

So if we take any subgroup  $G \leq GL_n(\mathbb{F})$ , its closure  $\bar{G}$  is a matrix Lie group.

**Example 4.34.** The closure of  $\mathbb{Q}^\times$  in  $\mathbb{R}^\times$  is the entirety of  $\mathbb{R}^\times$ .

**Proposition 4.35.**  $GL_n(\mathbb{R})$  is a closed subgroup in  $GL_n(\mathbb{C})$ .

*Proof.* Consider the continuous map  $f : GL_n(\mathbb{C}) \rightarrow M_n(\mathbb{R})$  sending  $A$  to  $\text{Im } A$ .  $GL_n(\mathbb{R}) = f^{-1}\{0\}$  so it's a closed subset.  $\square$

Lecture 19

**Proposition 4.36.**  $GL_n(\mathbb{C})$  is isomorphic to a closed subgroup in  $GL_{2n}(\mathbb{R})$ .

This tells us that a topological group  $G$  is a matrix Lie group with respect to  $\mathbb{F} = \mathbb{R}$  if and only if it's one with respect to  $\mathbb{F} = \mathbb{C}$ .

*Proof.* We will produce an injective group homomorphism  $\phi : GL_n(\mathbb{C}) \hookrightarrow GL_{2n}(\mathbb{R})$  which is a homeomorphism onto its image. Then  $\phi$  is an isomorphism of topological groups between  $GL_n(\mathbb{C})$  and  $\text{Im}(\phi)$ .

For  $M \in GL_n(\mathbb{C})$ , we define  $\phi(M)$  to be the block matrix

$$\begin{pmatrix} \text{Re } M & \text{Im } M \\ -\text{Im } M & \text{Re } M \end{pmatrix} \in GL_{2n}(\mathbb{R})$$

It is immediate that  $\phi$  is continuous, injective, and  $\phi(I) = I$ . Let's check it's a homomorphism.

Let  $M, N \in \text{GL}_n(\mathbb{C})$ , and write  $M = A + iB$  and  $N = C + iD$ , where  $A, B, C, D$  are in  $M_n(\mathbb{R})$  (why can't we assume they're in  $\text{GL}_n(\mathbb{R})$ ?). Then

$$\phi(M)\phi(N) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \begin{pmatrix} C & D \\ -D & C \end{pmatrix}$$

and

$$\phi(MN) = \phi((AC - BD) + i(BC + AD))$$

Expanding, we see that both sides are the same. So it remains to check that  $\text{Im } \phi \subseteq \text{GL}_{2n}(\mathbb{R})$  is closed.

Let  $J \in \text{GL}_{2n}(\mathbb{R})$  be the matrix

$$\begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$$

We will show that  $\text{Im } \phi = \{M \in \text{GL}_{2n}(\mathbb{R}) \mid MJ = JM\}$ , which is a closed subset.

Writing  $M = A + iB \in \text{GL}_n(\mathbb{C})$  with  $A, B \in M_n(\mathbb{R})$ , we can compute

$$\phi(M)J - J\phi(M) = \begin{pmatrix} A & B \\ -B & A \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} - \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ -B & A \end{pmatrix}$$

which we can expand to get 0.

Now suppose  $P \in \text{GL}_{2n}(\mathbb{R})$  satisfies  $PJ = JP$ . Then  $P$  is of the form

$$\begin{pmatrix} A & B \\ -B & A \end{pmatrix}$$

for some matrices  $A, B \in M_n(\mathbb{R})$ . Let  $M = A + iB \in M_n(\mathbb{C})$ . If  $M$  is invertible, then  $\phi(M) = P$ , so  $P \in \text{Im } \phi$ .

Multiplying the equation  $PJ = JP$  on the left and right by  $P^{-1}$ , we see that  $P^{-1}J = JP^{-1}$ . So  $P^{-1}$  must also be of the form

$$\begin{pmatrix} C & D \\ -D & C \end{pmatrix}$$

for some matrices  $C, D \in M_n(\mathbb{R})$ . Let  $N = C + iD \in M_n(\mathbb{C})$ . Then we can check that  $N$  is an inverse to  $M$ . □

Lecture 20

**Lemma 4.37.** *Let  $G \leq \text{GL}_n(\mathbb{F})$  be a closed subgroup, and  $H \leq G$  a closed subgroup. Then  $H \leq \text{GL}_n(\mathbb{F})$  is a closed subgroup.*

*Proof.* Subgroups of subgroups are subgroups, so  $H \leq \text{GL}_n(\mathbb{F})$  is a subgroup. Similarly closed subsets of closed subsets are closed subsets. So  $H \leq \text{GL}_n(\mathbb{F})$  is closed; together, these imply it's a closed subgroup. □

**Corollary 4.38.** 1. *Any closed subgroup  $G \leq \text{GL}_n(\mathbb{R})$  is a closed subgroup on  $\text{GL}_n(\mathbb{C})$ .*

2. *Any closed subgroup  $H \leq \text{GL}_m(\mathbb{C})$  is isomorphic to a closed subgroup in  $\text{GL}_{2m}(\mathbb{R})$ .*

*Proof.* 1. Let  $G \leq \text{GL}_n(\mathbb{R})$  be a closed subgroup; since  $\text{GL}_n(\mathbb{R}) \leq \text{GL}_n(\mathbb{R})$  is a closed subgroup, by the lemma,  $G \leq \text{GL}_n(\mathbb{C})$  is a closed subgroup.

2. Let  $H \leq \text{GL}_m(\mathbb{C})$  be a closed subgroup, and  $\phi : \text{GL}_m(\mathbb{C}) \rightarrow K$  an isomorphism, where  $K \leq \text{GL}_{2m}(\mathbb{R})$  is some closed subgroup. Then  $H \cong \phi(H)$ , and  $\phi(H) \leq K$  is a closed subgroup. So by the lemma,  $H \cong \phi(H) \leq \text{GL}_{2n}(\mathbb{R})$ , where  $\phi(H) \leq \text{GL}_{2n}(\mathbb{R})$  is a closed subgroup. □

## 4.4 Matrix exponentiation

**Definition 4.39.** For  $A \in M_n(\mathbb{F})$ , its exponential, written  $e^A$  or  $\exp(A)$ , is defined to be the infinite sum

$$e^A := \sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

**Remark 4.40.** For  $n = 1$ , this is the usual exponential.

**Theorem 4.41.** The power series for  $\exp$  converges for all  $A$ . Moreover

1. The power series for  $\exp : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  converges absolutely, and uniformly on compact subsets of  $M_n(\mathbb{F})$ .
2. The power series for its partial derivatives (in the components of  $A$ ) of  $\exp$  all converge absolutely, and uniformly on compact subsets.

In particular, this implies that  $\exp : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  is continuously differentiable.

This essentially says “this power series behaves as well as it possibly can”. You’ve seen the proof of this for  $n = 1$ ; the general case is longer but involves similar analysis.

**Example 4.42.** If  $A = 0$ ,  $e^A = I + 0 + 0 + \dots = I$ .

If  $A = I$ ,

$$e^A = I + I + \frac{1}{2}I + \frac{1}{6}I + \dots = eI$$

Here is a more interesting example:

**Example 4.43.** Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

Let’s compute  $e^A$ . We can see inductively that

$$A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

Then

$$\begin{aligned} e^A &= \sum_{i=0}^{\infty} \begin{pmatrix} \frac{1}{i!} & \frac{i}{i!} \\ 0 & \frac{1}{i!} \end{pmatrix} \\ &= \begin{pmatrix} e & e \\ 0 & e \end{pmatrix} \end{aligned}$$

There is also a logarithm for matrices, though we won’t use it nearly as much as the exponential.

**Definition 4.44.** For  $A \in M_n(\mathbb{F})$ , its logarithm,  $\log(A)$  is defined to be the infinite sum

$$\log(A) := \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(A - I)^k}{k} = (A - I) - \frac{(A - I)^2}{2} + \frac{(A - I)^3}{3} - \dots$$

Unfortunately, this does not converge for all  $A$ ; however when  $A$  is close enough to  $I$ , all the same properties as we had for the exponential hold:

**Theorem 4.45.** *There is an open neighbourhood  $U$  of  $I$  in  $M_n(\mathbb{F})$  such that the power series for  $\log$  converges for all  $A \in U$ . Moreover*

1. *The power series for  $\log$  converges absolutely on  $U$ , and uniformly on compact subsets of  $U$ .*
2. *The power series for its partial derivatives (in the components of  $A$ ) of  $\log$  all converge absolutely, and uniformly on compact subsets of  $U$ .*

*In particular, this implies that  $\log : U \rightarrow M_n(\mathbb{F})$  is continuously differentiable.*

**Example 4.46.** *Consider the case  $n = 1$ . Then this is the usual power series for  $\log$ ; this converges for  $z$  if  $|z - 1| < 1$ .*

**Example 4.47.** *Let*

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

*Let's compute  $\log(A)$ . Note that  $(A - I)^k = 0$  for  $k \geq 2$ , so only the first term of the sum is non-zero, so the series converges, and we find that  $\log(A) = B$ , where*

$$B := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

*In fact, we can compute that  $e^B = A$ ; so in this case  $e^{\log(A)} = A$ .*

When both sums converge,  $\exp$  and  $\log$  are inverse to each other.

**Proposition 4.48.** *Suppose the power series for  $\log(A)$  is absolutely convergent.  $\exp(\log(A)) = A$ . Similarly if the power series for  $\log(\exp(B))$  is convergent,  $\log(\exp(B)) = B$ .*

We won't prove this as we won't use it, but the proof amounts to writing the composition  $\exp(\log(A))$  as a double power series in  $A$  and showing that this power series is just  $A$ .

**Proposition 4.49.** *Let  $A, B \in M_n(\mathbb{F})$ . If  $AB = BA$ , then  $e^{A+B} = e^A e^B$ .*

*Proof.*

$$\begin{aligned} e^{A+B} &= \sum_{k=0}^{\infty} \frac{1}{k!} (A+B)^k \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \left( A^k + \binom{k}{1} A^{k-1} B + \dots + B^k \right) \\ &= \sum_{k=0}^{\infty} \sum_{l=0}^k \frac{1}{k!} \binom{k}{l} A^{k-l} B^l \\ &= \sum_{i,j=0}^{\infty} \frac{1}{i! j!} A^i B^j \\ &= \left( \sum_{i=0}^{\infty} \frac{1}{i!} A^i \right) \left( \sum_{j=0}^{\infty} \frac{1}{j!} B^j \right) \\ &= e^A e^B \end{aligned}$$

Here we use the fact that  $AB = BA$  to go from the first to the second line. Also note that when we rearrange these infinite series, we needed to use the fact that these series absolutely converge.  $\square$

**Remark 4.50.** The condition that  $AB = BA$  is necessary; see problem sheet.

**Corollary 4.51.** For any  $A \in M_n(\mathbb{F})$ ,  $e^A e^{-A} = e^0 = I$ .

In particular,  $e^A$  is always invertible. So  $\exp$  is really a map  $\exp : M_n(\mathbb{F}) \rightarrow \text{GL}_n(\mathbb{F})$ .

For  $n = 1$ , this just says that  $e^\lambda e^{-\lambda} = 1$  for all  $\lambda \in \mathbb{F}$ .

*Proof.*  $A$  and  $-A$  always commute, so  $e^A e^{-A} = e^{A-A} = e^0 = I$ . □

**Example 4.52.**  $A$  and  $A^k$  always commute, so  $A$  and  $e^A$  also commute. This implies the identity:

$$e^{Ae^A} = e^A \cdot e^{e^A}$$

**Lemma 4.53.** Let  $A, B \in M_n(\mathbb{F})$ , and suppose that  $A \in \text{GL}_n(\mathbb{F})$  is invertible. Then

$$e^{ABA^{-1}} = Ae^B A^{-1}$$

*Proof.* Note first that  $(ABA^{-1})^k = AB^k A^{-1}$ . So

$$\begin{aligned} e^{ABA^{-1}} &= \sum_{k=0}^{\infty} \frac{1}{k!} (ABA^{-1})^k \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} AB^k A^{-1} \\ &= Ae^B A^{-1} \end{aligned}$$

□

**Lemma 4.54.** For any  $A \in M^n$ ,  $(e^A)^T = e^{A^T}$ .

*Proof.* Note that  $(A^T)^k = (A^k)^T$  for all  $k$ . So

$$e^{A^T} = \left[ \sum_{k=0}^{\infty} \frac{1}{k!} (A^k)^T \right]^T = (e^A)^T$$

□

**Example 4.55.** Suppose  $A$  is an upper triangular matrix:

$$A = \begin{pmatrix} \lambda_1 & & ? \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

Then  $A^k$  is of a similar form:

$$A^k = \begin{pmatrix} \lambda_1^k & & ? \\ & \ddots & \\ 0 & & \lambda_n^k \end{pmatrix}$$

So

$$e^A = \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} \lambda_1^k & & ? \\ & \ddots & \\ 0 & & \sum_{k=0}^{\infty} \frac{1}{k!} \lambda_n^k \end{pmatrix}$$

which is also upper triangular. From this description though we can immediately see the trace, determinant and eigenvalues, and in fact

$$\det(e^A) = e^{\lambda_1} \dots e^{\lambda_n} = e^{\text{Tr}(A)}$$

**Lemma 4.56.** For any  $A \in M_n(\mathbb{F})$ , we have the identity

$$\det(e^A) = e^{\text{Tr}(A)}$$

*Proof.* Write  $A = PBP^{-1}$ , where  $P \in \text{GL}_n(\mathbb{C})$  (even if  $\mathbb{F} = \mathbb{R}$  we can do this) and  $B$  is in Jordan normal form (upper triangular will suffice).

$$\begin{aligned} \det(e^A) &= \det(e^{PBP^{-1}}) \\ &= \det(Pe^B P^{-1}) \\ &= \det(e^B) \end{aligned}$$

and

$$\begin{aligned} e^{\text{Tr}(A)} &= e^{\text{Tr}(PBP^{-1})} \\ &= e^{\text{Tr}(B)} \end{aligned}$$

So from the previous example,  $\det(e^B) = e^{\text{Tr}(B)}$ , so it follows that  $\det(e^A) = e^{\text{Tr}(A)}$ .  $\square$

Recall that  $\exp : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$  is differentiable. For any  $A \in M_n(\mathbb{F})$ , by differentiability of  $\exp$ , the function  $\gamma_A(t) := e^{tA}$  is a continuously differentiable function  $\mathbb{R} \rightarrow \text{GL}_n(\mathbb{F})$ .

**Lemma 4.57.** For any  $A \in M_n(\mathbb{F})$ , the derivative of  $\gamma_A$  at 0 is  $\dot{\gamma}_A(0) = A$ .

*Proof.*

$$\begin{aligned} \dot{\gamma}_A(0) &= \lim_{t \rightarrow 0} \frac{e^{tA} - e^0}{t} \\ &= \lim_{t \rightarrow 0} \frac{\sum_{k=0}^{\infty} \frac{1}{k!} t^k A^k - I}{t} \\ &= \lim_{t \rightarrow 0} \sum_{k=1}^{\infty} \frac{1}{k!} t^{k-1} A^k \\ &= A \end{aligned}$$

$\square$

**Lemma 4.58.** For any  $A \in M_n(\mathbb{F})$ ,  $\gamma_A(t) = e^{tA}$  is a continuous homomorphism  $\gamma_A : \mathbb{R} \rightarrow \text{GL}_n(\mathbb{F})$ .

*Proof.* Since  $\exp$  is continuous,  $\gamma_A$  is too, so we just need to check  $\gamma_A$  is a homomorphism.

Let  $s, t \in \mathbb{R}$ . Then

$$\begin{aligned}\gamma_A(s+t) &= e^{(s+t)A} \\ &= e^{sA} \cdot e^{tA} \\ &= \gamma_A(s) \cdot \gamma_A(t)\end{aligned}$$

where we are using the fact that  $sA$  and  $tA$  commute. So  $\gamma_A$  is a homomorphism.  $\square$

Lecture 21

**Corollary 4.59.** *For any  $t_0 \in \mathbb{R}$ , the derivative of  $\gamma_A$  at  $t_0$  is  $\dot{\gamma}_A(t_0) = A \cdot e^{t_0 A} = A \cdot \gamma_A(t_0)$*

*Proof.* Since  $\gamma_A$  is a homomorphism,

$$\gamma_A(t) = \gamma_A(t - t_0) \cdot \gamma_A(t_0)$$

Differentiating both sides at  $t = t_0$  and using the  $t_0 = 0$  case, we find that

$$\dot{\gamma}_A(t_0) = \dot{\gamma}_A(0) \cdot \gamma_A(t_0) = A \cdot e^{t_0 A}$$

$\square$

In particular, this tells us the derivative of  $\exp$  at 0. Recall that this is a linear map  $d\exp_0 : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$

**Corollary 4.60.** *The derivative of  $\exp$  at 0,  $d\exp_0 : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ , is the identity.*

*Proof.* By the chain rule, the derivative of  $\gamma_A(t) = e^{tA}$  at  $t = 0$  is  $d\exp_0(A)$ . So  $A = d\exp_0(A)$  for all  $A \in M_n(\mathbb{F})$ , i.e.  $d\exp_0$  is the identity map.  $\square$

**Proposition 4.61.**  *$\exp$  restricts to a continuously differentiable homeomorphism from an open neighbourhood of 0 in  $M_n(\mathbb{F})$  to an open neighbourhood of  $I$  in  $GL_n(\mathbb{F})$ , with continuously differentiable inverse.*

*Proof.*  $\exp$  sends 0 to  $I$ , and is differentiable at 0, and its derivative at 0,  $d\exp_0$ , is an invertible linear map. The result then follows from the inverse function theorem.

(Alternatively we could use  $\log$  to construct an inverse.)  $\square$

**Example 4.62.** *For  $n = 1$ , we can take  $U = (-1, 1)$ :*

$$\exp|_{(-1,1)} : (-1, 1) \rightarrow (e^{-1}, e^1) =: V$$

*is a homeomorphism, with inverse  $\log$ , which is continuously differentiable.*

**Example 4.63.** *Recall that we showed that any continuous homomorphism  $\phi : (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$  is of the form  $\phi(t) = \lambda t$  for some  $\lambda \in \mathbb{R}$ .*

**Claim 4.64.** *Any continuous homomorphism  $\psi : (\mathbb{R}, +) \rightarrow (\mathbb{R}^\times, \cdot)$  is of the form  $\psi(t) = e^{\lambda t}$  for some  $\lambda \in \mathbb{R}$ .*

*Proof of claim.* Let  $\psi : (\mathbb{R}, +) \rightarrow (\mathbb{R}^\times, \cdot)$  be a continuous homomorphism.

$\psi(0) = 1 > 0$ . So by the intermediate value theorem,  $\psi(t) > 0$  for all  $t \in \mathbb{R}$ , as otherwise there would be some  $s \in \mathbb{R}$  such that  $\psi(s) = 0$ , a contradiction.

Let  $\phi : \mathbb{R} \rightarrow \mathbb{R}$  be defined to send  $t \in \mathbb{R}$  to  $\log(\psi(t))$ . This is defined since  $\psi(t)$  is always positive.

Then  $\phi$  is continuous, and we'll show it's a homomorphism: let  $s, t \in \mathbb{R}$ . Then

$$\begin{aligned}\phi(s+t) &= \log(\psi(s+t)) \\ &= \log(\psi(s) \cdot \psi(t)) \\ &= \phi(s) + \phi(t)\end{aligned}$$

So  $\phi$  is a continuous homomorphism  $(\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$ , so  $\phi(t) = \lambda t$  for all  $t$ , for some  $\lambda$ .

So  $\psi(t) = e^{\phi(t)} = e^{\lambda t}$ . □

Here we used the fact that  $\log$  was defined on all of  $\mathbb{R} > 0$ , which isn't true for  $n > 0$ ; similarly if we wanted to study continuous homomorphisms  $\mathbb{R} \rightarrow \mathrm{GL}_1(\mathbb{C}) = \mathbb{C}^\times$ , we'd need to e.g. carefully choose a branch of logarithm.

Lecture 22

**Theorem 4.65.** Let  $\gamma : \mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$  be a homomorphism of topological groups.

Then  $\gamma(t) = e^{tA}$ , for some  $A \in M_n(\mathbb{F})$ .

In particular, this implies that  $\gamma$  is continuously differentiable, which is not at all obvious. So this theorem should be very surprising.

*Proof.* First, choose open neighbourhoods  $U$  of 0 in  $M_n(\mathbb{F})$  and  $V$  of  $I$  in  $\mathrm{GL}_n(\mathbb{F})$  such that  $V = \exp(U)$  and  $\exp$  restricts to a homeomorphism

$$\exp|_U : U \rightarrow V$$

By continuity of  $\gamma$ , there's some  $\delta > 0$  such that  $\gamma|_{[-\delta, \delta]}$  lies in  $V$ . Let  $\beta : [-\delta, \delta] \rightarrow U$  be defined by

$$\beta(t) := \exp|_U^{-1}(\gamma(t))$$

i.e.  $\beta$  is defined so that  $\gamma(t) = e^{\beta(t)}$ , whenever  $|t| \leq \delta$ .

**Claim 4.66.**  $\beta(r\delta) = r\beta(\delta)$  for all  $r \in [-1, 1]$ .

*Proof of first claim.* We first prove this when  $r$  is rational. Let  $t = p/q \in [0, 1]$  be rational. Then

$$\begin{aligned}e^{\beta(\delta)} &= \gamma(\delta) \\ &= \gamma(\delta/q)^q && \text{since } \gamma \text{ is a homomorphism} \\ &= \left(e^{\beta(\delta/q)}\right)^q \\ &= e^{q\beta(\delta/q)}\end{aligned}$$

which is in  $V$ , so  $q\beta(\delta/q) = \beta(\delta)$ , i.e.

$$\beta(\delta/q) = \frac{1}{q}\beta(\delta)$$

Similarly

$$\begin{aligned}e^{\beta(p\delta/q)} &= \gamma(p\delta/q) \\ &= \gamma(\delta/q)^p && \text{since } \gamma \text{ is a homomorphism} \\ &= \left(e^{\beta(\delta/q)}\right)^p \\ &= e^{p\beta(\delta/q)}\end{aligned}$$

So

$$\beta(p\delta/q) = (p/q)\beta(\delta)$$

So the claim is true for  $r \in [-1, 1]$  rational.

For general (possibly irrational)  $r \in [-1, 1]$ , we choose some sequence of rational numbers  $t_i = p_i/q_i \in [-1, 1] \cap \mathbb{Q}$  converging to  $r$ ; then since  $\beta$  is continuous,  $\beta(t_i\delta) \rightarrow \beta(r\delta)$ .

However  $\beta(t_i\delta) = t_i\beta(\delta) \rightarrow r\beta(\delta)$ . So  $\beta(r\delta) = r\beta(\delta)$  for all  $r \in [-1, 1]$ , proving the claim.  $\square$

**Claim 4.67.** *For any  $t \in \mathbb{R}$ , we have that*

$$\gamma(t) = e^{t \cdot \frac{1}{\delta} \beta(\delta)}$$

This claim then implies that the theorem holds, for  $A = \frac{1}{\delta} \beta(\delta)$

*Proof of second claim.* Let  $t \in \mathbb{R}$ . Then there is some large integer  $N > 0$  such that  $|t/N| \leq \delta$ , i.e.  $t/N = r\delta$  for some  $r \in [-1, 1]$ . So by the previous claim,

$$\gamma(t/N) = \gamma(r\delta) = e^{e\beta(\delta)}$$

Therefore

$$\begin{aligned} \gamma(t) &= \gamma(t/N)^N \\ &= e^{rN\beta(\delta)} \\ &= e^{t \cdot \frac{1}{\delta} \beta(\delta)} \end{aligned}$$

since  $t/N = r\delta$ .  $\square$

$\square$

A good exercise is to work through this in the case  $n = 1$  and either  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{C}$ ; when  $\mathbb{F} = \mathbb{R}$  we already proved this.

**Corollary 4.68.** *Let  $G \leq \text{GL}_n(\mathbb{F})$  be a closed subgroup. Then any continuous homomorphism  $\gamma : \mathbb{R} \rightarrow G$  is of the form  $\gamma = \gamma_A$  for some  $A \in \text{GL}_n(\mathbb{F})$ .*

*Proof.* Since  $G \leq \text{GL}_n(\mathbb{F})$ , any continuous homomorphism  $\gamma : \mathbb{R} \rightarrow G$  is in particular a continuous homomorphism  $\mathbb{R} \rightarrow \text{GL}_n(\mathbb{F})$ , so the result follows from the theorem.  $\square$

**Example 4.69.** *Let's find all the continuous homomorphisms  $\gamma : \mathbb{R} \rightarrow \text{SL}_n(\mathbb{R})$ .*

**Claim 4.70.** *For  $B \in M_n(\mathbb{R})$ , we have that  $e^B \in \text{SL}_n(\mathbb{R})$  if and only if  $\text{Tr}(B) = 0$ .*

*Proof of claim.*  $\det e^B = e^{\text{Tr}(B)}$  by an earlier lemma, so this is 1 if and only if  $\text{Tr}(B) = 0$  (since  $\text{Tr}(B)$  is a real number).  $\square$

So  $\gamma_A(t) = e^{tA} \in \text{SL}_n(\mathbb{R})$  if and only if  $\text{Tr}(tA) = 0$ , so  $\gamma_A$  is a continuous homomorphism  $\mathbb{R} \rightarrow \text{SL}_n(\mathbb{R})$  if and only if  $\text{Tr}(A) = 0$ .

## 4.5 The Lie algebra of a matrix Lie group

**Definition 4.71.** Let  $G \leq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup, and  $A \in G$ . Then the tangent space  $T_A G$  of  $G$  at  $A$  is the subset of  $\dot{\gamma}(0) \in M_n(\mathbb{F})$ , for  $\gamma : \mathbb{R} \rightarrow G$  a continuously differentiable ( $C^1$ ) path sending 0 to  $A$ . (Note we don't require  $\gamma$  to be a homomorphism).

Note that this subset was assigned to matrix Lie groups inside  $\mathrm{GL}_n$ , not an abstract matrix Lie group.

**Proposition 4.72.** For any  $A \in G$ ,

$$T_A G = A \cdot T_I G = T_I G \cdot A$$

We'll soon see that  $T_I G$  is much more important than any other  $T_A G$ .

*Proof.* Let  $M$  lie in  $T_I G$ . Then  $M = \dot{\gamma}(0)$  for some continuously differentiable function  $\gamma : \mathbb{R} \rightarrow G$  sending 0 to  $I$ . Consider  $A \cdot \gamma : \mathbb{R} \rightarrow G$ , sending  $t$  to  $A \cdot \gamma(t)$ . This is continuously differentiable and sends 0 to  $A$ .

Therefore  $\frac{d}{dt}|_{t=0} A\gamma(t) = A\dot{\gamma}(0)$  lies in  $T_A G$ . This shows  $A \cdot T_I G \subseteq T_A G$ . The same argument with  $A^{-1}$  shows  $A^{-1} \cdot T_A G \subseteq T_I G$ , so  $T_A G = A \cdot T_I G$ , and a similar argument shows that  $T_A G = (T_I G) \cdot A$ .  $\square$

**Proposition 4.73.** For any  $A$ ,  $T_A G$  is a real vector space.

Even if  $\mathbb{F} = \mathbb{C}$ ,  $T_I G$  might not be a complex vector space- there are examples where this happens on the problem sheet.

*Proof.* We prove this first when  $A = I$ , the general case then follows from the previous proposition.

Let  $M, N \in T_I G$ , and  $\lambda \in \mathbb{R}$ . We need to show that  $M + N \in T_I G$  and  $\lambda M \in T_I G$ . Pick continuously differentiable paths  $\alpha, \beta : \mathbb{R} \rightarrow G$  sending 0 to  $I$ , with  $\dot{\alpha}(0) = M$  and  $\dot{\beta}(0) = N$ .

Let  $\gamma(t) = \alpha(\lambda t)$ . Then  $\gamma : \mathbb{R} \rightarrow G$  is a continuously differentiable path with  $\gamma(0) = I$  and  $\dot{\gamma}(0) = \lambda \dot{\alpha}(0) = \lambda M$ , so  $\lambda M \in T_I G$ .

Now let  $\delta(t) = \alpha(t)\beta(t)$ . Then  $\delta : \mathbb{R} \rightarrow G$  is a continuously differentiable path with  $\delta(0) = I$  and, using the product rule,

$$\dot{\delta}(0) = \dot{\alpha}(0)\beta(0) + \alpha(0)\dot{\beta}(0) = MI + IN = M + N \in T_I G$$

$\square$

**Definition 4.74.** For  $G \leq \mathrm{GL}_n(\mathbb{F})$  a closed subgroup, its dimension is the dimension of  $T_A G$ , as a real vector space.

Note that it's not yet obvious that if  $G, H \leq \mathrm{GL}_n(\mathbb{F})$  are isomorphic, then they have the same dimension.

Recall the commutator of two matrices  $A$  and  $B$  is  $[A, B] := AB - BA$ .

**Proposition 4.75.** Let  $G \leq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup, and let  $M, N \in T_I G$ . Then  $[M, N] \in T_I G$ .

*Proof.* Pick continuously differentiable paths  $\alpha, \beta : \mathbb{R} \rightarrow G$  sending 0 to  $I$ , with  $\dot{\alpha}(0) = M$  and  $\dot{\beta}(0) = N$ .

For  $s, t \in \mathbb{R}$ , let  $\delta_s(t) = \alpha(s)\beta(t)\alpha(s)^{-1}$ . This defines a continuously differentiable function  $\mathbb{R}^2 \rightarrow G$ .

For fixed  $s$ ,  $\delta_s$  is a continuously differentiable path  $\mathbb{R} \rightarrow G$  so  $\dot{\delta}_s(0) \in T_I G$ . We can compute this derivative with the product rule:  $\dot{\delta}_s(0) = \alpha(s)\dot{\beta}(0)\alpha(s)^{-1} = \alpha(s) \cdot N \cdot \alpha(s)^{-1}$ .

Allowing  $s$  to vary,  $\dot{\delta}_s(0)$  defines a continuously differentiable path  $\mathbb{R} \rightarrow T_I G$ . So its derivative at 0 must also lie in  $T_I G$ , i.e.

$$\left. \frac{d}{ds} \right|_{s=0} \dot{\delta}_s(0) \in T_I G$$

Note that by the chain rule,  $\left. \frac{d}{ds} \right|_{s=0} \alpha(s)^{-1} = -\dot{\alpha}(0)$ . Then using the product rule,

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} \dot{\delta}_s(0) &= \left( \left. \frac{d}{ds} \right|_{s=0} \alpha(s) \right) \cdot N \cdot \alpha(-0) + \alpha(0) \cdot N \cdot \left( \left. \frac{d}{ds} \right|_{s=0} \alpha(-s) \right) \\ &= \dot{\alpha}(0) \cdot N \cdot \alpha(0) - \alpha(0) \cdot N \cdot \dot{\alpha}(0) \\ &= M \cdot N \cdot I - I \cdot N \cdot M \\ &= [M, N] \end{aligned}$$

which therefore lives in  $T_I G$ . □

**Proposition 4.76.** *For matrices  $A, B, C$  in  $M_n(\mathbb{F})$ , the Jacobi identity holds:*

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

*Proof.* Problem sheet. □

**Definition 4.77.** A (real) Lie algebra is a pair  $(V, [\cdot, \cdot])$ , where  $V$  is a real vector space, and  $[\cdot, \cdot] : V \times V \rightarrow \mathbb{R}$  is a bilinear map, such that

- $[\cdot, \cdot] : V \times V \rightarrow \mathbb{R}$  is antisymmetric, meaning  $[A, B] = -[B, A]$  for all  $A, B \in V$ .
- The Jacobi identity holds:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

We will frequently drop  $[\cdot, \cdot]$  from the notation and refer to the Lie algebra just as  $V$ .

A homomorphism of Lie algebras from  $V$  to  $W$  consists of a linear map  $f : V \rightarrow W$ , which respects the brackets, meaning that  $f([A, B]_V) = [f(A), f(B)]_W$  for all  $A, B$  in  $V$ .

**Definition 4.78.** For a closed subgroup  $G \leq \mathrm{GL}_n(\mathbb{F})$ , its Lie algebra is defined to be the vector space  $T_I G$ , with  $[\cdot, \cdot]$  given by the commutator of matrices.

**Notation 4.79.** We've been using capital letters to denote matrix Lie groups (both abstract ones, like  $G$ , and the standard ones, like  $\mathrm{GL}_n(\mathbb{R})$  and  $SU(2)$ ). A common convention is to use lowercase Fraktur font to denote Lie algebras. You can type in this font in LaTeX using

`\mathfrak{...}`

So the Lie algebra of  $\mathrm{GL}_n(\mathbb{R})$  is written  $\mathfrak{gl}_n(\mathbb{R})$ , the Lie algebra of  $SU(2)$  is written  $\mathfrak{su}(2)$ , and often the Lie algebra of  $G$  or  $H$  is written  $\mathfrak{g}$  or  $\mathfrak{h}$  respectively.

The next goal we'll work towards is to characterise  $T_I G$  in an alternative way.

**Proposition 4.80.** Let  $\gamma : \mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$  be a continuously differentiable function sending 0 to  $I$ , and let  $t \in \mathbb{R}$ . Then

$$e^{t\dot{\gamma}(0)} = \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n$$

In particular, the limit on the right hand side exists.

*Proof.* We can write  $\gamma(t) = e^{\beta(t)}$  for  $|t| \leq \delta$ , for some  $\delta > 0$  and some continuously differentiable function

$$\beta : [-\delta, \delta] \rightarrow M_n(\mathbb{F})$$

Since the derivative of  $\exp$  at 0,  $d\exp_0$ , is the identity map  $M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ , by the chain rule,  $\dot{\beta}(0) = \dot{\gamma}(0)$ . Let  $M = \dot{\beta}(0) = \dot{\gamma}(0)$ .

Then:

$$\begin{aligned} \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n &= \lim_{n \rightarrow \infty} \left(e^{\beta\left(\frac{t}{n}\right)}\right)^n && \text{since for } n \text{ large, } |t/n| < \delta. \\ &= \lim_{n \rightarrow \infty} \left(e^{\frac{t}{n}\dot{\gamma}(0) + o\left(\frac{t}{n}\right)}\right)^n && \text{using the Taylor expansion for } \gamma \\ &= \lim_{n \rightarrow \infty} e^{t\dot{\gamma}(0) + no\left(\frac{t}{n}\right)} \\ &= e^{t\dot{\gamma}(0)} \end{aligned}$$

and in particular, this limit exists. □

**Example 4.81.** Suppose  $n = 1$ , and  $\gamma(t) = 1 + t$ . Then the proposition says that

$$e^t = \lim_{n \rightarrow \infty} \left(1 + \frac{t}{n}\right)^n$$

which we already knew.

**Proposition 4.82.** Let  $G \leq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup, and let  $M \in \mathfrak{g} = T_I G$ . Then  $e^M \in G$ .

*Proof.* Let  $\gamma : \mathbb{R} \rightarrow G$  be a continuously differentiable curve sending 0 to  $I$ , with  $\dot{\gamma}(0) = M$ . Then

$$e^M = \lim_{n \rightarrow \infty} \gamma\left(\frac{t}{n}\right)^n$$

and the limit on the right hand side exists. Each  $\gamma(t/n)^n$  lives in  $G$ , and since  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  is a closed subset, the limit lives in  $G$  too. □

**Corollary 4.83.** The continuous homomorphisms  $\gamma : \mathbb{R} \rightarrow G$  are exactly those given by  $\gamma_A(t) = e^{tA}$ , for some  $A$  in  $\mathfrak{g} = T_I G$ .

*Proof.* We first prove any such continuous homomorphism is of this form.

Any continuous homomorphism  $\gamma : \mathbb{R} \rightarrow G$  is in particular a continuous homomorphism to  $\mathrm{GL}_n(\mathbb{F})$  (since  $G \leq \mathrm{GL}_n(\mathbb{F})$ ), so by the earlier theorem,  $\gamma(t) = \gamma_A(t) = e^{tA}$  for some  $A \in M_n(\mathbb{F})$ . But  $\gamma_A$  is always continuously differentiable, so  $\dot{\gamma}_A(0) \in \mathfrak{g}$ . But this derivative is  $A$ , so  $A \in \mathfrak{g}$ .

Conversely, if  $A \in \mathfrak{g}$ , then  $tA \in \mathfrak{g}$  for all  $t \in \mathbb{R}$ , so by the previous proposition,  $\gamma_A(t) = e^{tA} \in G$  for all  $t$ . We also showed earlier that  $\gamma_A$  defines a continuous homomorphism  $\mathbb{R} \rightarrow \mathrm{GL}_n(\mathbb{F})$ . Since it lands in  $G$  when  $A \in \mathfrak{g}$ , this implies that  $\gamma_A$  is a continuous homomorphism  $\mathbb{R} \rightarrow G$ . □

This corollary says there is a bijection:

$$\mathfrak{g} \leftrightarrow \{ \text{continuous homomorphisms } \mathbb{R} \rightarrow G \}$$

sending  $A \in \mathfrak{g}$  to  $\gamma_A$ , and in the other direction sending  $\gamma$  to  $\dot{\gamma}(0)$ .

**Remark 4.84.** We could have defined  $\mathfrak{g}$  to be the set of continuous homomorphisms  $\mathbb{R} \rightarrow G$ . However with this definition it would be less clear how to define the bracket  $[\cdot, \cdot]$ , nor even why  $\mathfrak{g}$  is a vector space.

**Remark 4.85.** Let  $\gamma, \delta : \mathbb{R} \rightarrow G$  two continuous homomorphisms, where  $G \leq \text{GL}_n(\mathbb{F})$  is a closed subgroup. Then they're continuously differentiable, and  $\dot{\gamma}(0) + \dot{\delta}(0) \in \mathfrak{g} = T_I G$ , which corresponds, by taking exponentials, to some continuous homomorphism  $\xi : \mathbb{R} \rightarrow G$  (by defining  $\xi(t) = e^{t(\dot{\gamma}(0) + \dot{\delta}(0))}$ ).

Recall that when we showed  $\mathfrak{g}$  (which we were calling  $T_I G$  back then) is a vector space, we showed that  $\eta(t) = \gamma(t) \cdot \delta(t)$  is a continuously differentiable map  $\mathbb{R} \rightarrow G$  with

$$\dot{\eta}(0) = \dot{\gamma}(0) + \dot{\delta}(0) \in T_I G = \mathfrak{g}$$

But since  $G$  might not be abelian,  $\eta$  isn't necessarily a homomorphism. So we might not have that  $\xi = \eta$ , but  $\xi(t) = e^{t\dot{\eta}(0)}$ . The formula in the corollary lets us express  $e^{t\dot{\eta}(0)}$  in terms of  $\eta$ :

$$\begin{aligned} \xi(t) &= e^{t\dot{\eta}(0)} \\ &= \lim_{n \rightarrow \infty} \left( \gamma\left(\frac{t}{n}\right) \cdot \delta\left(\frac{t}{n}\right) \right)^n \end{aligned}$$

Lecture 25

**Theorem 4.86.** Let  $G \leq \text{GL}_n(\mathbb{F})$  be a closed subgroup.

There are open neighbourhoods  $U$  of 0 in  $\mathfrak{g} = T_I G$  and  $V = \exp(U)$  of  $I$  in  $G$  such that  $\exp$  restricts to a homeomorphism  $\exp|_U : U \rightarrow V$ .

Note that the earlier theorem was a special case of this; it was the case that  $G = \text{GL}_n(\mathbb{F})$ .

*Proof.* (Non-examinable.) First, we pick a complementary (real) vector subspace  $W$  to  $\mathfrak{g}$  in  $M_n(\mathbb{F})$ , i.e. a subspace  $W \leq M_n(\mathbb{F})$  such that  $\mathfrak{g} \oplus W = M_n(\mathbb{F})$  (i.e.  $\mathfrak{g}$  and  $W$  together span  $M_n(\mathbb{F})$ , and their intersection is  $\{0\}$ ).

Define a function  $\widetilde{\exp} : \mathfrak{g} \oplus W \rightarrow \text{GL}_n(\mathbb{F})$  to send  $(X, Y)$  (where  $X \in \mathfrak{g}$  and  $Y \in W$ ) to  $e^{-X} \cdot e^Y$ .

Note that since  $X$  and  $Y$  might not commute, this isn't necessarily the same as  $\exp$ , which sends  $(X, Y)$  to  $e^{X+Y}$ . So  $\widetilde{\exp}$  is a modified version of the exponential map.

Note that  $\exp$  and  $\widetilde{\exp}$  agree on  $\mathfrak{g}$ , so crucially if the theorem is true for  $\widetilde{\exp}$ , it's true for  $\exp$  too. So we'll prove the theorem with this modified version of the exponential map,  $\widetilde{\exp}$ .

**Claim 4.87.** The derivative of the modified exponential map at 0,  $d\widetilde{\exp}_0 : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ , is the identity.

The proof of this claim is similar to the proof that  $d\exp_0$  was the identity.

*Proof of claim.* Let  $(X, Y) \in \mathfrak{g} \oplus W = M_n(\mathbb{F})$ .

Let  $\gamma(t) = e^{tX} \cdot e^{tY} = \widetilde{\exp}(X, Y)$ . Then

$$\begin{aligned}\dot{\gamma}(0) &= \left( \frac{d}{dt} \Big|_{t=0} e^{tX} \right) \cdot e^{0 \cdot Y} + e^{0 \cdot X} \cdot \left( \frac{d}{dt} \Big|_{t=0} e^{tY} \right) \\ &= X \cdot I + I \cdot X \\ &= X + Y\end{aligned}$$

But by the chain rule,  $\dot{\gamma}(0) = d\widetilde{\exp}_0(X, Y)$ . So  $d\widetilde{\exp}_0(X, Y) = X + Y$ .  $\square$

Similarly to with  $\exp$ , by the inverse function theorem, there's a neighbourhood  $\tilde{U}$  of 0 in  $M_n(\mathbb{F})$  and a neighbourhood  $\tilde{V} = \widetilde{\exp}(\tilde{U})$  of  $I$  in  $\text{GL}_n(\mathbb{F})$  such that

$$\widetilde{\exp}|_{\tilde{U}}: \tilde{U} \rightarrow \tilde{V}$$

is a homeomorphism.

**Claim 4.88.** *There's a neighbourhood  $Z$  of 0 in  $\tilde{U} \subseteq M_n(\mathbb{F})$  such that*

$$\widetilde{\exp}^{-1}(G) \cap Z = \mathfrak{g} \cap Z$$

*In words, this says that for  $(X, Y) \in Z \subseteq \mathfrak{g} \oplus W$ , we have that  $\widetilde{\exp}(X, Y) \in G$  if and only if  $Y = 0$ .*

Assuming this second claim for now, let's prove the theorem, with  $U = \mathfrak{g} \cap Z$ .

By the first claim,  $\widetilde{\exp}|_Z: Z \rightarrow \text{Im}(\widetilde{\exp}|_Z)$  is a homeomorphism to a neighbourhood of  $I$  in  $\text{GL}_n(\mathbb{F})$ .

Restricting to  $U := Z \cap \mathfrak{g}$ , which is an open neighbourhood of 0 in  $\mathfrak{g}$ , this implies that

$$\widetilde{\exp}|_U: U \rightarrow \text{Im}(\widetilde{\exp}|_U)$$

is a homeomorphism. The right hand side is, by the second claim,  $G \cap \text{Im}(\widetilde{\exp}|_Z)$ , which is  $G \cap (\text{some open neighbourhood of } I)$ , so it's an open neighbourhood of  $I$  in  $G$ . This prove the theorem. It remains to prove this second claim.

*Proof of claim.* Suppose the claim isn't true. This means that for every neighbourhood  $Z$  of 0 in  $M_n(\mathbb{F})$ , there's some  $(X, Y) \in \mathfrak{g} \oplus W = M_n(\mathbb{F})$  with  $Y \neq 0$  and  $\widetilde{\exp}(X, Y) = e^X \cdot e^Y \in G$ .

This means there's a sequence of vectors  $(X_i, Y_i) \in \mathfrak{g} \oplus W$  converging to 0, such that  $Y_i \neq 0$  for all  $i$ , and such that  $\widetilde{\exp}(X_i, Y_i) \in G$  (i.e.  $e^{X_i} \cdot e^{Y_i} \in G$ ) for all  $i$ .

$-X_i \in \mathfrak{g}$  too, so  $e^{-X_i} \in G$ , so  $e^{-X_i} \cdot e^{X_i} \cdot e^{Y_i} = e^{Y_i} \in G$  too.

The idea now is to use this sequence  $Y_i$  to find some non-zero  $Y \neq 0 \in W$  such that  $e^{tY} \in G$  for all  $t$ . However this implies that  $Y \in \mathfrak{g}$ , but  $\mathfrak{g} \cap W = \{0\}$ , so this is a contradiction.

Since each  $Y_i \neq 0 \in W$  is non-zero, each  $Y_i/|Y_i| \in S^d$  (for  $d = \dim(W) - 1$ ), where  $S^d = \{y \in W \mid |y| = 1\}$ . These  $Y_i/|Y_i|$  are the  $Y_i$ , "normalised".

By sequential compactness of  $S^d$  after passing to some subsequence,

$$\frac{Y_i}{|Y_i|} \rightarrow Y \in W$$

for some  $Y \in S^d \subseteq W$  with  $|Y| = 1$ , so in particular  $Y \neq 0$ .

We'll show that  $e^{tY} \in G$  for all  $t$ , implying  $Y \in \mathfrak{g}$ , a contradiction as  $Y \neq 0 \in W$ .

Fix  $t \in \mathbb{R}$ . Choose integers  $m_i$  such that  $m_i|Y_i|$ ; we can do this by choosing the  $m_i$  such that

$$m_i|Y_i| \leq t \leq (m_i + 1)|Y_i|$$

for all  $i$ .

Now consider  $e^{m_i Y_i}$ . We have that  $e^{m_i Y_i} = (e^{Y_i})^{m_i} \in G$  since  $e^{Y_i} \in G$  for all  $i$ . But

$$m_i Y_i = (m_i |Y_i|) \frac{Y_i}{|Y_i|} \rightarrow tY$$

since the first term  $m_i |Y_i| \rightarrow t$  and the second term  $Y_i/|Y_i| \rightarrow Y$ .

This implies that  $e^{m_i Y_i} \rightarrow e^{tY}$ . So since  $G \subseteq \mathrm{GL}_n(\mathbb{F})$  is a closed subset,  $e^{tY} \in G$  for all  $t$ , so  $Y \in \mathfrak{g}$ , a contradiction since  $Y \neq 0 \in W$ , thus proving the claim.  $\square$

$\square$

Some corollaries, then some examples.

**Corollary 4.89.** *Let  $G \leq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup. Then for all  $A$  in  $G$ , there is an open neighbourhood  $U$  of  $A$  in  $G$  which is homeomorphic to an open subset of  $\mathbb{R}^d$ , where  $d$  is the dimension of  $G$ .*

*Proof.* When  $A = I$ , this follows from the theorem, since  $\mathfrak{g}$  is homeomorphic to  $\mathbb{R}^d$ .

When  $A \neq I$ , multiplication by  $A$ ,  $\cdot A : G \rightarrow G$ , is a homeomorphism, and sends an open neighbourhood of  $I$  (which is homeomorphic to an open subset of  $\mathbb{R}^d$ ) to an open neighbourhood of  $A$  in  $G$ .  $\square$

**Corollary 4.90.** *Let  $G \leq \mathrm{GL}_n(\mathbb{F})$  be a closed subgroup. Then  $G$  is a discrete topological space if and only if its dimension is 0.*

*Proof.* ( $\Leftarrow$ ) If the dimension of  $G$  is 0, every  $A \in G$  has a neighbourhood homeomorphic to  $\mathbb{R}^0$ , which is just a point. This means that all points in  $G$  are open, so  $G$  has the discrete topology.

( $\Rightarrow$ ) If  $G$  has the discrete topology, there's a neighbourhood  $U$  of  $I$  in  $G$ , homeomorphic to an open neighbourhood of  $\mathbb{R}^d$ , where  $d$  is the dimension of  $G$ .  $U \subseteq G$  is open, and  $G$  is discrete so  $U$  is too, so  $d = 0$ .  $\square$

**Example 4.91.** *For any closed subgroup  $G \leq \mathrm{GL}_n(\mathbb{F})$ , making the open neighbourhood in the theorem smaller if necessary, there's an open neighbourhood  $U$  of  $I$  in  $G$ , which is homeomorphic to a ball in  $\mathbb{R}^d$ .*

*In particular,  $U$  is path-connected.*

*Consider the topological group  $(\mathbb{Q}, +)$ . No open neighbourhood of 0 in  $\mathbb{Q}$  is path-connected. So  $(\mathbb{Q}, +)$  isn't isomorphic to a closed subgroup of any  $\mathrm{GL}_n(\mathbb{F})$ , i.e.  $(\mathbb{Q}, +)$  isn't a matrix Lie group.*

**Example 4.92.** *Define*

$$G := U(1) \times U(1) = \left\{ \begin{pmatrix} z & 0 \\ 0 & w \end{pmatrix} \mid z, w \in \mathbb{C}, |z| = |w| = 1 \right\}$$

*Let  $\gamma : \mathbb{R} \rightarrow G$  be the continuous homomorphism sending  $t \in \mathbb{R}$  to the matrix*

$$\gamma(t) := \begin{pmatrix} e^{it} & 0 \\ 0 & e^{i\sqrt{2}t} \end{pmatrix}$$

*You could replace  $\sqrt{2}$  with your favourite irrational number if you'd like. Then  $\gamma$  is injective, but no open neighbourhood of  $I \in \mathrm{Im}(\gamma)$  is path-connected.*

*So  $\mathrm{Im}(\gamma) \leq G$  is a subgroup but it's not closed.*

*Compare this to the earlier parts of the course: images of (group/ring/module) homomorphisms are sub(groups/rings/modules), but this doesn't hold for matrix Lie groups.*

**Example 4.93.** Let's compute the Lie algebra of  $O(n)$ ,  $\mathfrak{o}(n) = T_I O(n)$ .

First, let  $A \in \mathfrak{o}(n)$ . Then  $tA \in \mathfrak{o}(n)$  for all  $t \in \mathbb{R}$  too. So  $e^{tA} \in O(n)$  for all  $t$  too. This means that

$$\begin{aligned} e^{tA} \cdot (e^{tA})^T &= I \\ &= e^{tA} \cdot e^{tA^T} \end{aligned}$$

So since the inverse of  $e^{tA^T}$  is  $e^{-tA^T}$ , we have that  $e^{tA} = e^{-tA^T}$ . Since  $\exp$  isn't injective, we can't conclude yet that  $tA = -tA^T$ .

For  $t > 0$  small enough,  $tA$  and  $-tA^T$  both lie in an open neighbourhood  $U$  of 0 in  $M_n(\mathbb{F})$  on which  $\exp$  is in fact injective. So for  $t > 0$  small enough, we do have that  $tA = -tA^T$ .

Dividing by  $t$ , we see that  $A = -A^T$ , meaning  $A$  is skew-symmetric.

Conversely, suppose  $A$  is skew-symmetric; we'll show that  $A \in \mathfrak{o}(n)$ .

$A$  and  $A^T = -A$  commute, so for all  $t \in \mathbb{R}$

$$\begin{aligned} e^{tA} \cdot (e^{tA})^T &= e^{tA} \cdot e^{-tA} \\ &= e^0 \\ &= I \end{aligned}$$

So  $e^{tA} \in O(n)$ .

So  $\gamma_A(t) = e^{tA} : \mathbb{R} \rightarrow O(n)$  is a continuous homomorphism, so  $A \in \mathfrak{o}(n)$ .

Altogether, what we've shown is that  $\mathfrak{o}(n) \leq M_n(\mathbb{R})$  is the set of skew-symmetric matrices.

**Definition 4.94.** Let  $G \leq GL_n(\mathbb{F})$  and  $H \leq GL_n(\mathbb{F})$  be closed subgroups, and let  $\phi : G \rightarrow H$  be a continuous homomorphism.

We define a function  $\phi_* : \mathfrak{g} \rightarrow \mathfrak{h}$  as follows.

For  $A \in \mathfrak{g}$ , let  $\gamma_A(t) = e^{tA}$ . So  $\gamma_A : \mathbb{R} \rightarrow G$  is a continuous homomorphism.

Now  $\phi \circ \gamma : \mathbb{R} \rightarrow H$  is a composition of two continuous homomorphisms, so it's a continuous homomorphism. So  $\phi(\gamma(t)) = e^{tB}$  for some unique  $B \in \mathfrak{h}$ . We define

$$\phi_*(A) := B$$

Another (equivalent) way of saying this is that  $\phi_*(A)$  is defined by the equation

$$\gamma_{\phi_*(A)} = \phi \circ \gamma_A$$

This description will also be useful.

Plugging 1 into this equation, using  $\gamma_{\phi_*(A)}(1) = e^{\phi_*(A)}$  and  $\phi \circ \gamma_A(1) = \phi(e^A)$  implies that the two maps  $\exp \circ \phi_*$  and  $\phi \circ \exp$ , from  $\mathfrak{g}$  to  $H$ , are the same:

$$\exp \circ \phi_* = \phi \circ \exp : \mathfrak{g} \rightarrow H$$

Lecture 26

**Lemma 4.95.**  $\phi_*$  is a linear map  $\phi_* : \mathfrak{g} \rightarrow \mathfrak{h}$ .

*Proof.* Let  $A, B \in \mathfrak{g}$ , and  $\lambda \in \mathbb{R}$ .

We first show that  $\phi_*(\lambda A) = \lambda \phi_*(A)$ .

Recall that for any  $t \in \mathbb{R}$ ,

$$\begin{aligned} e^{t\phi_*(\lambda A)} &= \phi(e^{t\lambda A}) \\ &= e^{(t\lambda)\phi_*(A)} \\ &= e^{t(\lambda\phi_*(A))} \end{aligned}$$

Since this is true for all  $t \in \mathbb{R}$ , this implies that  $\phi_*(\lambda A) = \lambda \phi_*(A)$ .

We next show that  $\phi_*(A + B) = \phi_*(A) + \phi_*(B)$ .

We know that for all  $t \in \mathbb{R}$ ,

$$\begin{aligned}\gamma_{A+B}(t) &= e^{t(A+B)} \\ &= \lim_{n \rightarrow \infty} \left( \gamma_A \left( \frac{t}{n} \right) \cdot \gamma_B \left( \frac{t}{n} \right) \right)^n\end{aligned}$$

Here we're using the earlier lemma which said that if  $\delta : \mathbb{R} \rightarrow G$  sends 0 to  $I$  and is continuously differentiable, then  $e^{t\dot{\delta}(0)} = \lim_{n \rightarrow \infty} \delta(t/n)^n$ , as well as the fact that if  $\delta(t) = \gamma_A(t) \cdot \gamma_B(t)$ , then  $\dot{\delta}(0) = A + B$ ; this is what we used when we proved that  $A + B \in \mathfrak{g}$ .

The same equation applied to  $\phi_*(A)$  and  $\phi_*(B)$  says that

$$\begin{aligned}\gamma_{\phi_*(A)+\phi_*(B)} &= \lim_{n \rightarrow \infty} \left( \gamma_{\phi_*(A)} \left( \frac{t}{n} \right) \cdot \gamma_{\phi_*(B)} \left( \frac{t}{n} \right) \right) \\ &= \lim_{n \rightarrow \infty} \left( \phi \left( \gamma_A \left( \frac{t}{n} \right) \right) \cdot \phi \left( \gamma_B \left( \frac{t}{n} \right) \right) \right)^n && \text{since } \gamma_{\phi_*(A)}(t/n) = \phi(\gamma_A(t/n)) \\ &= \lim_{n \rightarrow \infty} \phi \left( \left( \gamma_A \left( \frac{t}{n} \right) \cdot \gamma_B \left( \frac{t}{n} \right) \right)^n \right) && \text{since } \phi \text{ is a homomorphism} \\ &= \phi \left( \lim_{n \rightarrow \infty} \left( \gamma_A \left( \frac{t}{n} \right) \cdot \gamma_B \left( \frac{t}{n} \right) \right)^n \right) && \text{since } \phi \text{ is continuous} \\ &= \phi(\gamma_{A+B}(t)) && \text{by the previous discussion} \\ &= \gamma_{\phi_*(A+B)}(t)\end{aligned}$$

□

**Lemma 4.96.** *Let  $G \leq \text{GL}_n(\mathbb{F})$  and  $H \leq \text{GL}_n(\mathbb{F})$  be closed subgroups, and let  $\phi : G \rightarrow H$  be a continuous homomorphism.*

*Then  $\phi_* : \mathfrak{g} \rightarrow \mathfrak{h}$  satisfies*

$$\phi_*([A, B]) = [\phi_*(A), \phi_*(B)]$$

*for all  $A, B \in \mathfrak{g}$ .*

*In other words,  $\phi_*$  is a homomorphism of Lie algebras.*

**Lemma 4.97.**  *$\phi_*$  is a linear map. Furthermore  $\phi_*([M, N]) = [\phi_*(M), \phi_*(N)]$ , for all  $M, N$  in  $\mathfrak{g}$ . In particular,  $\phi_*$  is a homomorphism of Lie algebras.*

*Proof.* We first show  $\phi_*(tA) = t\phi_*(A)$  for  $t \in \mathbb{R}$ .

Let  $\lambda_t : \mathbb{R} \rightarrow \mathbb{R}$  be the homomorphism given by multiplication by  $t$ . Then  $\gamma_{tA} = \gamma_A \circ \lambda_t$ , and so

$$\gamma_{\phi_*(tA)} = \phi \circ \gamma_A \circ \lambda_t = \gamma_{\phi_*(A)} \circ \lambda_t = \gamma_{t\phi_*(A)}$$

So  $\phi_*(tA) = t\phi_*(A)$ .

We next show  $\phi_*(A + B) = \phi_*(A) + \phi_*(B)$ . We know that

$$\gamma_{A+B} = \lim_{n \rightarrow \infty} (\gamma_A(t/n) \gamma_B(t/n))^n$$

and similarly

$$\gamma_{\phi_* A + \phi_* B}(t) = \lim_{n \rightarrow \infty} (\phi(\gamma_A(t/n))\phi(\gamma_B(t/n)))^n$$

which equals  $\phi(\gamma_{A+B}(t))$  since we can take the  $\phi$  outside of the brackets since it's a homomorphism, and outside the limit since it's continuous. Together this says that  $\gamma_{\phi_*(A+B)} = \phi \circ \gamma_{A+B} = \gamma_{\phi_*(A) + \phi_*(B)}$ .

We next show  $\phi_*$  respects the Lie bracket. Let  $A, B \in T_I G$ . Then letting  $\delta_s(t) = \gamma_A(s)\gamma_B(t)\gamma_A(-s) = e^{sA}e^{tB}e^{-sA}$ , we have that  $\dot{\delta}_s(0) = e^{sA}Be^{-sA}$ , which, as  $s$  varies, defines a  $C^1$  path in  $\mathfrak{g}$  whose derivative at 0 is  $[A, B]$ .

We define  $\beta_s(t) = \gamma_{\phi_*(A)}(s)\gamma_{\phi_*(B)}(t)\gamma_{\phi_*(A)}(-s)$ , so the path  $\dot{\beta}_s(0)$  is a  $C^1$  path in  $\mathfrak{h}$  whose derivative at 0 is  $[\phi_*(A), \phi_*(B)]$ . Then since  $\phi \circ \delta_s = \beta_s$ ,  $\phi_*(\dot{\delta}_s(0)) = \dot{\beta}_s(0)$ , so taking derivatives at  $s = 0$  implies the result.  $\square$

**Lemma 4.98.** *Let  $\phi : G \rightarrow H$  and  $\psi : H \rightarrow K$  be continuous homomorphisms between closed subgroups of  $\mathrm{GL}_n(\mathbb{F})$ . Then  $\psi_* \circ \phi_* = (\psi \circ \phi)_*$ .*

*Proof.* Let  $A$  lie in  $\mathfrak{g}$ . Write  $\gamma_B$  for the homomorphism given by  $t \mapsto e^{tB}$ . Then  $\phi_*(A)$  is determined by the equation  $\gamma_{\phi_* A} = \phi \circ \gamma_A$ , and  $\psi_*(\phi_*(A))$  is determined by the equation  $\psi_*(\phi_*(A)) = \psi \circ \gamma_{\phi_*(A)} = \psi \circ \phi \circ \gamma_A$ .

$(\psi \circ \phi)_*(A)$  is determined by the equation  $\gamma_{(\psi \circ \phi)_*(A)} = \psi \circ \phi \circ \gamma_A$ . So  $(\psi \circ \phi)_*(A) = \psi_*(\phi_*(A))$ .  $\square$

**Corollary 4.99.** *If  $G, H \leq \mathrm{GL}_n(\mathbb{F})$  are closed subgroups which are isomorphic as topological groups, then their Lie algebras  $\mathfrak{g} = T_I G$  and  $\mathfrak{h} = T_I H$  are isomorphic.*

*Proof.* Let  $f : G \rightarrow H$  and  $g : H \rightarrow G$  be continuous homomorphisms which are inverse to each other. Then  $f_*$  and  $g_*$  are inverse isomorphisms.  $\square$

## 4.6 Abelian matrix Lie groups (Non-examinable)

Similarly to how we can classify finitely-generated abelian groups, we can classify connected abelian matrix Lie groups. This is what we will do in this section.

**Lemma 4.100.** *Let  $\phi : G \rightarrow H$  be a homomorphism between matrix Lie groups. Then  $\ker \phi$  is a closed topological subgroup of  $G$  and hence is itself a matrix Lie group, and*

$$T_I \ker \phi = \ker(\phi_* : T_I G \rightarrow T_I H)$$

*Furthermore if  $\phi$  is surjective,  $\phi_*$  is also surjective.*

*Proof.* Since  $\phi$  is continuous, if  $(x_i)_i$  is a sequence in  $\ker \phi$  converging to some  $x$  in  $G$ , we must have that  $\phi(x) = I$  and so  $x$  lies in  $\ker \phi$  too. So  $\ker \phi$  is a closed subgroup of  $G$ . Viewing  $T_I G$  as the set of continuous homomorphisms  $\gamma : \mathbb{R} \rightarrow G$ ,  $\ker \phi$  consists of those  $\gamma$  such that  $\phi \circ \gamma$  is constant, which holds if and only if  $\gamma$  lies in  $\ker \phi$ , if and only if  $\gamma$  lives in  $T_I \ker \phi$ .  $\square$

**Lemma 4.101.** *Let  $G$  be a matrix Lie group, and  $K \trianglelefteq G$  a discrete normal subgroup.*

*Then  $G/K$  is a topological group, the quotient map  $\pi : G \rightarrow G/K$  is a continuous homomorphism, and there is a neighbourhood  $U$  of  $I$  in  $G$ , such that  $\pi|_U$  is a homeomorphism onto its image.*

*Proof.* Multiplication  $G \times G \rightarrow G$  sends  $K \times K$  to  $K$  and so determines a continuous map  $(G \times G)/(K \times K) \rightarrow G/K$ .  $(G/K) \times (G/K) = (G \times G)/(K \times K)$  so the product for  $G/K$  is continuous; a similar argument shows the inverse is continuous. So  $G$  is a topological group.

Choose a compact neighbourhood  $C$  of  $I$  in  $G$  such that  $\pi|_C$  is an injection (this exists since  $K$  is discrete). Then  $\pi|_C$  is a continuous bijection onto its image, and  $C$  and  $\pi(C)$  are both Hausdorff, so  $\pi|_C$  is a homeomorphism onto its image. Now take  $U$  to be the interior of  $C$ .  $\square$

We now have a version of the first isomorphism theorem.

**Proposition 4.102.** *Let  $\phi : G \rightarrow H$  be a surjective homomorphism of matrix Lie groups, with discrete kernel. Then  $G/\ker \phi$  and  $H$  are isomorphic topological groups. In particular,  $G/\ker \phi$  is a matrix Lie group.*

*Proof.* Since  $\phi$  respects the equivalence relation defined by  $\ker \phi$ ,  $\phi$  descends to a continuous homomorphism  $\tilde{\phi} : G/\ker \phi \rightarrow H$ . It is injective, and surjective by assumption.

Since  $\ker \phi$  is discrete,  $\phi_*$  is an injection, and since  $\phi$  is surjective,  $\phi_*$  is also surjective. So  $\phi_*$  is an isomorphism, and so there is a neighbourhood  $U$  of  $I$  in  $G$  such that  $\phi|_U$  is a homeomorphism onto its image.

By the previous lemma, there is a neighbourhood  $V$  of  $I$  in  $G/\ker \phi$  such that  $\tilde{\phi}|_V$  is a homeomorphism onto its image. So  $\tilde{\phi}^{-1}$  is continuous in a neighbourhood of  $I$ .

Let  $A \in H$ . Then since multiplication by a fixed element is a homeomorphism,  $\tilde{\phi}^{-1}$  is continuous in a neighbourhood of  $A$ . So  $\tilde{\phi}^{-1}$  is continuous, so  $\tilde{\phi}$  is a homeomorphism.  $\square$

**Example 4.103.** *Let  $G = \mathbb{R}^2/\mathbb{Z}^2$ , and let  $\phi : \mathbb{R} \rightarrow G$  send  $t$  to  $(t, \pi t)$ .*

*$\phi$  is a continuous homomorphism, and it's injective. However  $\phi$  isn't a homeomorphism onto its image: take a small open neighbourhood  $U$  of  $0$  in  $\text{Im } \phi$ . Then  $U$  contains no path-connected neighbourhood of  $0$ . However every neighbourhood of  $0$  in  $\mathbb{R}$  contains a path-connected neighbourhood of  $0$ . So these spaces can't even be homeomorphic.*

*What this tells us is that it's not always true that  $G/\ker \phi$  is isomorphic to  $\text{Im } \phi$  when  $\ker \phi$  is discrete.*

**Lemma 4.104.** *Let  $G$  be an abelian matrix Lie group. Then the commutator vanishes on  $\mathfrak{g}$ .*

*Furthermore  $\exp : \mathfrak{g} \rightarrow G$  is a homomorphism of topological groups.*

*Proof.* Consider  $A, B \in \mathfrak{g}$ . Let  $\delta_s(t) = e^{sA}e^{tB}e^{-sA}$ , a path in  $G$  for each  $s$ . Recall from earlier that  $\dot{\delta}_s(0) \in \mathfrak{g}$  for all  $s$ , and this is differentiable in  $s$  with

$$\left. \frac{d}{ds} \right|_{s=0} \dot{\delta}_s(0) = [A, B]$$

Since  $G$  is abelian,  $\delta_s(t) = e^{tB}$ , so  $\dot{\delta}_s(0) = B$  for all  $s$ , so  $[A, B] = 0$ .

Choose  $A, B \in \mathfrak{g}$ . Since  $[A, B] = 0$ ,  $A$  and  $B$  commute, so  $e^{A+B} = e^A e^B$ , so  $\exp$  is a homomorphism, which we already know is continuous.  $\square$

**Lemma 4.105.** *Let  $G$  be a connected abelian matrix Lie group. Then  $\exp : \mathfrak{g} \rightarrow G$  has discrete kernel.*

*Proof.* Suppose not. Then there is a sequence  $(x_i)_i$  in  $\ker \exp$  such that  $x_i \rightarrow x$  for some  $x \in \ker \exp$ , such that  $x_i \neq x$  for all  $i$ . By replacing  $x_i$  with  $x_i - x$  if necessary, we may assume  $x = 0$ .

So  $\exp(x_i) = I$  for all  $i$ , and  $x_i \rightarrow 0$ . But we know that  $\exp$  is a homeomorphism and in particular injective on some neighbourhood of  $0$ , so this is a contradiction.  $\square$

**Lemma 4.106.** *Let  $G$  be a connected abelian matrix Lie group. Then  $\exp : \mathfrak{g} \rightarrow G$  is surjective.*

*Proof.* We know there is some neighbourhood  $U$  of  $I$  which is in  $\text{Im}(\exp)$ . Since  $G$  is connected, it is generated by any neighbourhood of  $I$  (proof is on the exercise sheet), so in particular any  $g \in G$  can be written as a product  $g = e^{A_1} \dots e^{A_k}$  for some  $A_1, \dots, A_k \in \mathfrak{g}$ .

Since  $\exp$  is a homomorphism, we have that  $g = e^{A_1 + \dots + A_k}$ .  $\square$

**Definition 4.107.** *A lattice is a discrete subgroup  $\Lambda$  of a finite-dimensional real vector space  $V$ .*

**Lemma 4.108.** *Let  $\Lambda \leq V$  be a lattice. Then there is a basis  $e_1, \dots, e_d$  of  $V$  such that  $\Lambda = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_k$  for some  $k \leq d$ . In particular,  $\Lambda$  is finitely-generated.*

*Proof.* We induct on the dimension of  $V$ . If  $V$  is 0-dimensional, this is clear. Similarly if  $\Lambda = 0$  this is clear, so assume  $\Lambda \neq 0$ .

Choose an inner product on  $V$ , and let  $e$  be a non-zero element of  $\Lambda$  which minimises  $|e|$  (this exists since  $\Lambda$  is discrete). Let  $V' = V/\langle e \rangle$ ,  $\pi : V \rightarrow V'$  the quotient map, and  $\Lambda' = \pi(\Lambda)$ . Then  $\Lambda'$  is a subgroup of  $V'$ , which we claim is discrete. Indeed if not, there is some non-zero sequence  $\lambda'$  in  $V'$  converging to 0. Choose  $\lambda_i \in \Lambda$  such that  $\pi(\lambda_i) = \lambda'_i$  for all  $i$ , and by adding multiples of  $e$  if necessary, we can assume  $\lambda_i$  converges to 0 too, which is a contradiction.

By the induction hypothesis, there is a basis  $e'_1, \dots, e'_d$  of  $V'$  such that  $\Lambda' = \mathbb{Z}e'_1 + \dots + \mathbb{Z}e'_k$ . Choose  $e_i$  such that  $\pi(e_i) = e'_i$  for each  $i$ . Then  $e, e_1, \dots, e_d$  is a basis for  $V$ , and  $\Lambda = \mathbb{Z}e + \mathbb{Z}e_1 + \dots + \mathbb{Z}e_k$ .  $\square$

We are now ready to prove the theorem.

**Theorem 4.109.** *Let  $G$  be a connected abelian matrix Lie group.*

*Then  $G$  is isomorphic to  $(\mathbb{R}/\mathbb{Z})^i \times \mathbb{R}^j$  for some  $i, j \geq 0$ .*

*$G$  is compact if and only if  $j = 0$ .*

The matrix Lie group  $(\mathbb{R}/\mathbb{Z})^n$  is called the *torus* of dimension  $n$ .

*Proof.* Consider the homomorphism  $\exp : \mathfrak{g} \rightarrow G$ . Since this is a surjective homomorphism with discrete kernel,  $G$  is isomorphic to  $\mathfrak{g}/\ker \exp$ .

$\ker \exp$  is a lattice in  $\mathfrak{g}$  so we can choose a basis of  $\mathfrak{g}$ , say  $e_1, \dots, e_d$  such that  $\ker \exp = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_k$  for some  $k$ .

Therefore  $G$  is isomorphic to  $(\mathbb{R}^d)/(\mathbb{Z}^k)$ , which is isomorphic to  $(\mathbb{R}/\mathbb{Z})^k \times \mathbb{R}^{d-k}$ .

The statement about compactness is because  $(\mathbb{R}/\mathbb{Z})^i \times \mathbb{R}^j$  is compact if and only if  $j = 0$ .  $\square$